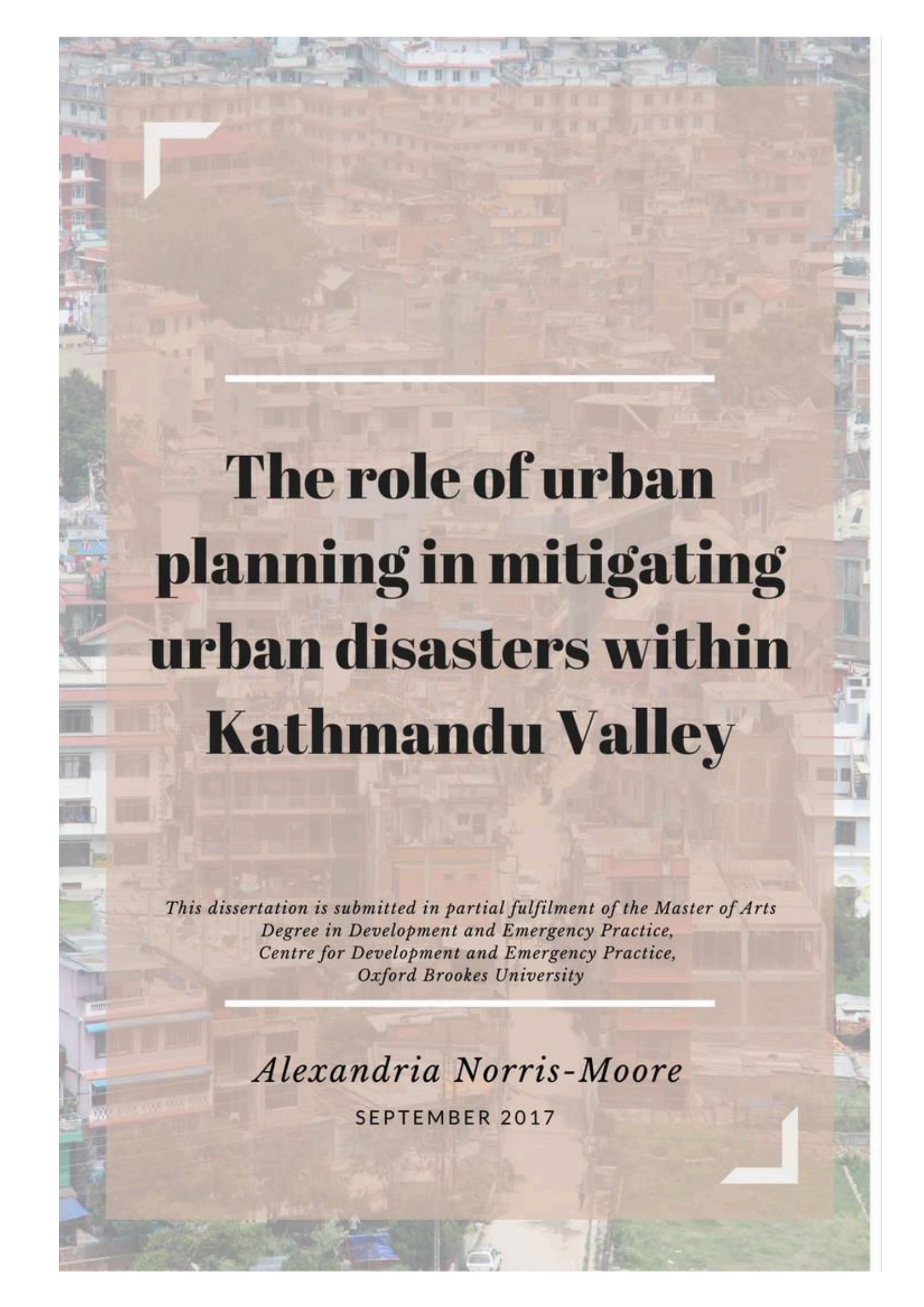




The role of urban planning in mitigating urban disasters within Kathmandu Valley

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ABSTRACT

The world suffers the loss of large numbers of people and infrastructure each year due to natural disasters. As people migrate more into cities natural disasters are having a greater impact on our urban spaces.

This study critically investigates the role of urban planning in mitigating urban disasters within Kathmandu Valley. Kathmandu Valley is geographically and inherently prone to natural disaster events. The amalgamation of unplanned urban development, irregular and substandard housing development, loss of open space, and decreased liveability has served to increase the vulnerability of Kathmandu Valley and its people. The earthquake that hit Nepal on 25 April 2015 took the lives of 8,693 Nepalese, and caused an estimated USD 7.0 billion damage to infrastructure (GoN, 2015a; Dahal, 2016). This paper argues that effective urban planning, smoother integration and implementation with disaster management plans, would have minimised the impact of the earthquake.

To investigate this research area an extensive literature review was undertaken with an analysis of Kathmandu Valley's land use and urban development and followed up by interviews with experts in the field.

By assessing the role land-use planning and management processes and provisions have to play in mitigating disasters within a context of high population and rapid urbanisation, like what was seen in Kathmandu Valley before 2015, this dissertation emphasises the need to prioritise urban planning in order for sustainable resilient urban development to be facilitated.

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My mom and dad, for everything. I wouldn't have got this far without you.

Front Cover Photo Credit: Norris-Moore, A. 2017. Kathmandu City, Nepal

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STATEMENT OF ORIGINALITY

This thesis is the result of my own independent work/investigation, except where otherwise stated.

Other sources are acknowledged by explicit references.

Signed.....

Alexandria Norris- Moore

Date: 25 September 2017

I hereby give consent for my thesis, if accepted, to be available for photocopying and for inter-library loan, and for the title and summary to be made available to outside organisations.

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RESEARCH CONSIDERATIONS

Limitations

Nepal provides some fundamental logistical challenges that are amplified by the limited time frame and location. Interviews were integral to the approach depending primarily on contacts and availability of interviewees. This required a flexible arrangement. The main limitation of the research relates to the extent of analysis and research that can be covered within the dissertation period of four months.

Ethics

This dissertation involved human participants. A Form TDE E1 showing ethics review approval, has been attached to this dissertation in **Appendix 4**.

All research participants were given appropriate information around the purpose, methods and intended uses of the research as well as what the risks and benefits of being a participant in the research entails. Participants were voluntary and free from coercion and undue influence. Their rights, dignity and autonomy has been respected and appropriately protected.

All interviewees signed a consent form which included a background note on the details of the research (**Appendix 3**). The research is independent and there are no foreseen conflicts of interest or partiality.

Exegesis

During the report, “urban planning” will be defined as a process ‘where people make decisions about intended future outcomes, how outcomes are to be accomplished, and how success is to be measured and evaluated’ (Emergency Management Australia, 2002:20) (See **Appendix 2** for detailed urban planning definition).

Exclusions

The scope of the study does not extend to a consideration of all historical planning legislation in Nepal nor does it review all government departments of Nepal because of lack of time and resources. As discussed in the Literature Review below, most previous work in this area has concentrated on disaster risk management.

‘Local governments’ as the authority at local level within Kathmandu before the 2015 constitution had no legal power and the politically accurate term for these authorities is ‘local bodies’.

ABBREVIATIONS

HRRP	Housing Recovery and Reconstruction Platform
GoN	Government of Nepal
MoFA	Ministry of Foreign Affairs
MoFALD	Ministry of Federal Affairs and Local Development
MoHA	Ministry of Home Affairs
MoPPWTM	Ministry of Physical Planning, Works and Transport Management
MoUD	Ministry of Urban Development
MPPW	Ministry of Physical Planning and Works
NPC	National Planning Commission
UN	United Nations

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Chapter One: Introduction

City blocks collapsed, century-old monuments were reduced to rubble, and apocalyptic cracks ran through the roads. How did we get to here?

Background

“At no time in human history have so many people lived in cities clustered around seismically active areas... poor land-use planning; environmental management; and a lack of regulatory mechanisms both increase the risk and exacerbate the effects of disasters.”¹

Increasingly cities around the world are facing impacts from large scale natural disasters (Ammann, 2012; Gencer, 2013). These cities are also facing a dominant phenomenon of urbanisation², especially within developing countries (Thapa et al, 2008; ASD, 2012). Climate change is likely to exacerbate the situation with migration influxes into cities and increases of natural disasters (Galvovic, 2010). There is now a growing worldwide concern about the significant impacts of natural disasters in many countries. International organisations are now focusing finances, resources and expertise on ways to reduce the risk of these disasters (Galvovic, 2010; HRRP, 2016). This imperative is keenly felt in the highly populated and urbanised cities in Asia, where disasters are increasingly effecting the lives of people.

Asia

Asia is home to 53 per cent of the world’s urban population. Additionally, sixteen of the world’s twenty-eight mega-cities are located in the continent (UNDESA, 2014). Asia is hosting the most people in an urban context, with half the world’s slum settlements within these mega-cities (Gencer, 2013). These are the people who are most vulnerable to disasters and tend to live in the fragile areas of the cities (Shah and Ranghieri, 2012; Gencer, 2013). Moreover, the region is considered to have the fastest rate of increased disasters for any region in the world (Shah and Ranghieri, 2012); in 2015, Asia accounted for more than 47 percent of the world’s 344 disasters, with 160 disasters in total (UNESCAP, 2015). Thus highlights the need for managing urban growth for resilience.

¹ Annan, K. 2002. Forward to *Living with Risk: A Global Review of disaster Reduction Initiatives*, (UN/ISDR). Quoted in: UN/ISDR, 2003. *Disaster Risk Reduction and Sustainable Development*. A background paper for the World Summit on Sustainable Development: <http://www.unisdr.org> (2006): 1.

² Urbanisation is defined as an increasing proportion of a population living in urban centres. (Satterthwaite, 2005; Gencer, 2013). It is important to note that government’s set their own population benchmarks standards to determine ‘urban’ and therefore it is difficult to accurately measure the world’s urban population.

Nepal

In 2006, Nepal was second amongst developing countries for frequency of large-scale disasters since 1985 (DFID, 2006; Jones et al, 2013). It is considered one of the most disaster-prone countries in the world, ranked 11th in 2012 among 150 countries (UNDP, 2012; Center for Excellence, 2015; UNDP/BCPR, 2004). Nepal also has one of the highest rates of urbanisation within Asia and the Pacific (ADB/ICMOD, 2006; Thapa et al., 2008; ASD, 2012). Due to its geological position, Nepal is prone to earthquakes, floods, landslides, fires, high winds and the impacts of global warming (Paudel et al., 2003; JICA, et al. 2012; MoUD, 2015). These hazard impacts are intensified when combined with the massive migrant movements to urban areas for economic and social benefits, and has led to large scale uncontrolled development (Sharma, 2006).

Figure 1: Map of Nepal



(Source: World Economic Forum, 2015)

Kathmandu Valley

Kathmandu, capital of Nepal, forms the core of the nation's most populous urban region, with Kathmandu Valley acting as the hub of Nepal's urbanisation (Thapa et al., 2008; MoUD, 2017; UNDP, 2015, HRRP, 2016). Kathmandu Valley has experienced exponential growth resulting in poorly planned infrastructure and safe housing, due to the lack of capacity to plan adequately and control

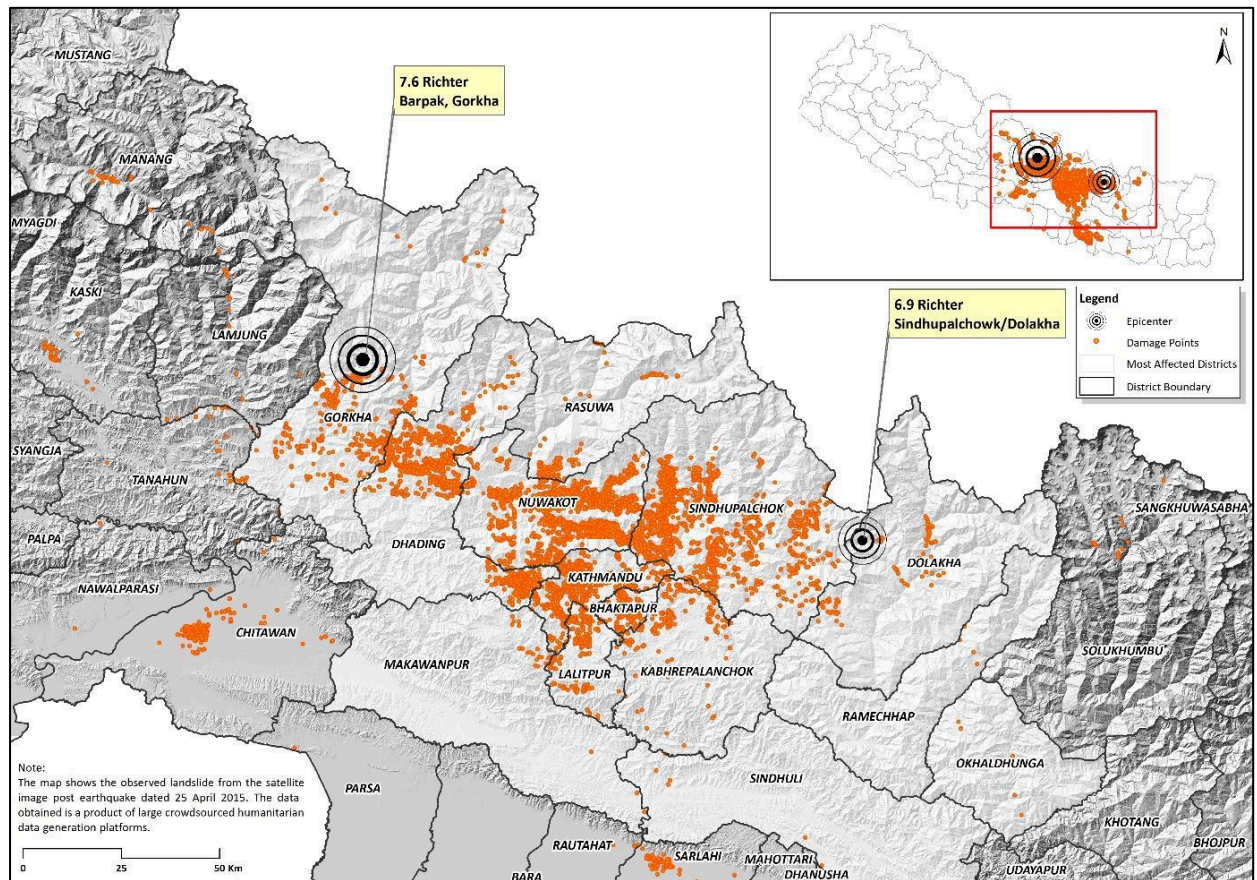
development (UNDP, 2012; WB, 2013; HRRP, 2016). This rapid unplanned and unsystematic urbanisation has resulted in increasing the vulnerability of the population in Kathmandu Valley, especially to environmental disasters (ASD, 2012). A UN report in 2012 identified that ‘the density of buildings, population and their extreme vulnerability in the valley are indicators of a large scale urban catastrophe waiting to happen with a level of destruction that is unprecedented’ (UNDP, 2012) (See **Appendix 1** for more details on background of Kathmandu Valley).

As a fast-growing valley due to economic, spatial, and mobility trends, several plans and concepts have been developed for the region (Thapa et al., 2007; MoUD, 2017). These will be examined as part of this study.

2015 Earthquake

The most recent disaster in Nepal, reflective on the urban context of Kathmandu Valley, was an earthquake with a magnitude of 7.8 that hit on April 25, 2015, with an epicentre in the Barpak Village of Gorkha district, 77km northwest of Kathmandu (UNOCHA, 2015; MoUD, 2015; HRRP, 2015). There were powerful aftershocks that followed including another powerful quake in Dolaka district on May 12, 2015 with a magnitude of 7.3 (UNOCHA, 2015; HRRP, 2016). **Figure 2** below shows the locations of the earthquakes. Although the earthquakes epicentres were not in Kathmandu Valley, the area itself suffered huge losses economically and physically, including through a multitude of injuries and deaths (HRRP, 2016; World Economic Forum, 2015). Studies since the earthquake highlight that poorly planned and developed infrastructure contributed to the amount of destruction to the built environment and also the deaths correlating to that destruction (UNDP, 2015; HRRP, 2016). This report will be looking into this in further detail.

Figure 2: Location of Damaged Buildings/Infrastructure from the Earthquakes in Nepal in 2015



(Source: UNDP, 2015)

The study done by the government's elected Housing Recovery and Reconstruction Platform (HRRP) in 2016, found that out of the 500,000 estimated houses that were fully damaged, 33% were within urban areas (HRRP, 2016; World Economic Forum, 2015). Of the 'urban areas' majority are within the Kathmandu Valley. A detailed analysis of the earthquake's devastation on housing within Kathmandu Valley can be found in the table below.

Table 1: Building Damage of Seven sectors of Studied Areas undertaken by HRRP

Note: A full assessment of damage across all Kathmandu Valley has not been undertaken

	Kathmandu Core (Kathmandu Core, Bhaktapur Core, Lalitpur Core, Thimi Core)	Peri-Urban (Gongabu)	Densely Settled area (Ghattekulo)	Historical Towns in Fringe (Panga, Khokana, Sankhu, Bungamati)	Total
Building Damage	27,843	1,363	25	5,063	34,294
Full	15,120	150	25	3,724	19, 019
Partial	12,723	1,213	0	1,339	15,275

(Source: HRRP, 2016)

Although **Table 1** does not include all urban areas in Kathmandu Valley or all infrastructure damage (water, roads, etc.), it does give an insight of the damage that occurred, and the study provided some findings as to why the damage was so prolific: in the core and historical towns in the fringe areas, the majority of the old houses were “built with unreinforced brick masonry were either completely collapsed or damaged due to vertical division and haphazard renovation and construction, lack of timely maintenance, poor strength of binding material and absence of tie beam around the house” (HRRP, 2016). As a result of poor building standards and urban governance, the Valley saw high death toll figures (HRRP, 2016). Modern houses within the Valley are mostly Reinforced Cement Concrete (RCC) within the peri-urban places. These were significantly damaged as a result of “poor construction, inadequate placement of rods and illegal construction” (HRRP, 2016). Apartments were also found to have questionable structural integrity and currently have been evacuated until this can be further investigated (HRRP, 2016).

Significance of the Study

The only way to eliminate disaster risk is to keep people out of harm’s way. (Glavovic, 2010, n.p.)

Urban planning provides for long-term strategies on how a city will grow factoring in sustainable principles like social, economic and environmental. In the last few decades, disasters have been occurring in urban contexts where there is not a prominent long-term planning strategy for that region. The lack of resilient disaster, land-use and strategic planning, political instabilities and poor implementation have resulted in poorly designed cities (UNHSP, 2004; Gencer, 2013; IPCC, 2001; Fillion et al., 2015).

Aims and Objectives of the research

This paper aims to highlight the importance of effective urban planning for disaster prone cities to instil urban resilience, manage vulnerability and overall provide long-term sustainable cities. The research focuses on Nepal Nepal's earthquake in 2015 provided an opportunity to examine the importance of strong, safe, and planned urban development.

The research will investigate how Kathmandu Valley has developed before the earthquake struck in 2015. Looking at factors such as geography, demographics, land use changes, politics, governance and policy and regulations. This purpose of this research is to investigate the role land-use planning and management processes and provisions have to play in mitigating disasters like what was seen in Kathmandu Valley in 2015.

Research Questions

Key Themes and Research questions

Theme	Research Question	Sub- Questions	Research Methodology
Urban Land Use	How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?	How has the link between unplanned urban development, irregular and substandard housing development, loss of open space and decreased liveability increased the vulnerability of Kathmandu to disasters?	Document analysis: Land Use Study Key informant interviews
Urban Development (Regulation and Policy)	What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?	How effective were urban planning documents in reducing damages in the 2015 earthquake? To what extent did the planning and development documents change the location, type, and intensity of land uses?	Document analysis: Urban Development Study Key informant interviews
Urban Governance	What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?	Who are the actors? What are the key institutions? What are the steps involved in decision-making? What are the power dynamics? Which are the different levels of decision-making?	Document analysis: Urban Development Study Key informant interviews
Lessons for Urban Planning of Kathmandu Valley	What is the role for planners in reducing disaster risks within a city? How can they help make a city resilient? (especially within context of developing countries)		Based on analysis of other thematic outcomes.

Synopsis

The structure of this report is as follows:

The second chapter gives a brief overview of relevant literature relating to concepts building the theoretical framework for the study; urban disasters, urban vulnerability, disaster risk reduction and mitigation of disasters through urban planning. Different approaches to conceptualisation of mitigating urban hazards are discussed, including their practical implications for hazard reduction strategies with regards to design and implementation. This section defines concepts relevant to mitigating urban hazards through urban planning strategies aiming at governance, spatial planning, legislation and coordination as the focus of social, physical and environmental protection for a city.

The third chapter provides details of the methodological approach undertaken to answering the research questions. The methodology section details the research purpose, approach and strategy for all the forms of data collection.

After framing the research focus within relevant literature, the fourth chapter applies the defined concepts to the case of Kathmandu Valley, Nepal. In this chapter, specific characteristics of urban planning in mitigating hazards within Nepal are explained, such as the supportive, legal arrangements in Nepal, the framework for plans and implementation. The chapter is broken up into three parts. First, Urban Land Use Study addressing geological, demographic, land use patterns and development activities factors within Nepal. The second part is an Urban Development Study drawing on Nepalese legislation, policy and plans while evaluating the legal frameworks and implementation issues. The third part continues exploring implications and implementation of urban planning within Kathmandu Valley through collected data from a list of key interviewees undertaken during a field visit to Kathmandu Valley in June to July 2017.

The paper concludes with highlighted key findings from the research tying together the research questions, the data uncovered, and the previous research and models as well as the arguments to look at further gaps and research that needs to be addressed.

Chapter Two: Literature Review

In this chapter, I will give a brief overview of the current literature in areas relevant to the topic; urban disasters, urban vulnerability, disaster risk reduction and mitigation of disasters through urban planning. Different approaches to conceptualisation of mitigating urban hazards are discussed. Including their practical implications for hazard reduction strategies with regards to design and implementation. Drawing on the urban planning definition (see Appendix 2), this section defines the concepts relevant for mitigating the causes, the consequences and recovery of the urban environment from calamities through the use of urban planning strategies. Integral to these strategies are the principles of governance, spatial planning, legislation and coordination as the fundamentals for social, physical and environmental protection for a city. This review highlights gaps (key concepts) in the literature to designate and illuminate areas of focus for the current study.

The resources for the literature review draws on a variety of sources. Including secondary data such as reports, action plans, and reflection papers of governmental entities, non-governmental organizations, private sector businesses, and the online news.

The Concept of Urban Disasters

In the current literature, though urban disasters are widely agreed to be a multi-dimensional phenomenon, there is no consensus on the most appropriate approach for identifying, mitigating and managing the impacts of urban disasters. This is critical, because when an urban disaster occurs, it disrupts the 'normal functioning of communities or societies, provoking large scale human, material, economic and/or environmental losses' (UNISDR, 2009, Nicholls, 2010).

Urban resilience

A common term that has become well known in the context of developing and designing cities and reducing hazard impacts is 'urban resilience' (100 Resilient Cities, 2015; Pickett and Cardenasso, 2014; Vale and Campanella, 2005). Resilience is seen as a risk reduction strategy and it is argued should be simultaneously developed across multiple overlapping dimensions: such as urban planning (Beck, 2008). Beck (2008) identifies that this includes elements such as the built environment, governance networks, social dynamics, and the production, supply and consumption chains (also see Resilience Alliance, 2007 and 100 Resilient Cities, 2015). Cities resilience also requires accounting for the variability of different disasters and their urban impacts: such as slow onset events (in an urban context sea-level rise) versus rapid onset events (in an urban context earthquake, hurricanes etc.). This

would include accommodating and understanding the differences between large-scale processes (e.g. global warming, economic migrations) vs localised and event specific impacts (Cutter et al., 2008). A resilient city is argued by Shah and Ranghieri (2012, p. 22) to be a city that can “cope with disaster and climate impacts now and in the future, thereby limiting the magnitude and severity of those impacts”.

Urban vulnerability

Birkmann (2006) points out that vulnerability reduction as part of a pre-disaster stage for cities should be a primary focus and not resilience for optimising the responses when a disaster occurs.

Vulnerability is argued to be different in the urban context and important to understand when considering mitigating within cities. Although it is argued that it is easier to provide for social and cultural services within a city context verse a rural context (WB, 2010) the counter argument is that as cities grow in population, the pressure on infrastructure, economy, environment and social networks increases (Shaw et al, 2009). Therefore, high population sizes and rapid growth can result in poorly managed development: higher fatalities and injuries due to more people in one small area (Shretha, 2013). Increased population and subsequent pressure on urban environments and structures increases the vulnerability of the people within the urban context and creates new dimensions of exposure to disasters. Filion (2015) highlights that these dimensions can be explored through the ‘interrelations between social and institutional dimensions of disaster’. Bohle (2001) categorises these as ‘external’ (structural context) and ‘internal’ (coping strategies people have access to) and as the parameters in which urban vulnerability should be understood. Shaw (2009, p. 9) identifies that urban vulnerability can be reduced by policies through being ‘prevented’ (control the source), ‘protected’ (build to withstand) and ‘controlled’ (land use planning and zoning). Campbell and Fainstein (2001, p. 2) highlight that this can be done through urban planning. Which they describe as an activity that has the capacity to ‘predict the consequences of its actions’ and potentially alter the course of events by addressing the vulnerability challenges of a city.

Disaster risk reduction

Disaster management has four interrelated stages: *mitigation* (reducing likelihood of disasters/hazards), *preparedness* (minimising impact of the hazard), *response* (taking action to deal with the actual disaster) and *recovery* (dealing with the result of the earthquake and the process of returning to ‘normal’) (Coppola, 2011; Topping, 2011; Cutter, 2001; Gencer, 2013). These actions are also referred to as disaster risk reduction (FEMA, 2002; Coppola, D. 2011). Disaster risk reduction is a relatively new theory which looks at the priorities around increasing community safety and the

reduction of losses resulting from one particular hazard (March and Leon, 2013). It is a term used for techniques focusing on preventing or minimising the effects of disaster. Shaw (2009, p. 14) qualifies that when looking at urban growths of cities, planning is 'seldom carried out with a view to reduce disaster risk'. Pelling (2003) insightfully states 'Urbanisation affects disasters just as profoundly as disasters affect urbanisation'.

Urban planning

Shaw (2009) argues that urban disasters are a direct result of lopsided policies: lack of urban planning and development processes and programs that do not take into account the resulting risks that they can generate. This is where urban planning approaches play a major role as it a process 'where people make decisions about intended future outcomes, how outcomes are to be accomplished, and how success is to be measured and evaluated' (Emergency Management Australia, 2002, p. 20). Urban planning approaches look at multiple concepts like urban growth, land development, infrastructure, economic development, quality of life, livelihoods, and projections. This includes considering the hazards contained within the context being planned for and how to mitigate the outcomes of that hazard.

Collaboration of all theories

Filion et al. (2015) advocates that natural disasters become crises when there is failure of public and private decision making that therefore causes disruption to the local fabric of the city. Whereas, disaster resilience is focused on the capacity of people to develop adaptive responses. Disaster vulnerability reflects the frequency and intensity of the large-scale risks. And disaster risk reduction is a process used to mitigate the resilience of an urban area (Filion et al., 2015).

The Intergovernmental Panel on Climate Change (IPCC) in 2012 identified collaboration as the important character missing between urban planning and risk reduction practices. Disaster risk reduction plans tend to focus on more specific solutions like reducing a risk to a specific type of hazard (e.g. earthquake resilient plan). Whereas urban planning approaches encompass multiple purposes. The argument that urban planning is a too forward thinking approach to mitigate sudden onset disasters that may arise (March and Leon, 2013), is counter-argued by the fact that not only will good urban planning incorporate the occurrence of such short-term disasters equally it is imperative to include them in the aftermath planning phase; such that the occurrence of disasters is planned for thereafter. If such planning is focused upon following an event or as the urban area is developing it will be possible to facilitate an urban community that will exist as a long-term sustainable community. Myers and Kitsuse (2000, p. 221) advocate that 'the very substance of urban planning is founded in

time, because the process of urban development unfolds over decades'. Mileti (1999, pp. 155-156) highlights further that if land use is managed appropriately with a sustainable hazard goal, disasters can be reduced to a point where governments, communities, individuals and businesses can mitigate their extent.

This paper therefore looks at how urban planning is considering disaster risk reduction techniques and how urban planning should be addressing disaster risk reduction strategies.

Urban Planning to mitigate disasters

Since the dawn of time, people across the world have lived with the realities of natural hazards: which are 'naturally occurring processes and events that have the potential to threaten people and things they value' (Tobin and Montz, 1997). The question central to the majority of scholarly works is 'How can city leaders and other policy-makers anticipate future disasters and take appropriate measures capable of alleviating their impacts?' (see: Allenby and Fink, 2005; Haines, 2009; Klien, Nicholls and Thomalla, 2003; Vugrin et al., 2010). Recently academic papers and research have begun examining the potential of urban land-use planning to reducing this exposure to and avoid the impacts of natural hazard events (see: Burby, 1996a; Burby et al., 1999; Godschalk et al., 1998; Mileti, 1999; Puszkina-Chevin et al., 2006.7; Smith, 2008). Although it is understood that the details and measures suitable to a city in an earthquake zone differ from those appropriate to a hurricane-prone urban area, fundamentally urban strategies and steps of preparedness, response and recovery to mitigate the natural hazards are the same (Gilbert, 2010). However, pre-event preparedness through urban planning has been proven to be the most beneficial step for post-disaster recovery and reconstruction (Glavovic, 2010; Rubin and Barbee, 1985; Schwab et al., 1998; Smith and Wenger, 2006).

Urban planners are one of the only professions that work across all the fields associated within developing cities (political, social, economic, ecological and environmental risks) (Glavovic, 2010). Urban planning has a means to advantageously modify the arrangements, functions and management of cities and therefore has a priority role in ameliorating natural disasters (March and Leon, 2013). Urban planners with their unique position, information and knowledge are strategically situated to address the challenges within the context of urban planning and disaster resilience and risk reduction management design. It will be argued within this paper that this is a distinctive and critical role urban planners can play in the humanitarian and development aid world, in particular with how it grasps with the issues of urban disasters.

Definition

Urban planning (in contemporary terms) is said to have emerged as a response to the deplorable living conditions in the industrialised urban Britain during the second half of the 19th century (Hall, 1996); 'linked to a broader reform movement that sought to redress the ill of unconstrained capitalism through changes to the politics, economy and geography of cities' (Yiftachel, 1998).

Urban planning seeks to 'favourably influence the spatial and functional organisation of cities and regions' (March and Leon, 2013). This includes 'broad-based strategies for managing urban change' (Kaiser and Godshalk: 2007: 367) through practices like developing master plans for future use that serve as frameworks for 'community consensus on future growth' (Kaiser and Godshalk: 2007: 366). A very important key element of urban planning is a focus on gathering, developing and applying various types of data or intelligence to create a foundation for decision making (Rydin, 2007). Planning outcomes are built upon a level of intelligence, planning, background and spatial modelling; where a problem is defined and the contexts are established (Fry and Wiliams, 2013). With this information, it is then adapted into legislative, agency, governmental and professional bases to ensure legitimacy and frameworks for action (March and Leon, 2013).

Urban disasters in urban planning reference

Despite efforts in many developing countries the planning of urban areas (both regional and metropolitan) is often weak and lacks authority (Bergh, S. 2004; IHC, 2009; Miraftab, 2004; Rondinelli et al., 1983). In response to this, some developing country governments have tried to establish regional authorities and public-private partnerships but only with a focus on major investments in urban infrastructure and not on mitigating hazards within the fabric of the city. Often these initiatives have failed too, because there is no consideration of the broader regional land use planning goals (for more information see: Plane, 1999; Bennet, 1998).

Burby (1998) espouses that you cannot have one single approach to dealing with a disaster because disasters in themselves impact multiple sectors, people, environment and infrastructure. Urban planning plays a role in not only directing new buildings and infrastructure away from identified hazards and vulnerabilities, but also through structural mitigations to reduce the physical development risk through design and retrofitting standards (Glavovic, 2010). Good urban planning can also ensure that vital lifeline systems (roads, utilities and other support facilities) meet the demand during an extreme event and that all members of the community (governments, businesses included) are able to work together to plan for, and cope with hazards (Glavovic, 2010).

Land Use Planning

Land Use Planning is looking at the spatial relationships between physical, social, economic and ecological then developing a set of approaches based on that knowledge (March and Leon, 2013). Poor land use planning can lead to urban agglomerations which are uncoordinated, inefficient and cause many conflicts in land use (UNDP, 2012).

One important aspect of land use planning is the consideration of hazards, though the spatial realm of planning, that may place the environment (physical and natural) at serious risk (UNDP, 2012; Masterson et al., 2014; March and Leon, 2013). Hazards are seen as spatial in their characteristics, as they occur across different scales, are influenced by natural and human made systems and occur within various spatial territories (March and Leon, 2013). Many disasters can have impacts across many different regions and likelihood of risk when in densely populated areas. Hence the integration of recognised hazards (risks and vulnerabilities) into the planning systems and then translated into the built environment 'through policies, strategies, projects and activities, making the planning process risk-sensitive' (UNDP, 2012) is so important. Land use planning is identified by the UNDP as one of the most effective ways to achieve disaster risk reduction.

Analysing land-use changes is required to address the complex issue of land-use changes and to understand land-use practices (Verburg et al., 2004; Van Daalen et al., 2002; Lambin and Geist, 2006). A variety of methods have been used to detect and evaluate the land-use and land-cover change due to urban expansions (Rahim, 2016). Bobek (1927) highlights morphogenetic approaches to looking at urban growth from a geographical perspective: the site and geographic location (Geography), historical development of the urban settlement (Land Use Patterns) and the urban settlement as part of the boarder landscape (Development Activities) (as cited in: Hofmeister, 2004; Saey, 2008). This approach is laid out by Hofmeister (2004) and will be used as part of the research analysis in this report.

Governance/ Politics

Many sectors within the development world are currently undergoing decentralisation in the current neo-liberal climate (Jones, S. et. al 2013). Decentralisation is a central component of neoliberal reforms and is often regarded as the answer if governments lack funds or incentives to provide services and infrastructure (Rondinelli, 1989). Rajan (2002) identifies that disaster management in developing countries is characterised by poor governance within the state and can be a direct result of mismanagement during the decentralisation process.

A UNHSP (2004) and Shaw (2009) both identify that one of the key steps for urban disaster risk reduction involves good governance. The role that urban planning plays with regards to governance and politics of city design is through balancing the concept of urban resilience (implementation of laws and legislation included) with the challenge that neoliberalism brings (Filion et al., 2015; Peck 2010). There is significant difference between the neoliberal values of the major states of the world and their political and economic ramifications on urban resilience (Filion, 2014).

Reviewing urban governance forms part of the Urban Development Study approach and will be included in the research analysis part of this report. This model is further explained in the research section.

Regulation and Policies

Research undertaken by Shaw (2009, p. 17) has found that many casualties in urban disasters are a direct correlation to quality of housing construction and infrastructure due to the extent of collapsed buildings. Within a developed country urban planning has developed legitimacy as a base of action through 'means of legislative, agency, governmental and professional fundamentals' (March and Leon, 2013). Shaw (2009) found that many developing countries don't have construction regulations that relate to their hazard profile and in cases where they exist the regulations are not appropriately enforced.

Coordination

As identified by Galvovic (2010) one of the blocks for translating sustainable hazard mitigation intentions into practicality is from collaboration of different agencies and departments of governments. Geis (2000) advocates that the best approach to create a disaster resistant community is by addressing the physical aspects of a community through building codes and land use plans but also by other social dimensions relating to the built environment, such as integration of all sectors.

Urban planning therefore plays a key role in hazard reduction by 'specifying a set of actions that engage participating organisations in the regular review of risk factors in their interdependent environment' to substitute the 'processes of continuing learning and reverse the destructive spiral that has led to an increasingly fragile and threatened environment' (Comfort et. al., 1999, p. 42).

Kathmandu Valley's devastation

The concepts and theories discussed through this literature review will now be applied to the case study of Kathmandu Valley. During 2015, Kathmandu Valley saw massive devastation through loss of life and infrastructure as a direct result of the collapse of its built environment. To determine how this occurred, this paper will draw on the defined concepts around mitigating urban disasters through urban planning strategies of land use planning, governance/politics, legislation and coordination. These will be applied through Bobek's (1927) morphogenetic approaches to looking at urban growth and Filion et al., (2015) theory of governance and politics playing a key role in how sustainable cities are designed. The analysis will include drawing on the different theories and concepts surrounding urban disasters illustrated through literature: urban resilience, urban vulnerability and disaster risk reduction, as a means of measuring the resilience, vulnerability and disaster risk methods within Kathmandu Valley.

Now more than ever, the literature highlights the importance of finding the most appropriate approach for identifying, mitigating and managing the impacts of urban disasters.

Chapter Three: Methodology

The research for this study is focused on Nepal and in particular the urban planning within Kathmandu Valley. Nepal being naturally prone to disasters due to its geographically location has extensive historical evidence of dealing with natural disasters. Most recently the earthquake in 2015. This was seen as one of the worst urban disasters seen in Asia.

Having framed the research focus within relevant branches of the literature, the research will apply the defined concepts to the case study of two main densely populated areas in Nepal: 'Kathmandu Metropolitan City' and 'Lalitpur Sub-Metropolitan City'. The metropolitan area of Kathmandu includes three nearby municipalities within the valley: Kirtipur, Madhyapur and Bhaktapur. These areas will also be considered in this study as they are viewed as urban³. See **Figure 3** below.

While this paper primarily focuses on the urban resilience of Kathmandu Valley with relation to the damages caused by the 2015 earthquake, it will address other natural hazards: like floods and landslides.

The study's initial focus is identifying the gaps in urban planning policy and legislation: many policies and legislation has been examined looking at gaps and linkages (**See Appendix 7, Appendix 8 and Appendix 12**). The focus is cognizant of the necessity to be aware of political structures, social norms, implementation, other sectoral development projects and relevant context information. Particularly, the interface between what is legislated and what occurs. Three government employee's primary stakeholder's perspectives and accounts of urban planning historically within Nepal were studied. Twenty-two humanitarian professional workers were interviewed around urban planning in Kathmandu to identify development and implementation issues.

³ Under the Local Self Governance Act 1999, urban centres in Nepal are classified as Metropolitan Cities, Sub-Metropolitan Cities and Municipalities.

The map displays the Kathmandu Metropolitan Area, divided into four administrative regions: Municipality (yellow), Kathmandu VDC (pink), Bhaktapur VDC (green), and Lalitpur VDC (brown). The map is color-coded to show the distribution of these regions. A legend at the bottom identifies the colors: yellow for Municipality, pink for Kathmandu VDC, green for Bhaktapur VDC, and brown for Lalitpur VDC. An inset map shows the location of the Kathmandu Metropolitan Area within Nepal.

Appropriateness

26

Research Introduction

Through the approach of historical and structural contexts, the legislation and interviews were analysed with an aim to provide an analysis of underlying reasons, opinions, and motivations around urban development within Kathmandu in relation to mitigating risk disasters. This research hopes to develop an insight into the issues around urban disasters to further progress ideas on potential quantitative research.

Research Design: Land Use Study

Research Purpose

The Land Use Study will draw on Bobek (1927) highlighted morphogenetic approaches to looking at urban growth from a geographical perspective (as cited in Hofmeister, 2004). These have been summarised into: Geography, Land Use Patterns and Development Activities. Analysis of Demographic changes through a historical approach will also be examined. A land use study is an analysis and record of the way in which land use has been used and is being used in an area. This study identifies the historical actions and development impacts on the Valley.

Research Approach

Background and Research papers on Nepal have been sourced through online journal platforms and correlated. Documents were also provided by international and national government organisations following interviews and have been included through the research analysis within the paper.

Research Strategy- Methods for Processing Data

The section is accordingly broken up based on Bobek (1927) approach and shown in the **Table 2** below.

Table 2: Land Use Study Structure

Land Use Study	
Geography & Culture	The geography and cultural history of Kathmandu Valley has been briefly examined to provide a background for the context to which this study is undertaken.
Demographics	A quick analysis of demographics within Kathmandu Valley is presented to give framework to the growth of the valley over time.
Changing Patterns of Land Use in the Kathmandu Valley	To appropriately examine the construction and development of Kathmandu Valley prior to the earthquake, it is important to consider the historical land use patterns and how they have impacted and guided development in the Valley.
Development Activities	Since the introduction of the Apartment Ownership Act in 1997, the private sector has taken a major role in development activities in the Valley. Information on the types of development are presented. The public sector development projects are also presented in this report.

Research Design: Urban Development Study

Research Purpose

The Urban Development Study will look at the urban growth in Kathmandu Valley, history of governance, study of urban planning and development regulation and policy and the sectoral approach towards disaster management. The aim is to measure the success of urban disaster planning within the Kathmandu region by comparing the results to related literature and accepted world practice and understanding.

Research Approach

Government policies and legislation have been collected through government websites (National Planning Commission, The Ministry of Home and Affairs and the Ministry of Urban Development and Planning) and documents provided by international humanitarian organisations. Research papers were sourced through online journal platforms and correlated documents were also provided by

international and national government organisations following interviews and have been included through the research analysis within the paper (See Appendix 7, Appendix 8 and 12 for a list of all regulations and policies reviewed).

Research Strategy- Methods for Processing Data

The Urban Development Study is broken up into the following areas shown in the **Table 3** below.

Table 3: Urban Development Study Structure

Urban Development Study	
Urban growth and planning in Kathmandu Valley	To understand the urbanisation history of Kathmandu Valley, a brief summation of Kathmandu Valley's urbanised history and the urban development legislation is presented.
History of Governance, legal arrangements in Nepal	This chapter draws on critical literature, regarding "Decentralisation" of the governance system and the consequences for urban planning in Kathmandu.
Study on Urban Planning and Development Regulation and Policy	Examination of the Physical Planning, Policies and the Building Code Regulations within Kathmandu while addressing the implementation tools and governance framework.
Sectoral approach towards Disaster Management	This section will look at the disaster management framework and policies and link them back to urban planning to analysis the urban planning within Kathmandu Valley around building resilience.
Analysis on issues on Urban Planning Administration	The administration and governance of Kathmandu Valley plays a vital role in effectiveness of the urban development regulation and polices. The administrative and governance functions addressed are categorised under Implementation, Monitoring and Coordination, and Resources.

Research Design: Interviews

Research Purpose

Interviews were the secondary data collection for the research. The purpose of the interviews was to gather data on the understanding and implementation of Nepal's urban planning legislation, prior to the earthquake in 2015. The overall purpose of the interviews was to gain a greater understanding of the government's preparedness and history of planning within Kathmandu Valley.

Research Approach

For this research, there was a requirement to interview respondents to collect further data that was clearly determinable through literature such as implementation and political situations. However, due to limited resources and timing of the field visit, it was not practical to interview all organisations or government departments working within the field of urban development. Organisations and government departments were decided based on their involvement in either urban development or disaster resilience within Nepal.

In selecting interviewees, the research has progressed through a 'snowball procedure' whereby people in specific organisations were targeted for contacts (Atkinson and Flint, 2001). This included contacting organisations by email and uploading the research proposal onto the online Urban Humanitarian Response Portal³.

The selection was based on availability, the detail of responses of respondents, their position within an authority and their level of professional experience. All the interviews were with people who had profound or extensive knowledge and experience about the area of research and the context. The interviews being undertaken in Nepal ensured the interviewees were familiar with the country and its political structure. **Table 7 on page 51** provides details on the interviewees, their nationality and professional work history.

Due to the expertise of the interviewees the interviews were conducted within a loose structure akin to a semi-structured interview⁴. This reflected the perception that they would not be engaged nor all have the same availability of time to answer a series of proposed questions.

³ Urban Humanitarian Response Portal was set up by ALNAP and UNHabitat. See <http://www.urban-response.org/about.aspx>

⁴ *Semi-structured interviewing is usually taken to mean that the same questions are asked of all those involved and that approximately equivalent interview time is allowed* (Gillham, 2005, n.p.). This was not the appropriate approach due to the expertise of the interviewees and the flow of the meetings.

The interviews all took place in person during the field visit to Nepal in June to July 2017. There were no language barriers as all interviewees spoke fluent English. In some cases, the interview was supplemented by follow up emails with questions and discussion points.

Research Strategy- Methods for processing the data

Questions were prepared in advance based on the research questions (**Appendix 15**). Key questions were asked around growth boundaries, building approvals, land use zoning, disaster risk mapping, sector communication and corporation, political structures and social norms to ensure that consistent and measurable data was collected. These key questions were asked to achieve specific cross-tabulated data to contribute to addressing the research findings. The key questions also ensured that the information collected was accountable due to multiple answers concluding similar responses. The semi- informal structure of the interviews allowed the capture of information that would not have been captured had the interviews been tightly structured.

The answers to key questions was then inserted into a matrix system to analyse common factors highlighted by the international and national humanitarian professionals and government employees.

Research Sections

The Research analysis section is divided into three Parts. **Part one** (Chapter four) is a Land Use Study that examines the historical context of urban planning in Kathmandu and the situation that existed before the earthquake. **Part two** (Chapter five) is an Urban Development Study that examine policies, legislation and urban governance. **Part three** (Chapter six) will provide the findings from the Professional interviews undertaken during a field visit to Kathmandu, Nepal in June-July 2017.

Due the word count limit of this dissertation, certain tables have been included in the Appendices and referenced accordingly.

Chapter Four: Land Use Study

Land use planning, may be defined as the ‘proper management of land resources’ (Serote, 2004). This involves using land in a consistent way with its natural qualities and involves interventions in the decisions by State and Local regulation and control of certain activities (UNDP, 2012). Land use planning, seen as the responsibility of the government or state (UNDP, 2012). As detailed in the literature review, the Land Use Study is an adoption of the Bobek (1927) morphogenetic approaches to looking at urban growth from a geographical perspective (cited in: Hofmeister, 2004).

Geography & Culture

Kathmandu Valley has for hundreds of years played an important role, economically, administratively, culturally and politically (Thapa et al., 2008). Kathmandu Valley is famous around the world for its rich cultural heritage and includes five United Nations Educational, Scientific and Cultural Organisation (UNESCO) World Heritage sites (UNHabitat, 2015; Thapa et al., 2008; MoUD, 2015; ASD, 2012; HRRP, 2016). Kathmandu’s long cultural history is evidenced by its’ many cultural and heritage sites such as temples, monasteries, palaces, monuments, historical ponds and finely embellished public wells (Thapa et al., 2008; HRRP, 2016).

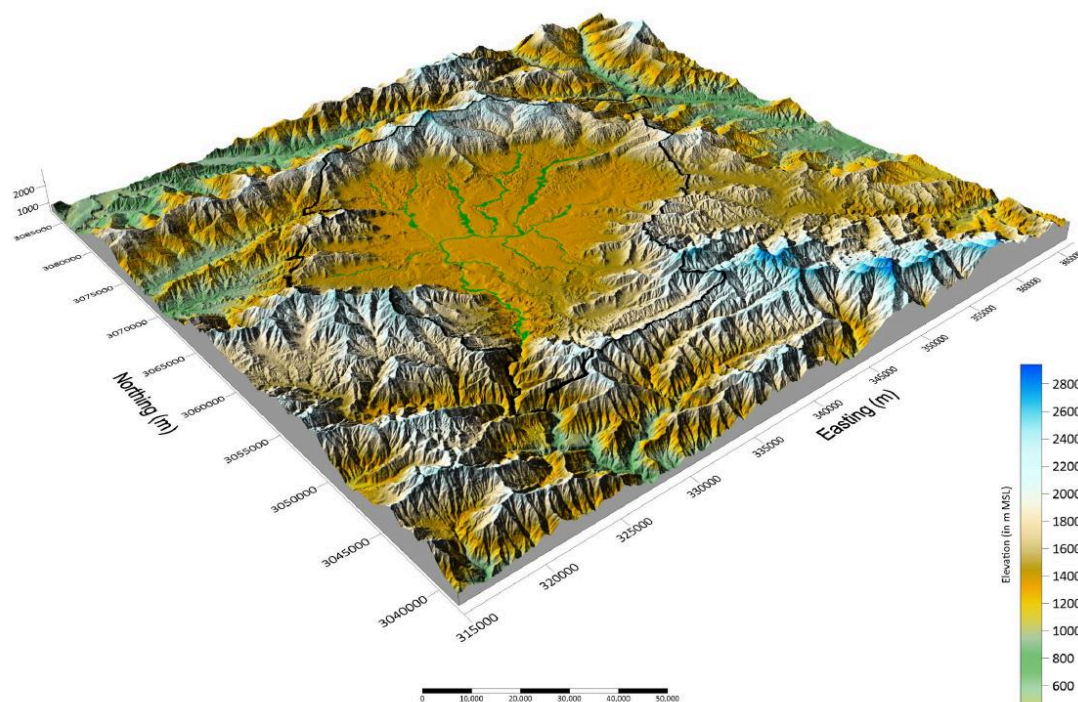
The valley itself is surrounded by the hills of the Mahabharat range. Forming a bowl-shaped valley floor and contains some of the oldest human settlements in the central Himalayas region (UNHabitat, 2015; Thapa et. al, 2008; Regmi, 2965; Wolfgang, 1976; Malla, 1978; HRRP, 2016). The river of Bagmati is the Kathmandu city’s main source of water for drinking and irrigation and it holds religious, cultural and social value to the people (Thapa et al., 2008; ADB/ICIMOD, 2006).

Kathmandu city is generally flat and has two principle landforms; alluvial and flood plains along the rivers and elevated river terraces (Thapa et al., 2008; Haack and Khatiwada, 2007). Historically, the Kathmandu city area would have been primarily agricultural land (Ranjitkar, 1983; Thapa et al., 2008; World Bank, 2016). With rapid urban expansion the majority of this land has been converted and produce is now grown outside the 27km Ring Road (Thapa et al., 2008; HRRP, 2016). The Valley has e growth following river banks into the flat landscapes (See **Figure 4 below**).

Demographics

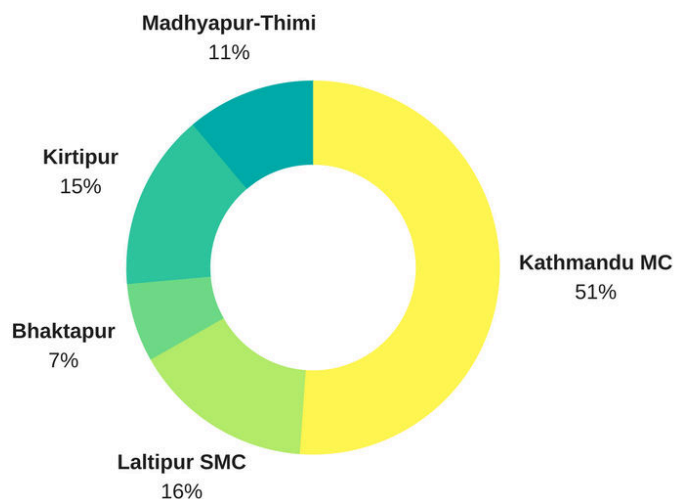
In 2011, at the time of census, the total population of Kathmandu Valley was 2,517,023, of which 1,465,254 was considered to be living in urban areas (the five municipalities) (UNHabitat, 2015; World Bank, 2013; UNDP, 2015; Central Bureau of Statistics, 2014). During the 1990s, Kathmandu Valley experienced rapid urbanisation growth of 6.6 per cent per annum which was considered among the highest in Asia Pacific region (UNHabitat, 2015; ADB/ICIMOD, 2006; UNDP, 2015). This was a direct result of migration, natural growth and expansions of rural towns amalgamating into urban areas (UNHabitat, 2015; UNDP, 2016). This shift from predominantly rural populations to high urbanised populations is similar to the transformation already experienced by today's more developed nations (Gouverneur, 2015; ASD, 2012).

Figure 4: 3D Perspective View of Kathmandu Valley



(Source: UNDP, 2015)

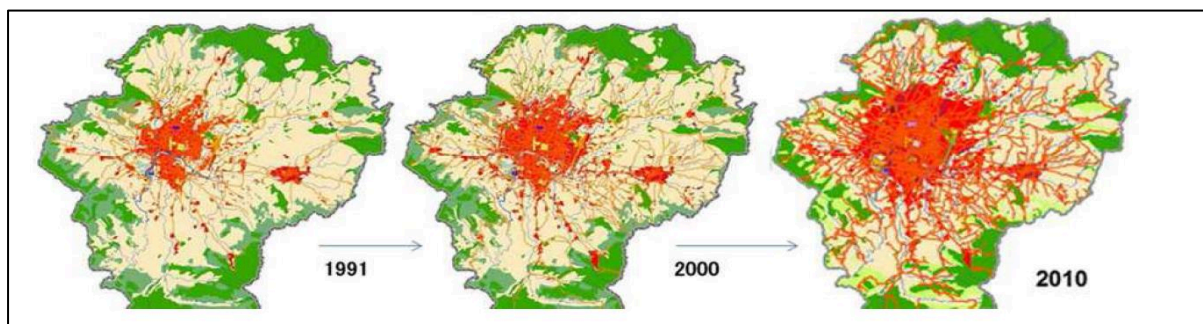
Table 4: Total urban population proportions of the Municipalities in Kathmandu Valley



(Source: KVDC and UNDP, 2013a, pp. 78-79; Central Bureau of Statistics, 2014)

Changing Patterns of Land Use in Kathmandu Valley

Figure 5: Land Uses in Kathmandu Valley: 1991, 2000 and 2010. Red representing growth.



(Source: KVDA and UNDP, 2013a.)

Urban land use

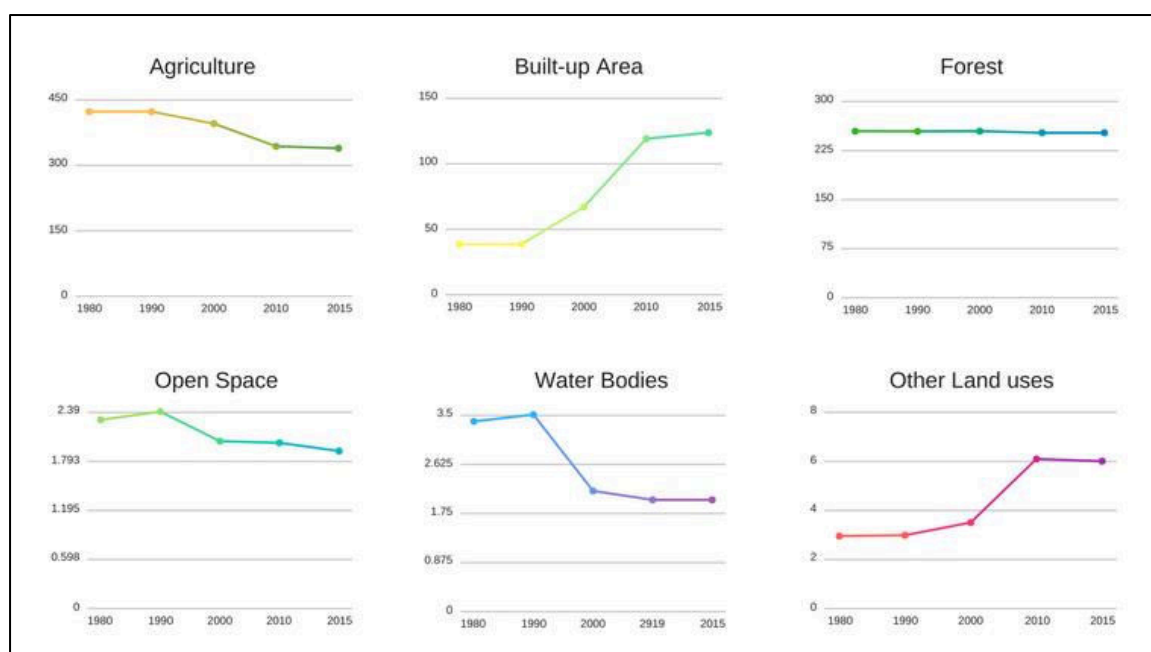
Kathmandu Valley has experienced a high urbanisation rate in recent years. The identification of the land use transition changes is important in analysing the driving forces behind those changes (JICA, et al. 2012). The 'urban expansion' and changes to the land use of Kathmandu can be linked to the Kathmandu Valley Urban Development Plans and Programmes which began in the 1990's and continued into the 2000's (KVDA, 2015; Thapa et al., 2008). The Kathmandu Valley Town

Development Committee (KVTDC, 2002) report identified that the built-up area in Kathmandu metropolitan region doubled between 1984 and 2000 growing from 5% to 10% per year. This caused extensive changes to the land use patterns of the Valley. Development activities occurred primarily in the farmlands located in the peripheral areas and were not well-planned due to economic reasons and weak development controls by the local governments (JICA et al., 2012; MoUD, 2015; ADB, 2010). The latest land use map registered in Kathmandu (2007) shows built up area covering a total of 38% of the Kathmandu Valley (Thapa et al., 2008). During 2000-2010, drastic changes were observed to the urban pattern, such as massive built-up areas expanding along major roads and encroaching on primal agricultural lands, forests and open spaces (Source: KVDA, 2015; UNDP, 2012; ASD, 2012). The drastic land use changes (and urban sprawl) over time can be seen in **Figure 5 above** and in **Figure 6 below**.

The 2011 JICA report registered that five to seven thousand buildings were constructed every year in the Kathmandu Valley (cited in: JICA et al., 2012 and backed up by Central Bureau of Statistics, 2014).

A review done by Pradip and Devendra in 2009, of all historical land use in Kathmandu Valley found that the cultivated areas within the Valley is susceptible to haphazard unrestricted urbanization development (Pradip and Devendra, 2009; WB, 2013, UNDP, 2015). The devastation of recent flooding and landslide events highlighted the need for better regulation within these areas (UNDP, 2015; HRRP, 2016).

Figure 6: Land Cover in Kathmandu Valley 1980 – 2015 (hectares)



Source: Figures taken from (KVDA and UNDP, 2013b, p. 13; UNDP, 2015, p. 14)

This rapid built-up expansion without appropriate land use planning played a direct result in the devastation caused during the earthquake through the failure to adhere to building codes and structural safety codes, to weak governance development controls (UNDP, 2015; HRRP, 2016). The rapid depletion of open spaces in the urban areas of Kathmandu Valley, due to urban expansions, especially into marginal lands, such as steep slopes and flood plains has significantly increased the vulnerability of the people and reduced the capacity for Kathmandu Valley to be resilient (Kumar and Pradha, 2015; MoUD, 2017; HRRP, 2015; ADB, 2010). Currently there is only 0.48% open space in Kathmandu city and 0.06% in Lalitpur (MoUD, 2017; UNDP, 2015).

The different risks within the Valley including risks to informal settlements, landslide, flood and earthquake areas are analysed further in **Appendix 1**.

Development Activities

Following the introduction of the *Joint Housing Ownership Act 2054* (1997) and the *Apartment Ownership Act* in 1997, the private sector started taking a prominent role in development activities, especially in Kathmandu Valley (JICA et al., 2012; HRRP 2016; ICIMOD/UNEP/MoEST, 2007). The *Apartment Ownership Act 1997* provided a legal framework for development sales and management of multi-storey buildings (KLHC, n.d.; HRRP, 2016). This involved three primary types of development: Land Development, Group Housing and Joint Housing (Apartments). **Appendix 5** provides further details on these different developments.

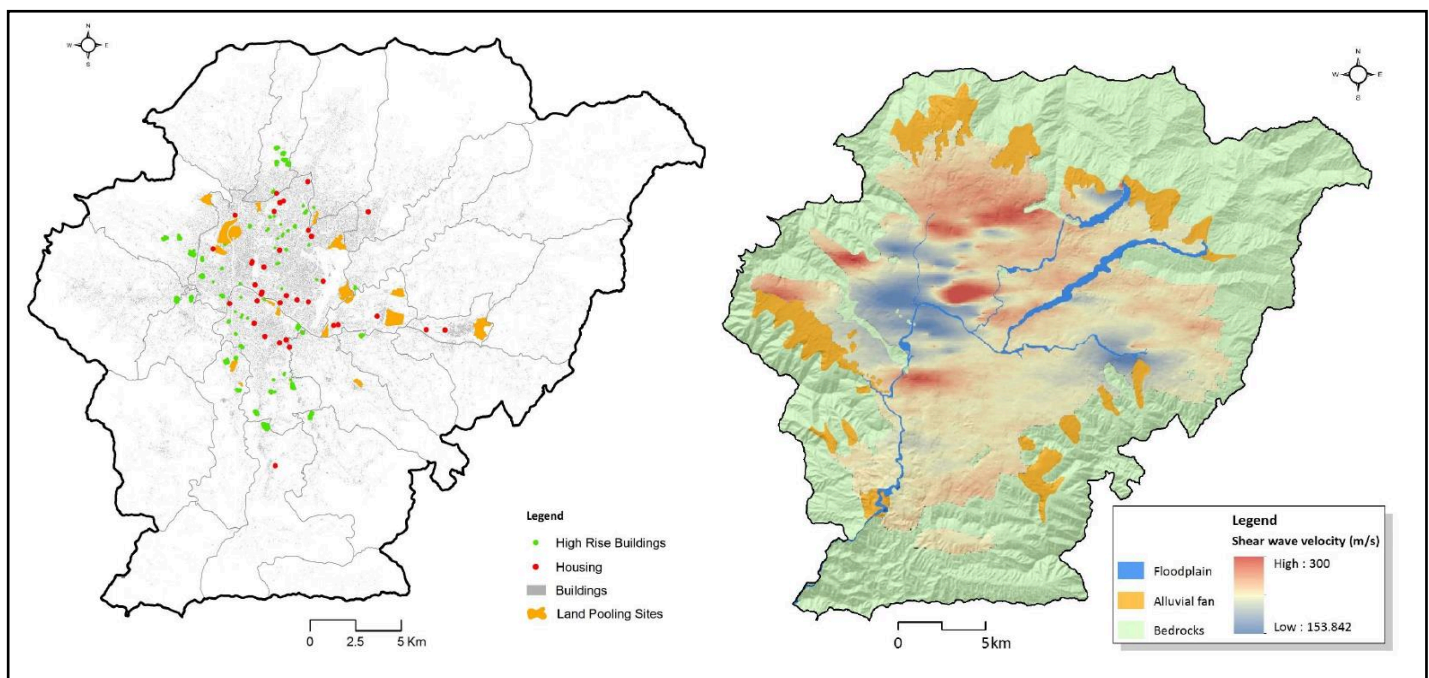
The massive spike in private development construction through land brokers and housing development companies lead to many ad hoc developments in the Valley: unplanned and uncontrolled (Pradip and Devendra, 2009). These constructions also occurred in risk areas (flooding, landslide and earthquake risks) and changed the land use of agricultural, forest and open space land (ADB, 2010). This included a massive increase of apartments, given fast tracked building permits as units provide a higher revenue for government (HRRP, 2016).

The public- sector projects undertaken around land development projects and programs within Kathmandu Valley can be categorised into three themes: Sites and Services Development, Guided Land Development and Land Pooling. These are projects undertaken by the Kathmandu Valley Development Authority (KVDA) (previously KVTDC) in accordance with the provisions of the *Town Development Act 1989*. **Appendix 6** provides details of these programs and an analysis of problems with the programs.

Before KVTDC, the land development programs were done by Municipalities through loans from the government (established through the Town Development Fund that was set up in 1988), which lead to other financial issues for the municipalities around repayments of loans (ASD, 2012).

Neither public or private development activities considered hazards in selection of development sites as Kathmandu Valley was going through a boom of urbanisation and the economy was the driving force. This has had and will have major repercussions. Construction in hazard risk areas results in buildings being affected by earthquakes are in high risk landslide or riverbed areas which are directly impacted in earthquakes. This is illustrated graphically in **Figure 9**.

Figure 9: Location of housing, high rise apartments and land pooling sites (left) compared to the seismic hazard analysis (right). The comparison highlights the lack of consideration of risks when locating urban development activities.



(Source: UNDP, 2015)

Chapter Five: Urban Development Study

History of Urban Growth and Planning in Kathmandu Valley

From the very beginning, Kathmandu Valley has been important to the people and the country. *The history of ancient Nepal is the history of Kathmandu Valley* (JICA et al., 2012, p. 7-2). Kathmandu Valley is significant as it is the centre of the concentration of peoples, history and culture of Nepal.

Recent rapid urban migration into Kathmandu Valley has been linked back to 1950s with the development of the Tribhuvan highway linking India to China (DoR, 2004; Thapa et al., 2008). This was then followed in the 1960s by the agglomeration of rural settlements of Kathmandu Valley into the city (Thapa et al., 2008). Following this, urban areas started to expand along the radial road encroaching on farmlands without providing appropriate infrastructure such as service roads, water supply, sewage and electricity (JICA et al., 2012; ASD, 2012; HRRP, 2016). Urban planning began with the development of the *Town Development Committee Act* in 1963 which created the legal framework for preparing and implementing urban development projects (Gyawali, 1997; Thapa et al., 2008). The first development plan for Kathmandu was in 1969 and the *Town Development Implementation Act* was promulgated in 1972 to implement the plan. The plan aimed to preserve the historical and cultural heritage, through a primary aim of preventing urban sprawl in the city's fringe areas (HMGN, 1969; Thapa et al., 2008).

In 1976, the KVTDC was formed to take on responsibility for overall planning and regulating of urban growth and developed the Kathmandu Development Plan (KVTDP) which drew from the original objectives of the 1969 plan (Thapa et al., 2008; UNDP, 2015). As rapid urbanisation development occurred in the city's rural periphery, the plans failed to address the city's long-term problems (Thapa et al., 2008): Resulting in traffic congestion, lack of drinking water and degrading of Kathmandu's urban environment and threatened cultural and heritage sites (Thapa et al., 2008; Adhikari, 1998; IUCN, 1999; ADB, 2003).

Due to political circumstances in the country during the 1990's many land use and development plans remained in draft form (Thapa et al., 2008). An enacted and implemented plan during this period was the 1991 'Kathmandu Valley Urban Development Plans and Programmes' which integrated plans for the Valley's development. It's focus however was on urban expansions focusing on land pooling and guided development programs. The plans had no focus on mitigating hazards or avoiding growth in high risk areas (UNDP, 2015). The plan did have successes such as stone paving of market centres; upgrades of pedestrian access; improvement in waste management and storm water drainage;

renovation of public parks and public toilets (Thapa et al., 2008). The focus on urban expansion has neglected the city leaving a legacy of roads in poor condition, layout and design, land use fragmentation, growth of unplanned housing in fringe areas, chronic drinking water problems, degraded air quality and significant construction of infrastructure and development within high risk areas of flooding, landslides and earthquakes (Thapa et al., 2008; UNDP, 2012; ADB, 2010).

History of Governance, legal arrangements in Nepal

Urban planning is usually integrated within governance structures (March and Leon, 2013) and therefore it is important to consider the governance structure within Nepal and how that has impacted or advanced urban planning approaches within Kathmandu Valley. Governance is the term given to 'the structures and processes by which societies share power' (Lebel et al., 2006).

In 1990, within Nepal there was a restoration of a multi-party democracy who adopted a liberal market led economic policy (MoF, 2005; Thapa et al., 2008). The international investment that flooded the countries' financial markets and industries had direct and indirect impacts on the city development process (Thapa et al., 2008). Internal problems and political turmoil within Nepal, that followed, was reflected in slow development and economic investment into the city (Thapa et al., 2008). The 'Peoples war' or 'Maoist Conflict' lead to frequent government and administration instability (Thapa et al., 2008). During the 'Maoist' Conflict (1996 to 2006) about 240,000 people were estimated to be displaced to urban centres and 12,000 people were killed (Sharma, 2006; UNDP, 2012). It has been acknowledged that 'countries that are in or emerging from violent conflict present special challenges, both in terms of increased vulnerability of the poor and weak and non-existent governance structures' (DFID, 2006, p. 7). Nepal following the conflict and ongoing political transition was suffering from a weakened governance structure. The lack of strong governance contributed significantly to consequences such as unmanaged industrialisation, unsustainable urbanisation and non-existent disaster management planning (ASD, 2012).

Decentralisation

Nepal has a long history of decentralisation and is currently still undergoing decentralisation in the current neo-liberal climate (Jones et al., 2013). The *1962 Constitution of Nepal* instilled the concept of decentralisation by the formation of the local bodies at zonal, district, city and village level (MoUD, 2016). The *Third Plan (1965-70)* adopted decentralisation as a policy and considered involving lower levels in decision making. However actual devolution of power and political rights did not occur. Not until the introduction of the new constitution in 2015 (MoUD, 2016). During the period of conflict in

1999 the government brought in the *Local Self-Governance Act 1999*. This was Nepal's decentralisation act, with the primary aim of giving local bodies the authority to raise funds by taking out loans and applying taxes. Following the conflict, the government identified a broader decentralised agenda, including the management and responsibility of development and the mitigation of natural disasters and hazards to be placed within districts and local levels. This occurred with very limited financial or technical support (Jones et al., 2013). This resulted in limited implementation or monitoring of urban development and natural disaster management.

Effective urban development strategies and land use planning must be implemented at local and regional level and will depend for their success on strong governance systems to enforce, monitor and evaluate their outcomes (Thapa et al., 2008). Until a decentralised agenda is supported, maintained and implemented, Nepal's governance will remain weak (Jones et al., 2013). As a consequence, the implementation of resilience strategies through urban planning will remain limited.

Study on Urban Planning and Development Regulation and Policy

This section of the study will examine the relevant regulations and policies evident prior to the earthquake and how effective they were in implementation and mitigating disasters in the built environment of Kathmandu Valley (See **Appendix 7** for a breakdown of all major Nepal regulations and policies). These urban sector related policies, acts, regulations, standards which have an impact on building urban resilience are addressed below. Physical Planning and Policy looks at all plans, maps and policies that guide the physical development of Kathmandu Valley. The Town Development Act and Building Code section considers the Regulations and Acts that govern construction and development of Kathmandu.

Physical Planning and Policy

Physical planning has evolved within Kathmandu over many years starting back in 1960's. This was briefly discussed under *History of Urban Growth in Kathmandu Valley* above. **Appendix 8** provides a detailed analysis of the key physical planning and policy documents which have attempted to plan for the impacts of growth within the Valley. Key issues found through the analysis was implementation and financing of the plans, however missing data on linking plans to disaster mapping, identification and risk reduction strategies was noted

Town Development Act (1988 & Regulations (2004), Building Act (1997), National Building Code (1996) and Kathmandu Valley Building by-Laws (2007)

The *Town Development Act 1988* defined provisions for land development to increase supply of planned urban plots (MoUD, 2016) and to facilitate the reconstruction, development and expansion of urban areas throughout Nepal.

Following the massive deaths and damages caused by the 1988 earthquake and the fires that followed in 1999, the Nepal government lead a Building Code Development Project with the primary objective to include regulations and design documents for the use of the planners and engineers. Enabling improvement of the seismic safety aspects of the existing buildings and to detail safer building design and construction practices to be introduced and enforced gradually in Nepal through the development of the Nepal *National Building Code* (NBC) (Parajuli et al., 2000; MoUD, 2016; MoUD, 2015; HRRP, 2016).

The development of the code involved wide seismic hazard mapping and risk assessment, the development of an implementation management plan that included institutional and legal framework requirements, studies of alternative building materials and technologies, and a comprehensive plan for training, public awareness and dissemination of the materials of the Code (Parajuli et al., 2000; MoUD, 2016; MoUD, 2015). The NBC has 23 volumes that cover different categories of buildings (See **Appendix 9**) (JICA et al., 2012; MPPW, 2009; NBC).

Though published in 1998, the NBC was not formally implemented in the Kathmandu Valley until 2003 through one Municipality Lalitpur, followed in a few years by the Kathmandu Metropolitan Municipality (MPPW, 2009). Therefore prior to this time although the NBC's existed they were not enforced with Kathmandu Valley.

Although the implementation of NBC is made mandatory by issue of instructions by the Ministry of Local Development (now Ministry of Federal Affairs and Local Development (MoFALD)). The Building Act and Building bylaws do not include NBC provisions and hence remains ineffective and not applied during the building permit process (MPPW, 2009).

There were four levels of application for building approval in Nepal prior to the earthquake: International State of Art (using international standards), Professional – Engineered Structures (standard codes requiring engineer approval), Rules of Thumb (pre-engineered for defined building types) and Advisory Guidance (non-engineered construction employing traditional materials and skills). These are explained in more detail in **Appendix 10**. The important level to analyse is the Rules

of Thumb, as most construction in Kathmandu Valley seems to fall under this application criterion. Rules of Thumb is a process that requires very little technical expertise to meet standards and is provided to a non-technical appraiser at local authority for approvals (Parajuli et al., 2000; UNCRD, 2008; KVDA, 2015b). These two factors are major issues for the safety of the building standard that have been constructed.

Although the urban development regulations are well-structured and designed. The implementation has been very limited. The KVDA (2015b) identified that ‘making all new buildings comply to the building code is an enormous and almost impossible task.’ This is because the NBC was perceived as too difficult to understand and comply with, and the perception that to comply would increase the cost of construction significantly (MoUD, 2015). Furthermore, Municipalities feel it is a burden on their resources to enforce the NBC. Thus, the local authorities and civil societies confusingly believing that constructing using the NBC will be costlier without any additional benefit to the house owners, and therefore opt out of following the NBC for construction (MoUD, 2015; MPPW, 2009). The repercussions of this can be evidently seen in the photos provided in **Appendix 17** from the 2015 earthquake.

Even though all building permits processes in the Municipalities enforce NBC compliance in theory with only limited generalised checklists for development, the checklists are often overlooked and easily manipulated (Giri, 2013; WB, 2013). There is currently no effective mechanism for field verification of approved drawings. Vulnerability of the urban areas in Kathmandu Valley had been rising for many years before the earthquake due to building permits issued which are not NBC compliant (Giri, 2013; HRRP, 2016; UNDP, 2015).

Kathmandu Valley Building By-laws (2007)

Prior to the formation of the KVTDA/KVDA there were many different building construction by-laws all comprising different rules and regulations across the valley. Dating back to the enactment of *Kathmandu Valley Town Development Act of 1976*, when building construction bylaws were first formulated. They were basic and with minimal standards and requirements controlling new construction in the Valley. The building bylaws were then updated in 1993. Each Municipality in the Valley (Kathmandu Metropolitan City (KMC), Lalitpur Sub-Metropolitan City (LSMC), Bhaktapur Municipality, Madhyapur Thimi Municipality, Kirtipur Municipality) and adjoining Village Development Committees (VDCs) had different bylaws.

The Kathmandu Valley Building By-laws (KVBB) were prepared by KVTDC (now KVDA) in 2007 (JICA, et al. 2012). It is a building control regulation that also encompasses general land use zoning maps for all

of the valley including the five municipalities (Kathmandu, Lalitpur, Bhaktapur, Kirtipur and Madhyapur-Thimi) (KVBB, 2007).

However, the machinery for development control and enforcement of the land use and zoning plans is not explicit within the KVBB (Source: KVBB, 2007). Preserved areas primarily focus on heritage sites and conservation areas within the valley. There are no hazard risk zones. Also, to date, most residential buildings (even in urban areas) do not have provision requirements in them for strength criteria. Building permits within Kathmandu Valley only consider compliance related to planning (ground coverage) and building by-laws (height, set-backs, provisions of waste disposal) (Parajuli et al., 2000; WB, 2013).

Analysis of Urban Planning and Development Regulation and Policy Study

A full analysis of the urban planning and development regulations and policies have been undertaken and can be found in **Appendix 11**. The analysis was developed from Philip Berke, Timothy Beatley and Richard Eisner's growth management tools for natural hazard mitigation (Berke et al., 1992) and from the 'land use management tool' to guide development in hazard-prone areas from Olshansky and Kartez, 1998. **Table 5** below shows the key findings.

Table 5: Findings

Key Findings
There are many missing links between implementation, documentation and lack of policy Land use planning has not appropriately integrated disaster risk areas with where 'urban areas' are expanding. 'Risk' zoning does not exist. Measures for safe retrofitting in the building codes is missing. No provision for governing change of approved use. Compliance enforcement is ineffective. There is no legislative process for acquisition of existing development or prevent future. development in hazardous areas. There are missing hazards in the special hazard resistance building standards. Subdivision ordinances are missing. There are missing taxation and fiscal policies that can cover additional public costs of building in hazardous areas and taxation for open space or non- intensive uses in hazardous areas. There is lack of dissemination programs or policies around issues with building in hazard risk areas. There are some provisions around set-back distances from the road centre for new buildings but repairs and renovations are exempt from these provisions which causes issues on its implementation and purpose.

Sectoral approach towards Disaster Management

Natural disasters in Nepal include earthquakes, floods, landslides, droughts and epidemics (Kumar and Pradhan, 2015; MoUD, 2015). According to Global Facility for Disaster Reduction and Recovery (GFDRR), more than 90 percent of Nepal's population lives in relatively high risk areas from two or more hazards, of which the unpredictable and highly probable one is earthquakes (Kumar and Pradhan, 2015). The 2015 earthquake highlighted the sever vulnerability of low-strength urban houses to earthquakes (MoUD, 2015) and lack of integration of disaster preparedness and urban planning. This section will address this further.

Appendix 12 presents the progression of disaster management related efforts in Kathmandu Valley including an analysis of the policies and regulation. Key findings are that regulations primarily focus on an emergency phase of a disaster (relief and response). It identified that implementation and lack of coordination between Ministries and local authorities is a prominent issue and it is evident in analysis of budget costs and spending in disaster management that it is primarily concentrated on post disaster response and recovery (Kumar and Pradha, 2015).

A study undertaken by the Nepalese Government following the earthquake in 2015 identified two key problems with regards to disaster management planning (cited in: Kumar and Pradha, 2015). Firstly, there is a significant lack of multi-hazard approaches that deals with different types of disasters. Secondly, significant lack of information on historical events such as flood inundation areas derived from impacts of climate change and return periods (cited in: Kumar and Pradha, 2015). On top of these issues is the linkages between other sectors especially regarding the relationship between mapping and planning, land use and building codes. Although all four exist and are quite established they are not incorporated to create an overall strategy on disaster resilience.

The UNDP in 2012 identified that the only way to bring any National Disaster Strategies forward was through land use planning being the most effective ways of achieving disaster risk reduction (UNDP, 2012). A land use planning approach to dealing with disasters was seen by the UNDP as a required approach in reducing disaster risks as part of Kathmandu's pursuit for sustainable development. The disaster risk policies and regulations shown in **Appendix 12** address major matters regarding building disaster resilient urban spaces. Legislation links to actual urban development from drafting, construction, approval process and the planning of the city's built environment is critical to ensure the safety and resilience of the city.

Analysis on issues of Urban Planning Administration

The formulation, revision, update and implementation of the NBC, resilience and disaster risk based plans, urban development strategies and plans are influenced by the political fragility and urban governance within Kathmandu Valley. This section will explore this a bit further.

Despite the efforts of DUDBC, the NBC and other relevant codes, plans and bylaws have not been able to be enforced as envisaged in *the Building Act 1998* (MoUD, 2015). Issues of urban governance within municipalities like Kathmandu Valley is vital and so far, unsuccessful (MoUD, 2017). Most of the issues relating to urban governance and administration can be categorised under Implementation, Monitoring and Coordination and Resources. In 2012, the institutional setup of the KVDA and MoUD was a direct response for the need to strengthening central oversight and metropolitan level planning and management in the Kathmandu Valley (WB, 2013). However, the impact of this new institutional arrangements was not felt by the time the earthquake hit in 2015 therefore it is not possible to examine their institutional capacities in detail.

Implementation

Various national and local institutions are involved in urban development activities of Kathmandu urban areas and the whole Valley. Apart from the KMC, LSMC and the three municipalities, there are several central government agencies working in Kathmandu Valley. **Appendix 13** analyses and identifies the most prevalent departments and institutions responsible for development activities in Kathmandu Valley. As addressed through the table in **Appendix 13**, there has been a lack of cooperation and coordination between institutions linked to unplanned urban development and infrastructure activities within Kathmandu Valley (WB, 2013; ADB, 2010). An example is the current arrangements with urban planning and infrastructure under MoUD and urban governance (local authorities) and administration under MoFALD (MoUD, 2017). ‘Fragmented institutional arrangement, uncoordinated mechanisms between MoFALD and MoUD and lack of technical expertise and capacity’ are major issues of concern when it comes to urban planning governance and administration within the Kathmandu Valley (MoUD, 2017; HRRP, 2016).

Monitoring

Prior to the earthquake, monitoring of safer building construction through planning and disaster risk reduction, was seen to be very ‘sketchy’ (MoUD, 2015) with permit application documents being ‘lost’ and ‘manipulated’ (Parajuli et al., 2000). The quality of monitoring mechanisms from national to local level is lacking as a direct result of poor institutional and technical support (MoUD, 2015; MoUD,

2017; HRRP, 2016; Parajuli et al., 2000). In the cases where the drawings were provided and approved, they did not comply with NBC and the construction on the site usually did not even comply with the original approved drawing and designs (MoUD, 2015; HRRP, 2016). A study of Kathmandu Metropolitan alone identified that 30% of building permits issued result in alternations and violation of the original approved construction (Awale, 2014). There have been no studies on how many buildings' have no approval. This indicates major flaws in multiple levels of monitoring of the NBC and the By-laws within Kathmandu Valley (HRRP, 2016, UNDP, 2015).

Coordination

As a result of urban policies not being formed through a coordination process, it is unclear who carries out the policy and therefore a lack of implementation is a direct result (ASD, 2012; WB, 2013). This was a critical issue before the earthquake. An example is the Ministry of Urban Development (MoUD)/Department of Urban Development and Building Code (DUDBC) and MoFALD using the *Building Act* and LGSA attempting to enforce the NBC in Municipalities yet separately from the Ministries despite there being a strong desire for them all to work together (MoUD, 2015; WB, 2013). Agencies and departments are acting independently without synchronizing with other activities and agencies and it has resulted in a very complex web of resource flow at implementation level (MoUD, 2015; UNDP, 2015). On the ground collaboration and implementation of the NBC between public, private and local institutions is also incredibly weak (MoUD, 2015). This has a major and direct impact on the construction standards within the valley.

Resources- Capacity

As highlighted earlier, Nepal has been experiencing a high pace of urbanisation, especially within Kathmandu Valley (ASD, 2012). The lack of institutional capacity to effectively manage urbanisation has seen a direct result of the KVTDC (now KVDA) not being able to control land development and construction (UNDP, 2012; HRRP, 2016). Safer building construction is both a technical and government issue and needs appropriate skills and resources to deal with it adequately (MoUD, 2015). This is a reflection of the on the ground capacity of the local authority institutions, professional engineers and architects (MoUD, 2015). Kathmandu Municipalities were found to lack basic institutional structure along with physical facilities and tools to bear the responsibility of managing construction in a safe hazard reduction way (MoUD, 2015; HRRP, 2016; UNDP, 2015). There has not been 'effective efforts on identification, mobilisation and allocation of internal and external resources for implementation of the NBC' as well as the Building By-Laws (MoUD, 2015; HRRP, 2016). This has resulted in no Municipality within the Valley appropriately managing or implementing the full cycle of

building regulations from gaining approval, inspection and enforcement to penalties (HRRP, 2016). There is also very poor databases and systems of regular updating and monitoring in all municipalities as they have not been institutionalised yet (MoUD, 2017).

The lack of resources and capacities of the Municipalities within Kathmandu Valley can be directly linked to the revenue base being very weak. Source revenue being an average 30% and the remainder being accrued from grants from central government (MoUD, 2017; ASD, 2010). Funding support for the relevant administration that institutionalise the documents is minimal (UNDP, 2012). Monitoring of the funding is also a major issue with sources, flow direction and outputs not being appropriately mapped or recording misleading information for all stakeholders, including government (MoUD, 2015).

All of these factors have left a valley that is largely unplanned and unmanaged with poor infrastructure.

Chapter Six: Interviews

“I think we were lucky in Kathmandu Valley as the Kids were on holidays when this earthquake happened which is why there was less deaths. Next time this will be different.” – PM 2

Having framed the findings from the research on politics, governance and implementation surrounding growth, design and political structures of Kathmandu Valley in part one and part two. Part three will investigate the professional expert’s perspectives in the field of urban development and gather data on the understanding and implementation of Nepal’s urban planning legislation and governance system, prior to the earthquake in 2015. The overall purpose of the interviews was to gain a greater understanding of the government’s preparedness and how the planning of Kathmandu Valley has come about.

The interviews being undertaken in Nepal ensured the interviewees were familiar with the country and its political structure. Details of interviewees can be found in the **Table 6** below.

Questions were prepared in advance based on the three key research questions (see **Appendix 15**). Key questions were asked around growth boundaries, building approvals, land use zoning, disaster risk mapping, sector communication and corporation, political structures and social norms to ensure that consistent and measurable data was collected.

To ensure consistency with the report, this section will be broken down into the three main themes and questions, linking to key factors found through the interview analysis (**Table 7** below). Each factor was founded on the specific questions that were asked. The layout and factors are shown in **Appendix 15**.

Appendix 16, provides key comments and quotes made by interviewees that back up each of the findings analysed in this section under each factor and interview question.

Table 6: Factors from the Interviews

Theme	Research Question	Sub- Questions	Research Methodology
Urban Land Use	How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?	How has the link between unplanned urban development, irregular and substandard housing development, loss of open space and decreased liveability increased the vulnerability of Kathmandu to disasters?	Document analysis: Land Use Study Key informant interviews
Urban Development (Regulation and Policy)	What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?	How effective were urban planning documents in reducing damages in the 2015 earthquake? To what extent did the planning and development documents change the location, type, and intensity of land uses?	Document analysis: Urban Development Study Key informant interviews
Urban Governance	What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?	Who are the actors? What are the key institutions? What are the steps involved in decision-making? What are the power dynamics? Which are the different levels of decision-making?	Document analysis: Urban Development Study Key informant interviews
Lessons for Urban Planning of Kathmandu Valley	What is the role for planners in reducing disaster risks within a city? How can they help make a city resilient? (especially within context of developing countries)		Based on analysis of other thematic outcomes.

Table 7: Details of professional interviews

Acronym	Job Title	Details
Arch	Architect (Nepalese)	- Over 5 years' experience as an architect in Nepal. - Over 10 years' experience as an architect in Nepal (previously worked for a government department). - Over 4 years' experience as an architect in Nepal - Nepalese- Over 20 years working and researching the built environment in Nepal, works for a research and teaching institute.
Eng	Engineer (Nepalese)	- Nepalese – Over 6 years' experience as an Engineer in Nepal. Prior to Earthquake was in a Municipality. - Nepalese – Civil Engineer working for local International NGO alongside government in developing urban resilience policy for Nepal. Background in mostly in Disaster Risk Policies and Strategy Action Plans. Prior, approximately 12 years as a Civil Engineer.
PM	Programme Manager (Nepalese)	- Nepalese – A Programme Leader for 7 years working on urban policies and their impacts on the disaster and recovery effort, with a focus on children. - Nepalese – Programme Leader for Disaster Risk Management and Climate Change in local NGO. Over 20 years' experience working in Nepal, mostly in the field of Disaster Management, Community Development and Risk Management. - Nepalese - A Doctor, with 5 years running a local International branch of an NGO focusing on Health. Works directly with Ministry of Health and has undertaken different reconstruction projects of Health Care facilities and Hospitals in Nepal since the Earthquake.
CC	Community Consultant (Nepalese)	- Nepalese- Responsible for issues around disaster management and communication- focus on Cross cutting issue. Previously worked in Ministry in Disaster management field. - Nepalese- Coordinator for International community contingency planning process in UN agency. - Nepalese- Responsible for Emergency response and preparedness of communities. Over 15 years' experience. - Nepalese- Senior Programme Officer for Urban Community Management. Over 10 years in local NGO.
DisMEe	Disaster Monitoring & Evaluation Expert	Nepalese- Disaster Monitoring, Evaluation and Knowledge Management & Communication Manager. Over 20 years' experience working in Nepal mostly in Monitoring and evaluation of Disaster Management.
UrbREM	Urban Resilience & Engagement Manager	Nepalese- Urban Resilience and Engagement Programme Manager for local NGO.
GEng	Government Engineer (Municipality)	Nepalese- 10 years' experience working in a Municipality as an engineer dealing with urban development through Solid Waste projects and different Project Management positions.
GArchP	Government Architect & Planner	- Nepalese- Over 20 years' experience working as an architect and planner in DoUDBC in a very senior position. - Nepalese- Over 10 years' experience working as an architect and planner in KVDA.
iNGOP	International NGO Planner	Westerner- 8 years' experience as a Planner in a Western country. Two years' experience working in Nepal as a Planner and Local Governance advisor.
iPM	International Programme Manager	Westerner - Project Manager for a Western Aid Program with 3 years' experience working on Sustainable Transport projects and coordination with reconstruction projects within Nepal.
iCC	International Community Consultant	Westerner- Over 5 years' work experience in Nepal, manager of Urban resilience program for an INGO.
HD	Humanitarian Director	- Nepalese – A Doctor, with 10 years Director of local International branch of an NGO focusing on Health. Works directly with Ministry of Health and has undertaken different reconstruction projects of Health Care facilities and Hospitals in Nepal since the Earthquake. - Westerner – 4 years working in Nepal on reconstruction within rural and urban settings. Background in Project Management. - Moroccan - Senior international development professional with over 15 years of experience. Previous work includes seismic retrofit methodologies for houses in Haiti and Nepal.
UNP	United Nations Planner	Nepalese- Worked for UN organisations for last 6 plus years in fields of coordination, planning, information management systems and evaluation. Previously worked in Government authorities as a Planner.

Urban Land Use: How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?

To address the first research question, the interviewee responses have been categorised under three Factors. Factor One analyses the responses relating to Geographical and Demographic impacts on the resilience of Kathmandu. Factor two looks at the changes to land use patterns that impacted the extent of the disaster earthquake caused in 2015. The third factor considers the disaster context of Kathmandu and how urban resilient the valley really was before the Earthquake.

Throughout all factors, a key theme that arose was the importance of social and cultural norms within the valley and how they have impacted the resilience of the area. This includes exclusions of foreigners until 1895 and social pressures and law pressures from the *Right to Land Act*.

Factor One- Geography and Demographics

Factor One, representing seven of the respondents, reflected that the the lack of geographically and demographic data and mapping impacted on the extent Kathmandu Valley can be urban resilient. This factor asserts that statistical data is missing within Nepal and that has effected the planning legislation and mapping before the earthquake. Responses to these questions are characterised by a scepticism to data ever being produced and strongly believe that this has affected the planning legislation and political will to make changes in the Valley. They are particular that “no long term strategic thinking” (ICC 1) is a direct result of the lack of statistical data being seen as valuable in the political situation in Nepal.

Factor Two- Change of Land Use Patterns in Kathmandu Valley

Factor Two, representing eleven of the respondents, is characterised by an emphasis on changes to land use patterns resulting in ad hoc, unplanned development in environmentally constrained areas, heightening Kathmandu Valley’s exposure to natural hazard impacts. They believe that a lack of mapping and management of land use changes in the valley has left the area at great risk to disasters and therefore not ‘urban resilient’. There is overwhelming agreement that Kathmandu Valley for the most part has not been planned, and when planning has been undertaken in some areas, this has been ignored due to enforcement and implementation issues. Another belief emerging because of these factors is development in the valley has been happening in hazard prone areas.

More than any other factor, there was strong opinions attached to the lack of planning for informal settlements. Not only do respondents view that settlements have gone unnoticed, but they are in the most dangerous areas within the valley (see PM 1 comments in **Appendix 16**).

Through this factor, respondents placed emphasis on the enforcement issues of environmental or risk mapping in the valley as the system supports a certain degree of developer flexibility. In addition, they view the geographical landscape of Kathmandu to be complex and constantly changing due to development, placing “Kathmandu Valley more at risk each time.” - **Arch 1**

The cluster response strongly supported the notion that development within the valley has not been planned and has adversely impacted the environment but also frustrated the process for future development planning. In addition, they view that planning is highly political and has been influenced by election cycles and political instability. Bold statements were made that urban planning has not existed in Kathmandu, just “ad hoc and reactive kind of planning” - **GEng 1**. They believe that with the huge demand for housing, and inadequate framing of the role of developers; most of the housing that was built before the earthquake was constrained, under-regulated and in some cases in inappropriate places.

Factor Three- Disaster impacts on urban resilience of Kathmandu

Factor Three, exhibits the different considerations of disaster’s impacting Kathmandu and how these have played a role in the resilience of Kathmandu. Overall the respondents agreed that the lack of planning for all major hydrological disasters in the valley is why the urban context is not resilient. They place high value on the damages to property and economic losses during disasters, not on how the land use changes have occurred in the valley as a direct result of construction planning, standards and monitoring over time.

How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?

The respondents highlighted lack of any environmental mapping determining development activities within the valley which could have made the valley more resilient, primarily by not placing development in high risk areas. Comments provided around how development in Kathmandu has occurred over time, illustrates that it has been ad hoc, unplanned, and irresponsible on location and standards. This was identified by interviewees as being a major factor in the impact of the devastation to the buildings following the 2015 earthquake.

Collectively from the questions asked around the theme of Urban Land Use, the findings show that Kathmandu prior to the earthquake was not an urban resilient place and that this directly impacted on the amount of damage.

Urban Development: What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?

Factor One: Unplanned urban development

“Kathmandu Valley has always been a rent seeking behaviour, just build and build with no checks. It wouldn’t matter if there were risk areas identified, they would be ignored.” – CC1

Factor One, addressed questions relating to the unplanned urban development within the valley and the links between ad hoc development and the devastation to the built environment resulting from disasters (Earthquakes, flooding, landslides and fires) within Kathmandu Valley.

There was strong agreement that growth within the valley has historically never been controlled and that development has happened on a mass scale as result of urbanisation and private development. Key respondents identified that the Risk Sensitive Land Use Plan (RSLUP), that is still to be approved, is the first attempt of the government to bring in zonings, growth boundaries and restricted development areas. There was support on the concept that if this plan had existed and been enforced many years before the earthquake, these planning strategies would have saved many lives and damage to infrastructure within Kathmandu Valley.

There were strong agreements from the interviewees that there has not been any attempt to restrict development in disaster risk areas, through zonings or any other urban planning approach. Most pointed towards lack of political support with developers dominating the construction field and Municipalities wanting more rent.

The responses evidently illustrated that the building permit system is seen to be inadequate within the valley. The respondents expressed concern that developers constructing dodgy, poorly designed buildings due to the lack of an appropriate building permit system which directly resulted in the damage that was caused in the earthquake. The focus on design of the permit system is further reflected in the fact that Kathmandu Valley experienced ‘tragic’ loss in poorly developed houses (HRRP, 2016). The interviewees took a broader perspective, stating that the building permit system has failed to provide safe and sufficient infrastructure to support the city which has resulted in poor people being pushed to the city’s outskirts and within risk areas. The respondents strongly expressed

negative attitudes to its quality, implementation and monitoring. They believe planners play a strong role in facilitating safe development and currently this is missing.

Factor Two: Urban Growth Boundaries

“[Urban Growth Boundaries] would have made a big difference on where development occurred, and we might not have so many buildings in areas of high risk.” – **GEng 1**

Factor Two, representing almost half the respondents, is typified by a highly negative attitude towards Kathmandu Valley using any planning approaches for urban growth boundaries to control growth and reduce risks. Respondents with roles as Community Consultants, Programme Managers and even Humanitarian Directors collectively were unaware of such planning approaches and therefore answers were expressed along the lines of ‘it wouldn’t work’. When further questioned, it became evident this was because of political and cultural situations in Nepal. They also collectively agreed that growth boundaries would help mitigate disasters which could have resulted in less losses in 2015. “The urban sprawl as you call it, has caused major problems in how to deal with disasters. ... The area is becoming more and more at risk... a boundary would help with lowering these risks.” – **Arch 2**

The respondents were concerned with the lack of hazard risk information and data in determining urban expansion, especially around development occurring in risk areas. They believe that when hazard risk information was produced (like maps, data analyses etc.) it was ignored and never connected to urban development activities.

Factor Three: Urban Planning ability to reduce damages in the 2015 earthquake

“Kathmandu needs to seriously consider an urban regeneration of Kathmandu as most buildings are still in trouble of future disasters.... This is a direct result of the lack of enforcement of building standards in the past... we need to do something now, or we will be in even further trouble.” –

GArch1

Factor Three is characterised by investigating further the interviewee’s professional opinion on the impact that the lack of planning for urban development increased damages in the 2015 earthquake. The responses to this factor were typified by a highly negative attitude towards development in the valley reflecting any urban planning legislation. A key example of this is “A study recently done by Ministry of Urban Development and collaboration with JICA showed that 94% of buildings in Nepal are unsafe and not up to building standards.” – **INGOP 1**. More than any other factor, the respondents agree there needs to be better controls over how, where and when development in the valley occurs.

This factor therefore places great emphasis on planning legislation incorporating disaster risk considerations. Overall the respondents highlighted that it is unclear the effect of Urban Planning on mitigating the risk of the earthquake as it hasn't been appropriately applied in the valley. However, most argued that if there had been more land use mapping and that if the NBC had been applied, the devastation would have been significantly less.

Factor Four: Effectiveness of Disaster Management

Factor Four, addresses the disaster management that has occurred in Nepal. This approach is to gather insight around how disaster management strategies have been undertaken and if they have been filtered back into urban development. As this is not a simple question, three (there is only two questions there) main questions were asked. The responses from the interviewees identified that some work has been done in this area, but it has been primarily around response and recovery. The opinions expressed were that there is a great deal of work that needs to happen in preparedness but also around 'urban resilience'. Collectively, respondents identified that disaster risk management has not been linked to urban development and its corresponding regulations, as the two fields are seen separate within Municipality structures and government policies.

There was a collectively deep antipathy towards disaster management policies before the earthquake. It was highlighted the antagonism between urban planning and disaster management planning. Not only do they view disaster management as a process that sits in silo to planning, they also view that there is still a link yet to be made by professionals in both these fields. This scepticism is further reflected in their belief that disaster management has in the past only focused on responses and relief during emergency phases of disasters and not the work that can be done beforehand. "There is no link. The damage that happened in the earthquake shows this" – **GEng 1**.

The respondents strongly agreed that 'site selection' is not a process undertaken through the building permit system and therefore not a consideration for any development approval given. There was scepticism expressed of this ever being a consideration in future development.

Factor Five: The Bigger Picture Thinkers

Factor Five, exhibits slightly different factors than the others. The respondents all expressed belief in enforcement being the major issue in restricting the role that urban planning could have played in mitigating the risks of disasters within Kathmandu Valley. They identify that Nepal has a 'reactive' response to disasters which has led most of the planning to be around recovery and response and not preparedness or risk reduction. The comments also highlight the need to focus on all disasters

and the changing attitude happened too late to make a difference in the damages caused by the 2015 earthquake.

What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?

Overall, respondents have identified that there has been significant lack of urban planning for the development that has occurred in the valley in the past. This lack of planning they linked to enforcement, implementation and political governance issues. They view the lack of planning as a key result of the devastation that happened in Kathmandu Valley from the earthquake and the reoccurring damages happening from flooding and landslide across the valley. The responses all stated that there has been no attempt in the past to control urban expansion from developing in risk areas. It is conclusive that even when hazard risk information was provided relating to land uses, this was not connected to urban development activities.

Urban Governance: What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?

To address the question around urban governance within Nepal and how this impacted the urban development management within Kathmandu Valley with regards to urban disaster risk reduction, there were five factors identified through the interviews. A summary of the key concepts identified are provided under this theme, including key comments made to different questions relating to urban governance.

Factor one: Historical Governance and Political Structures and their impacts

“If [urban development or disaster risk legislation is] not passed, it probably won't get passed. The parliament passes things very quickly when it wants something passed and it will sit on things it doesn't...” – iNGOP 1

Factor one, considers the history of governance and political structures of Nepal and Kathmandu Valley in general and how this impacted the design, planning and implementation of urban development and land use planning within the valley. This factor is characterised by a far more sceptical attitude towards planning and political systems and a greater emphasis on the integration of the two sectors.

The answers provided indicate that political systems within Nepal have significantly impacted the limited development control within the valley from control of budgets, and lack of delegation of

power. This cluster strongly believes that development has been under-regulated, stating that lack of planning regulation is directly linked to the frustrating political processes. The instability of governance and reactive behaviours of politicians were highlighted as the key issues.

“With the historical political system, we have, the government and the Nepalese people are very reactive, it’s going to be a hard push to move this to proactive for all disasters in Nepal.” – **CC 2**

Factor two: Key Government Actors

“Governments of all levels are not directly linking with ground realities they are trying to tackle... therefore there is no actual key government actors” – **PM 2**

Factor two, is designed around highlighting who the key government agencies are for disaster resilience within Kathmandu Valley. The responses provided exhibit similarities with historical urban governance, sharing a broadly negative view on coordination and budgeting issues impacting the resilience prospects of the valley. They place great emphasis on the lack of identification of who takes the lead role and who is a responsible entity. They also advocate the lack of coordination and accountability of Ministries in tackling the issue of disaster risk management and urban planning which is linked to the amount of damage that was seen following the earthquake.

Factor three: Key decision makers and implementation

Factor three, draws on from factor one and two to determine who the actual decision makers are around urban development and resilience and historically how those actors have implemented urban planning. The responses primarily identify a lack of community engagement in decisions surrounding the future planning and construction of their communities. Key examples of this is “Field survey done by UNDP for community integration found that there has been no serious engagement from Municipality representation to their people, this is a representation of historical contexts to which development has been occurring in the valley” – **GArchiP 2**. Humanitarian professional workers identified that the communities they worked with in the urban contexts, for the most part were unaware of any building controls.

The respondents unanimously agree that the implementation of the NBC and the Building By-laws have been inefficient and in some cases downright ignored. This was identified as a direct result of perception of the codes being too strict, too hard to implement, lack of belief in the codes and enforcement issues at a local level.

Factor four: Monitoring

Factor four, represented almost half of the respondents and identified issues of monitoring urban development in Kathmandu Valley. The responses highlight that there is very limited monitoring and the biggest issue is the capacities of local authorities. Many of the statements are indicative of lack of capacity at local level to ‘undergo required monitoring and evaluation’ – **DisMEe 1**. These factor asserts that inspections following building permits with which there were issues were not done which resulted in many constructions happening that did not reflect the approved permit and were in some cases, respondents identified, as not meeting seismic standards of the NBC.

Factor five: Coordination

“Coordination between agencies and organisations (public and private) is a major issue when it comes to reducing risks in urban contexts” – **CC 2**.

Factor five, is drawn from responses to three questions all relating to coordination of urban governance. This factor places great emphasis of coordination being the key when looking at impacts that cause devastation of urban development following a natural disaster. Many of the statements are indicative of a lack of coordination between all sectors. The reasons identified were around political tensions between different Ministries, regarding where responsibilities and funding was allocated. Most respondents agree that this lack of coordination is a significant problem for urban development to being resilient to natural disasters. This discourse around coordination maintains that historically there has been very little sectoral development projects or even communication between responsible entities of the projects. This lack of connection and coordination has resulted in urban development’s restrictive ability to mitigate risks within the valley.

The responses addressed issues identified around municipalities having poor history of coordination, especially in the planning processes within Kathmandu Valley. An important issue was expressed through the Engineers interviewed, that there is currently no accountability for local bodies to MoUD as they sit under MoFALD therefore even if MoUD produce an Urban Policy, the Municipalities can choose to adopt it or not.

Factor six: Resource and Capabilities

Factor six, draws on findings around the local institutional and federal governance resources and capabilities and how that has impacted the implementation and coordination of urban development within Kathmandu Valley. The responses overwhelmingly illustrated there is a huge short fall in the

capacities of local institutions to implement any development controls due to financial situations and lack of knowledge and resources. In addition, they view planning for urban development and disasters as highly political factors as stated earlier and that more emphasis should be placed on building capacities of local institutions so changes don't have to be directed by central government before they can happen.

What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?

The findings from the interviewees identify that political instability over the past 20 years within Nepal has significantly impacted on the amount of control over urban development and disaster risk management. The identification, communication and coordination of key government actors' responsibilities and requirements have not been clearly defined through government. Which has resulted in a confusing political and governance arrangement. When trying to address environmental issues through urban development, the clear roles and responsibilities of urban governance is critical. With no clear government departments with accountabilities to manage risks collaboratively, the past development activities seen in the valley will continue.

The findings that community engagement is poor within the valley is a significant issue and needs to be addressed to ensure that community are brought along the journey for urban resilience and understand the importance of urban development control. Historically, if a community feels disengaged, it will feel it has no ownership around their community. This directly impacts the way the community uses the built environment space.

Lessons for Urban Planning of Kathmandu Valley

The following question was asked to all interviewees to identify any gaps that may exist: *In your opinion, what has been the biggest issue for urban planning and development in mitigating disasters within KV?* The responses summarise what has already been highlighted through the three factors political turmoil, rapid unplanned urbanisation, lack of coordination and implementation and the lack of the urban planning legalisation in integrating disaster risk qualities. Overall, the biggest issue is the lack of control of how and where growth development occurred.

Conclusion

‘Cities present an opportunity for everyone to commit to share resources and spaces in a way that ensures [long term] protection of the planet and its natural resources.’ – The Government of Nepal (MoUD, 2016). The 2030 Agenda for Sustainable Development includes a Sustainable Development Goal addressing urban areas, aiming to *‘make cities and human settlements inclusive, safe, resilient and sustainable’*.⁵ Looking at this as a benchmark, the question addressed in this paper is how has Kathmandu stood up to being an inclusive, safe, resilient and sustainable city when viewed through the lens of urban planning and development? As it is this lens that builds the city’s resilience.

In a country that has evidently experienced significant and multiple disasters over time, it is surprising the limited life policies and building standards seem to have. As previously addressed, the BC was developed following the 1988 earthquake to enforce safer building design. However, as the Urban Governance section demonstrated, this has not resulted in the implementation of action, with many buildings in Nepal having been declared unsafe. The question then is this: How did we get here? How did a country with so much history of devastation due to natural disasters end up in another event where the devastation was so prevalent as the extent of what was seen in 2015, despite all the initiatives that have been put forth since 1988?

The overall three key findings are:

1. Disaster Management in Kathmandu Valley is all about response. There is little to no real disaster management which is preparedness around building infrastructure so that it can withstand earthquakes and bounce back quickly (resilient infrastructure).
2. There is lack of urban planning maturity. Urban Planning is still young in the country and historically has been ignored.
3. Urban Governance: Historical political will around making urban resilient cities is a bigger question and needs to be addressed through further research.

For Kathmandu to achieve a more resilient urban space there needs to be an overall integrated approach of disaster management through providing the conditions of physically, socially, economically, environmentally and culturally safe and resilient urban spaces. (A list of recommendations can be found in **Appendix 18**).

⁵ Sustainable Development Goals: <https://sustainabledevelopment.un.org/?menu=1300>

Question 1: How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?

Nepal's geographical and cultural factors play a significant role in how Kathmandu Valley has grown over time and the changes in land use patterns throughout the area. This also includes the impacts of ad hoc development not complying with planning regulations and lack of capacities of institutions to enforce regulations. On top of this is the problem of how little planning regulations and policy address hazards and natural disasters. The largely unplanned and unmanaged development evident through historical analysis has left the Kathmandu Valley with major shortages in infrastructure and services, high vulnerability to earthquakes and one of the worst air qualities of any urban agglomeration in the world (WB, 2013b).

From the Land Use Study and the Urban Development Study it was found that unregulated development has resulted in not only loss of agriculture and forest land but development in hazardous area without appropriate building controls. This was evident in the amount of damage to buildings during the earthquake in Kathmandu including nearly complete destruction of Bungamati (main city in Lalitpur District).

Without a long-term perspective for the area, Kathmandu Valley's urban development has rapidly expanded and increased with no coordination and lack of vision. This has resulted in ad hoc development with no consideration of development risks from the land upon which the construction occurs, nor to the development construction type and techniques, nor to the possible escape routes during a disaster (such as fire flood, earthquake or landslide).

The historical urban expansions on the fringe areas of the valley without any development control, the prevalence and growth of slums, inadequate transport, as well as degrading environmental conditions, all aggrandized the extent of damage wrought by the earthquake in 2015.

Question 2: What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?

Critical review and analysis of the data presented in the research sections of this report reveal that there are varying degrees of impacts on the development of housing within urban areas of Kathmandu Valley. From the Land Use Study and the Urban Development Study it was determined that within Nepal there has been a long-standing attempt, at least conceptually and in policy, to further develop the role of infrastructure to reduce the likelihood and impacts of disasters. This has however, been difficult to translate into practice.

It has been well recognized that the single most important development in reducing earthquake losses in the world has been the incorporation of seismic design provisions into the building codes (MPPW, 2009). With 98% of buildings in Nepal being built by owner-builders who follow the advice of local craftsmen and not the NBC provisions, is reflected in the extent of the disaster impact that occurred in Kathmandu Valley through 2015 earthquakes but also the recent flooding and landslide disasters (Parajuli et al., 2000).

Overall there has been no updated strategy and plan to guide urban development in the valley since the 2020 Kathmandu Valley Long Term Development Plan of 2002. This plan lacks necessary implementation and financing plans (WB, 2013). Also, data and technological tools to support the planning process at the metropolitan scale are inadequate. Similarly, at a local level plans and regulations are outdated and largely ad hoc. Building by-laws and the NBC are not being enforced, resulting in increased development in high risk areas. HRRP (2016) study found that the reason for the collapse of the buildings that killed more than a dozen people within Kathmandu Metropolitan was due to 'ignorance of the building bylaws and failure to comply with the NBC'.

Interviews identified that development and construction has not complied with any of the building standards and that this directly impacted the amount of damage that occurred during the earthquake. To evidence this theory, one interviewee drew from a study undertaken by MoUD and JICA which found that "94% of buildings in Nepal are unsafe and not up to building standards" (INGOP 1). If this figure is even close to the truth, it's conjunction with much of the urban growth being spread into risk areas results in not only unsafe buildings, but these structures being constructed on unsafe land. As one Government Architect Planner expressed vividly, "we need to do something now, or we will be in even further trouble" (GArchP 2).

Major challenges to the role *urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley* can be linked to private investment-driven development, national development policy, implementation of building codes and by-laws, and high rates of urbanisation without basic infrastructure services and land management. As well as lack of funding allocation to local governments to appropriately carry out measures to mitigate these problems. This links into the last part of this analysis: Urban Governance.

Question 3: What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?

The major issues raised in this report relate mostly with Urban Governance. From the Land Use Study and the Urban Development Study it was found that urban areas in Nepal have increased with natural

expansion, over a period in which the government has been transition from a conflict era as well as decentralising. This has impacted on the implementation, coordination and monitoring of the regulation and policies.

Historically there has been a spatially fragmented and asymmetric local governance structure with limited resources, inadequate coordination between themselves and the entire governance system of Nepal, outdated plans and planning techniques, and inadequate enforcement. As well as inadequate financing and financial instruments. These factors have all played a major role in how the Nepalese governments have functioned but also on the complexity that exists around identifying steps of who is involved and who has the power to make Kathmandu a resilient city through the means of urban planning and development.

The legal basis and institutional capacity to enforce land use and zoning laws as well as environmental standards has still to be created. Government policies on roads and transportation, local development, fiscal and economy, industry, tourism, agriculture, health and education, etc., have largely influenced the nature and type of urbanisation in the country. This is due to an unfinished and partially decentralised system that lacks supportive leadership from central government and shifts too much financial burden and responsibility to subordinate units at district and local level, without the transfer of sufficient resources. In addition, these authorities often lack experience and capacity to raise enough funds, especially in rural areas, and are therefore unable to cover needs of the population.

The overall finding is that the key urban development regulations that do exist (NBC and Building By-Laws) are not effectively being implemented resulting in massive development occurring across the Valley that is unregulated and unsafe. Coordination was also highlighted by respondents as a key issue with **INGOP 1** highlighting accurately: “The Valley has many municipalities with interrelated infrastructure, however due to urbanisation of unplanned settlements and lack of coordination, major issues have arisen.” The best example was provided by **GEng 1**, who provided contextual information around the arrangement of Municipalities being only accountable to Ministry of Federal Affairs and Local Development. Therefore, any urban development policy designed and implemented by the Ministry of Urban Development (or any other Ministry for that matter), Municipalities have no accountability to actual implement it themselves. This is a major fault of the established governance arrangements around the Ministries of Nepal. When there is a Ministry that is responsible for the development and implementation of building standards but has no power or capacity to actual implement it, building standards become essentially useless.

Nepal is at risk from further earthquakes and many other natural disasters. The experience of the 2015 earthquake underlines the importance of urban planning to prepare Kathmandu Valley for future disasters. This involves stronger coordination and communication between sectors, greater emphasis on implementation of planning documents and a 'bigger picture' plan streamlining resilience across all factors of urban development.

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Introduction

Appendix 1: Background information of Kathmandu Urban Development Situation before the Earthquake

The research has been applied to the two main densely populated areas in Nepal: 'Kathmandu Metropolitan City' and 'Lalitpur Sub-Metropolitan City'. The two cities have grown together and are so interlinked there is no longer a clear geographical and socio-cultural boundary (Thapa et al., 2008). The two urban centres will be considered together as part of this study and here forth referenced as 'Kathmandu'. The metropolitan area of Kathmandu includes three nearby municipalities within the valley: Kirtipur, Madhyapur and Bhaktapur. These areas will also be considered in this study as they are viewed as urban. Kathmandu Valley in the census year of 2011 was 2,468,316 with annual growth projections of 4.63% per annum (UNDP, 2015; Central Bureau of Statistics, 2014).

Table 1: Total urban Population of the Municipalities in Kathmandu Valley

Kathmandu Valley	Area (Ha)	Census	Density
		2011	2011
Total Urban Area	9,669	1,426,641	148
Kathmandu Metropolis	4,945	975,453	197
Lalitpur Sub-metropolis	1,515	220,802	146
Bhaktapur Municipality	656	81,748	125
Kirtipur Municipality	1,476	65,602	44
Madhyapur-Thimi Municipality	1,076	83,036	77

(Source: KVDC and UNDP, 2013a, pp. 78-79; Central Bureau of Statistics, 2014)

In 2013, piped water supply meet less than half of the total demand and the Sisdol landfill site was (and still is) at capacity (WB, 2013). Solid waste is often disposed on riverbanks and in open dumps where there is major flooding and poor informal settlements. There is no public transport system in the Valley and citizens rely on private and largely unregulated small bus Companies (WB, 2013). This is evident of the urban development situation that exists within the Valley.

Cultural Heritage

The Traditional "Guthi" Trust are responsible for management of public historical land. Over time due to the excessive pressure of commercial activities from developers in the private sector. Public historical land has therefore been slowly being leased out to private companies who are building big

commercial centres on the lands (Pradip and Devendra, 2009). This is leading to a gradual erosion of the cultural heritage land use areas in Kathmandu Valley.

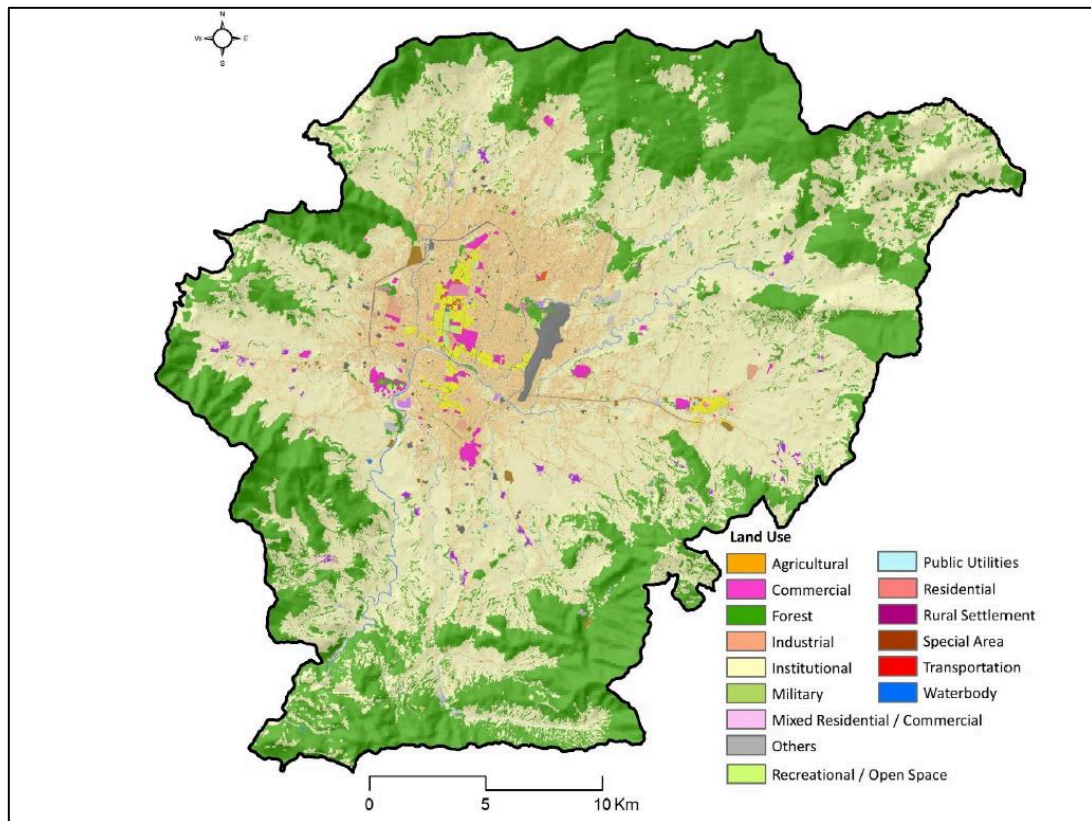
Transport

Kathmandu Valley is prominent for recording a much higher road density than the official world development indicator (WB, 2012; Thapa et al., 2007; KVDA, 2015). This is because the only means of transportation within the city is by road which are often unlinked, narrow, crooked and poorly maintained (Thapa et al., 2007; UNDP, 2012; Pradip and Devendra, 2009). Inadequate road infrastructure is a direct result of urban development and planning taking place without adequate provision of transport infrastructure demands and therefore consequently impacts on the nature and composition of the traffic levels in the city (Thapa et al., 2007; UNDP, 2012; ICIMOD/UNEP/MoEST, 2007). This issue played a major role in the devastation Kathmandu Valley experienced during the earthquake from access issues during recovery phase to the destruction of the roads due to poor construction and lack of urban spaces for safety points (HRRP, 2016).

Land ownership

It is worth noting that land ownership in Nepal is quite political. Land ownership papers are more valuable than citizenship papers. This is so far embedded socially that individual land owners have complete rights over their plot of land. This makes regulation of development via zoning only possible through incentives and deterrants (PAHAR, 2003). This links into the findings of the Land Use Study regarding changing of land-use can occur by the owner deciding it will change the land from agriculture to residential or commercial due to economic initiatives, and there is no regulations to mitigate this change of use.

Figure 1: Land Use Map of Kathmandu Valley 2015

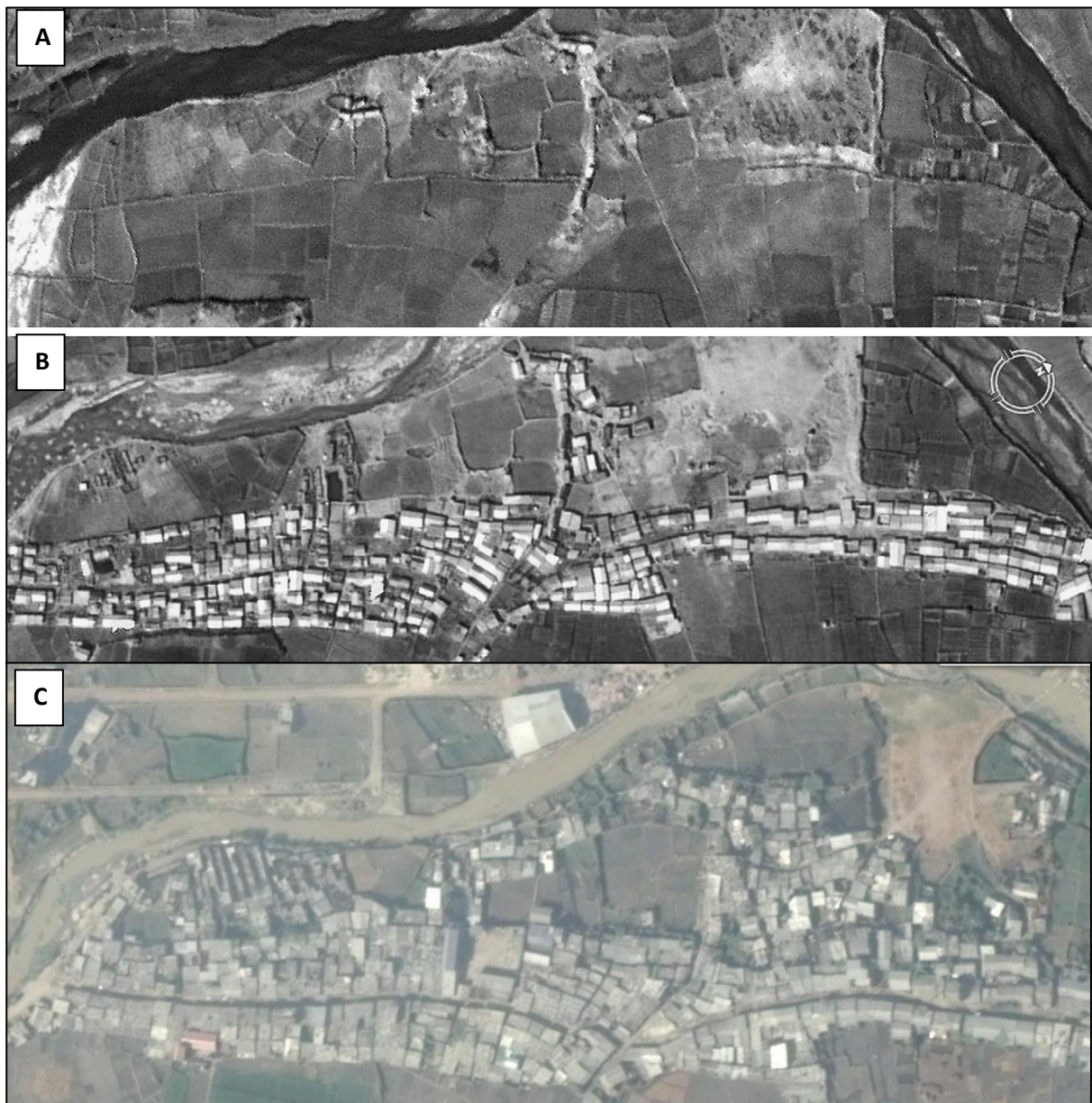


(Source: UNDP, 2015)

Informal settlements

Although there is no 'official' documented slums or squatter settlements within land use planning in Nepal, Kathmandu Valley recorded 17 squatter settlements in 1985 (2134 inhabitants) and 61 settlements (11,862 inhabitants) in 2000 (Pradhan and Perera, 2005; Thapa et al., 2007; ASD, 2012; ADB, 2010). The rapid growth in settlements is predominately occurring in high risk areas within Kathmandu Valley (MoUD, 2016; ASD, 2012; ADB, 2013; ICIMOD/UNEP/MoEST, 2007). With a recorded 29 squatter settlements within river banks (2,031 households) in 2012 with high risk of flooding (MoUD, 2016). This growth near river banks is evident in **Figure 2** which shows a squatter settlement on the banks of Manohara River near the Tribhuvan International Airport runway from 2000, 2007 and now 2017. This settlement was majorly impacted through recent flooding events in 23 July 2002 and 20 July 2011 (Nehren et al., 2013; Binod, 2002).

Figure 2: Squatter settlements on the banks of Manohara river system (400m away from South-East Corner of Tribhuvan International Airport runway).



A: IKONOS image 2000 (Source: Thapa, R.B. 2007) B Google Earth image 2007 (Source Thapa, R.B. 2007). C: Google Earth image, 2017.

Earthquake vulnerability

Determining the extent of the seismic risk is by seismicity ratings and the vulnerability of structures and infrastructure in correlation with high density of built environments (UNDP, 2012). In 2012, the United Nations Development Programme (UNDP) reported that the buildings and structures within Kathmandu indicated high to very high vulnerability with regards to damage from earthquakes

(UNDP, 2012; MPPW, 2009). This has been found to be indicative of the land use changes and the building codes standards within Kathmandu leading to this vulnerability (UNDP, 2015).

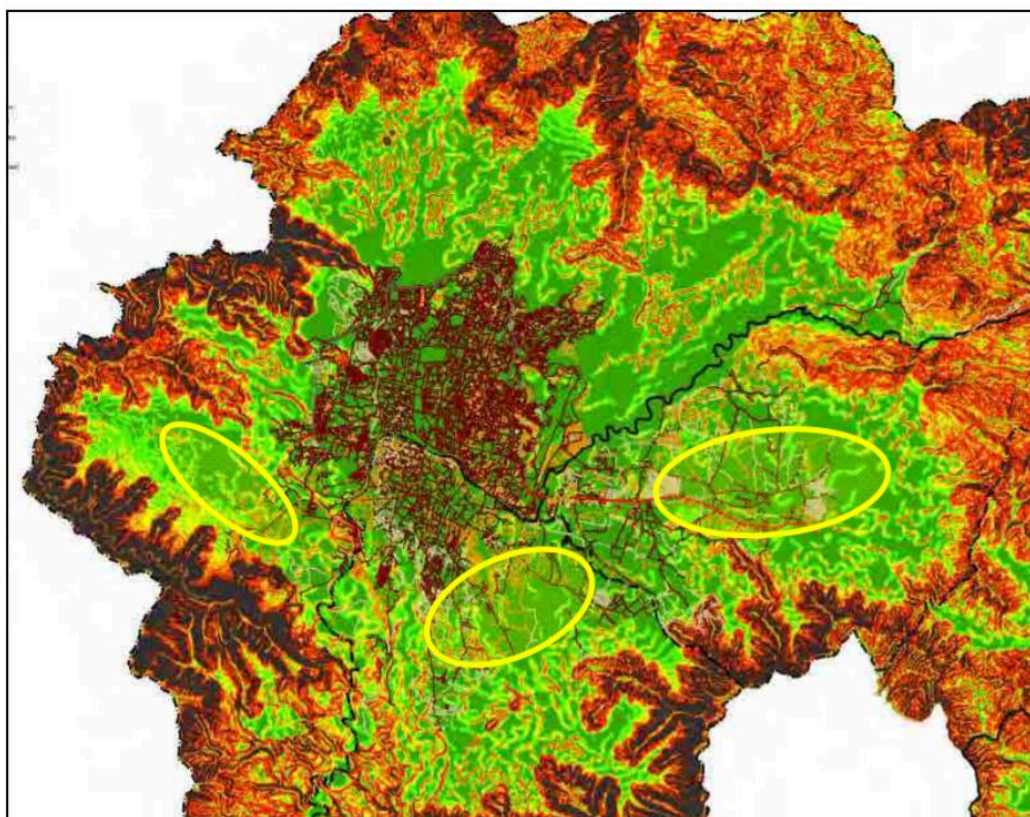
River Beds- Flooding and Earthquake Hazards

It was proposed in the JICA report in 2002 to the Nepal Government that riverbed areas are not firm enough to resist earthquakes and for this reason these areas should not be suitable for urban development (JICA et al., 2012). It was also highlighted that these large areas could be used as evacuation areas after a disaster. However, in 2007 the *Kathmandu Valley Building By-Laws 2007* (KVBB) identified the riverbeds within Kathmandu Valley to be designated urban expansions zone's which are categorised a 'to accommodate future expansions of growth' (JICA et al., 2012). This caused an increase of urban development along these river beds that suffer flooding problems during monsoons, especially the Sangla Khola, Dhobi Khola, Manahara Khola, Hanumante Khola, Nakhu Khola and Bishumati rivers (JICA et al., 2012).

Landslides

Urban development in steep sloped areas of Kathmandu Valley are at high risk of landslides during heavy rains and earthquakes (JICA et al., 2012). Geomorphologically, Kathmandu Valley is divided into two main areas: firstly, mountainous areas around the valley and low gentle sloping land in the centre of the valley. In the mountainous areas, the landslide hazard is quite high with significant possibility of slope failure as found by the JICA Report in 2002 (JICA et al., 2012). The low-medium slope areas in the centre narrow zones of the valley have potential hazard based on the relative height of the terrace surface. The **Figure 3** (JICA et al., 2012) below shows the land inclination of Green being 0 to 4 and yellow 4 to 8 degrees. In general land inclination of more than 8 degrees is not suitable for urban development (JICA et al., 2012). These risk areas have not been reflected in the zoning of Kathmandu Valley which is mostly urban and site selection suitability is also not considered as part of the Building Code requirements to check suitability of land. This is also reflective in **Table 2**.

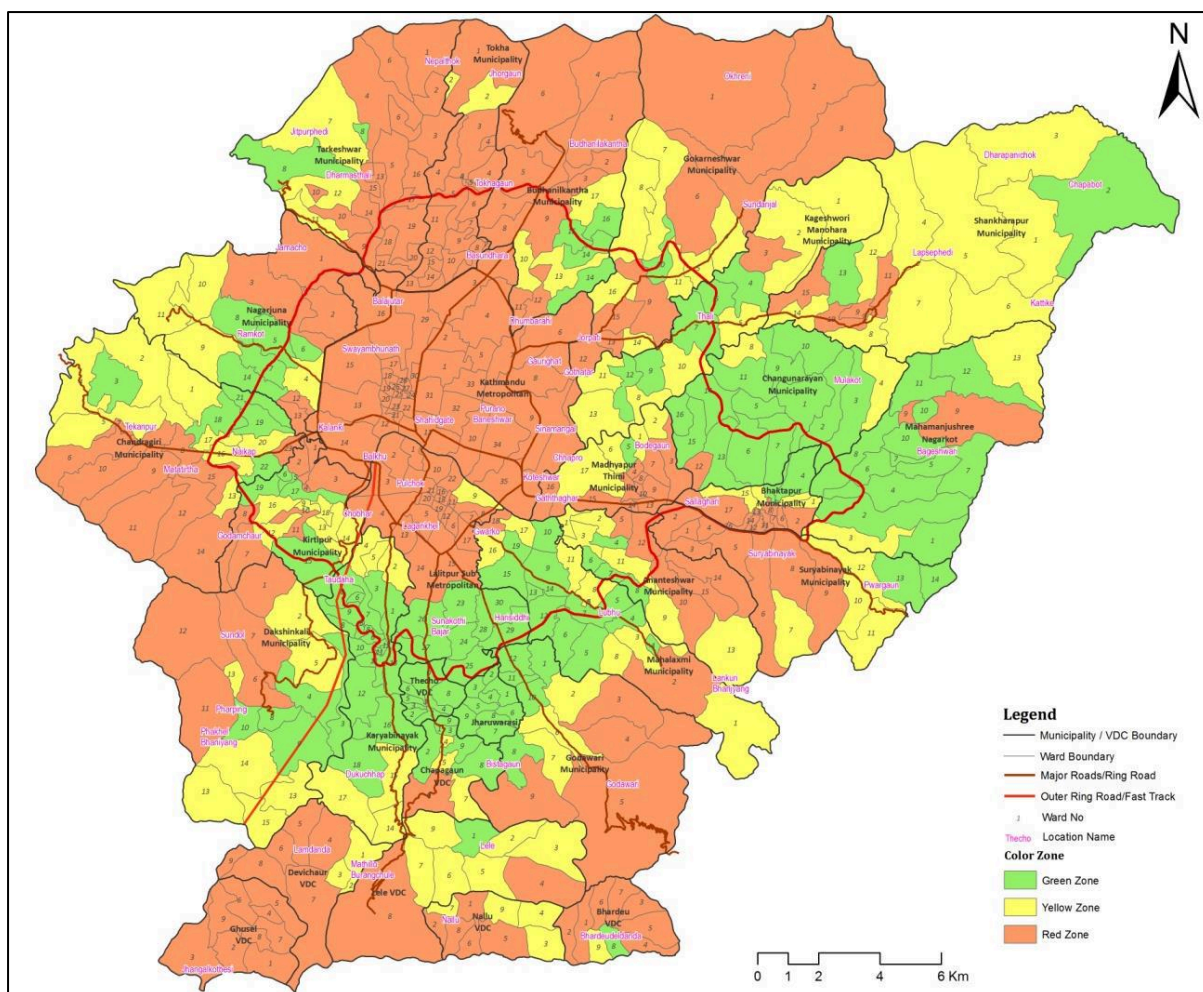
Figure 3: Risk areas showing possible Areas for Large-scale Urban Development proposed in 2012 (3 circled areas).



(Source: JICA et al., 2012)

Table 2: Colour Zones and Constraints (2015)

Colour Zone	Area in Ha (2015)							
	Built-up on Constraint	Non-Built-up on Constraint	Total Constraint Area	Built-up on Constraint Free	Non Built-up on Constraint Free	Total Constraint Free Area	Grand Total	Non- Built up Constraint Free Area Percent
Red	4,116	20,612	24,728	4,746	8,017	12,763	37,491	21.4%
Yellow	800	12,334	13,134	1,549	10,852	12,401	25,535	42.5%
Green	59	2,287	2,347	561	6,253	6,815	9,162	68.3%
Total	4,975	35,234	40,209	6,856	25,122	31,979	72,187	34.8%



(Source: UNDP, 2015)

Appendix 2: Urban Planning Definition

Urban Planning involves determining and drawing up plans for the future physical arrangement and condition of a community (Farlex, n.d.). March and Leon (2013) describe it as a practical process of “seeking advantageous future physical, social, economic and ecological outcomes in cities and regions, drawing a range of action modes and delivery mechanisms together”. Hall and Twedwr-Jones (2011) and Hopkins (2001) allude to its highly procedural nature which they identify to be: action agendas, policies, visions (of desired future outcomes), designs, master plans and strategies, decision making systems to modify the directions and plans over time.

Planning is meant by the land uses, infrastructure, and real estate investments that can be used to improve the existing urban areas, but more importantly spur, plan for and direct future urban growth while considering external factors (economic, environment, social, etc.). Planning also includes the coordination of different sectors, stakeholders and community. Strategic planning has the ability to not only harness the economic energy that already exists in informal settlements but also to address the physical limitations and risks that characterise informal settlements. This includes hazard risk reductions (fires, floods and landslides) and what is termed as ‘social infrastructure’ planning (schools, health clinics, roads and transport systems, water and sewage capture and distribution as well as open community spaces and commercial areas).

Burby et al. (1999) pinpoint the power of planning to deal with disasters, in particularly in terms of the social and organisational aspects being drawn into planning approaches. This can reduce losses by ‘affecting both the location and the design of urban development’ and by ‘helping create a knowledgeable constituency of citizens who support hazard mitigation programs’ (March and Leon, 2013). Peck (2010) further highlights that urban planning plays the role of balancing these two principles for the achievement of reducing risk in cities. This is through participatory decision making, transparency and accountability and implementation of sustainable development in managing urban risks (Shaw et al., 2009).

Methodology

Appendix 3: Interview Background Note and Consent Form

Nepal Urban Planning Research- June-July 2017 Information Sheet

You are being invited to take part in a research study. Before you decide whether or not to take part, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully.

What is the process of the research?

Planning legislation provides for long-term strategies on how an area will grow factoring in sustainable principles like social, economic and environment. In the last few decades, disasters have been occurring in urban contexts where there is usually a prominent long-term planning strategy for that region. However, lack of resilient disaster relief planning and impact of major international reconstruction methods and applications have resulted in no long-term planning strategy being considered as part of the recovery and reconstruction process.

The world loses large numbers of people and infrastructure each year due to natural disasters. As people migrate more into cities for economic, environmental and social reasons; natural disasters are having a greater impact on our urban spaces. Nepal is geographically and naturally prone to disasters with an average of 900 natural disasters a year. The link between unplanned urban development, irregular, substandard housing development, loss of open space and decreased liveability has increased the vulnerability of Kathmandu to disasters. This has left Kathmandu to be declared one of the most earthquake-vulnerable cities in the world.

The purpose of this research is to investigate the effectiveness of urban planning on the preparedness, recovery and reconstruction efforts within Kathmandu before, during and following the earthquake in April and May 2015. The research aims to highlight the importance of planning legislation for disaster prone communities in tackling vulnerability and providing long-term sustainable communities.

The research information will be gathered through interviews, informal discussions and research of planning legislation and projects within Nepal and internationally. The interviews and informal discussions will be undertaken during the research visit to Nepal in June- July 2017. The final research dissertation will be completed by September 2017.

Do I have to take part?

It is up to you to decide whether or not to take part. If you do decide to take part you will be given this information sheet to keep and be asked to sign a consent form. If you decide to take part you are still free to withdraw at any time and without giving a reason.

What will happen to me if I take part?

The discussion will only take a maximum one hour of your time to which the interview may be recorded by a voice recording device as a means of collecting information. Alternatively, the interview may just be recorded by form of written notes.

What are the possible benefits of taking part?

The benefit of taking part in the study is to assist in furthering our understating of the importance of planning legislation for disaster prone urban communities in tackling vulnerability and providing long-term sustainable communities.

Will what I say in this study be kept confidential?

All information collected regarding the person undertaking the interview will be kept strictly confidential. Confidentiality, privacy and anonymity will be protected in the collection, storage and

publication of the research material. This will be through encrypted password protected storage device.

The University policy on Academic Integrity highlights that data generated in the course of the research must be kept securely in paper or electronic form for a period of ten years after the completion of a research project.

What will happen to the results of the research study?

The results of the research study will be used in the final Masters Dissertation for the Masters of Development and Emergency Practice. The research paper may be published on the university website. Anyone involved in the study will be provided an electronic copy of the research upon completion in October 2017.

Who is organising and funding the research?

This research is being conducted as a student of Oxford Brookes University under the Faculty of Technology, Design and Environment in the School of Architecture.

Who has reviewed the study?

The research has been approved by the University Research Ethics Committee, Oxford Brookes University.

Contact for Further Information

Alexandria Moore

alexandriakiaya@gmail.com

If you have any concerns about the way in which the study has been conducted, you can contact the Chair of the University Research Ethics Committee on ethics@brookes.ac.uk.

Thank you for taking time to read this information sheet.

CONSENT FORM

Nepal Urban Planning Research

Alexandria Norris-Moore

Student- Masters of Development and Emergency Practice

16034229@brookes.ac.uk

Please initial box

1. I confirm that I have read and understand the information sheet for the above study and have had the opportunity to ask questions.
2. I understand that my participation is voluntary and that I am free to withdraw at any time, without giving reason.
3. I agree to take part in the above study.

☐☐☐

Please initial box

4. I agree to the interview being audio recorded

Yes

No

☐☐

5. I agree to the use of anonymised quotes in publications

☐☐

6. I agree that my data gathered in this study may be stored (after it has been anonymised) in a specialist data centre and may be used for future research.

☐☐

Name of Participant

Date

Signature

Name of Researcher

Date

Signature

Appendix 4: Ethics Form

Faculty of Technology, Design and Environment

Ethics Review Form E1

This form should be completed by the Principal Investigator / Supervisor / Student undertaking a research project which involves human participants. The form will identify whether a more detailed E2 form needs to be submitted to the Faculty Research Ethics Officer.

Before completing this form, please refer to the University **Code of Practice for the Ethical Standards for Research Involving Human Participants**, available at <http://www.brookes.ac.uk/Research/Research-ethics/>, and to any guidelines provided by relevant academic or professional associations.

It is the Principal Investigator / Supervisor who is responsible for exercising appropriate professional judgement in this review. Note that all necessary forms should be fully completed and signed before fieldwork commences.

Project Title: The role of planning legislation in urban areas before and after the Gorkha Earthquake, 2015

Principal Investigator / Supervisor: Cathrine Brun

Student Investigator: Alexandria Norris-Moore

		Yes	No
1.	Does the study involve participants who are unable to give informed consent? (e.g. children, people with learning disabilities, unconscious patients)	☐	☒
2.	If the study will involve participants who are unable to give informed consent (e.g. children under the age of 16, people with learning disabilities), will you be unable to obtain permission from their parents or guardians (as appropriate)?	☐	☒
3.	Will the study require the cooperation of a gatekeeper for initial access to groups or individuals to be recruited? (e.g. students, members of a self-help group, employees of a company, residents of a nursing home)	☐	☒
4.	Are there any problems with the participants' right to remain anonymous, or to have the information they give not identifiable as theirs?	☐	☒
5.	Will it be necessary for the participants to take part in the study without their knowledge/consent at the time? (eg, covert observation of people in non-public places?)	☐	☒
6.	Will the study involve discussion of or responses to questions the participants might find sensitive? (e.g. own drug use, own traumatic experiences)	☐	☒

7.	Are drugs, placebos or other substances (e.g. food substances, vitamins) to be administered to the study participants?	☐	☒
8.	Will blood or tissue samples be obtained from participants?	☐	☒
9.	Is pain or more than mild discomfort likely to result from the study?	☐	☒
10.	Could the study induce psychological stress or anxiety?	☐	☒
11.	Will the study involve prolonged or repetitive testing of participants?	☐	☒
12.	Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?	☐	☒
13.	Will deception of participants be necessary during the study?	☐	☒
14.	Will the study involve NHS patients, staff, carers or premises?	☐	☒

If you have answered 'no' to all the above questions, send the completed form to your Module Leader and keep the original in case you need to submit it with your work.

If you have answered 'yes' to any of the above questions, you should complete the Form E2 available at <http://www.brookes.ac.uk/Research/Research-ethics/Ethics-review-forms/> and, together with this E1 Form, email it to the Faculty Research Ethics Officer, whose name can be found at <http://www.brookes.ac.uk/Research/Research-ethics/Research-ethics-officers/>

If you answered 'yes' to any of questions 1-13 and 'yes' to question 14, an application must be submitted to the appropriate NHS research ethics committee.

Signed:

Catharine Brown

Principal Investigator
/Supervisor

Signed:

Student Investigator

Date: 30.05.17

Urban Land Use

Appendix 5: Private Sector Development Activities

Type of Development	Details
<i>Land Development</i>	This is the process of developers purchasing certain portion of land and changing that landform for purpose of housing, and then selling those plots. Purchases have to build their own house.
<i>Group Housing</i>	This is the process when develops purchase land, build housing units, with minimal infrastructure in most cases, and sell the land by the name of the blocks
<i>Joint Housing (Apartment)</i>	This is the process by which Developer buy the land build a building of two or more floods, develop join building infrastructure and common facilities and then sell each housing unit individually.

(JICA, et al. 2012).

Appendix 6: Public Sector Development Activities

The Table below provides further details.

Program	Details
<i>Sites and Services Development</i>	This form of land development evolved in the 1960s and 1970s due to the growing need of affordable housing with Kathmandu Valley. The development concept is primarily construction affordable low-cost housing on serviced land.
<i>Guided Land Development (GLD)</i>	The GLD program involve providing access or service roads through parcels of privately owned land in the Kathmandu Valley. By the late 1990s, more than 320km out of 475km road access was opened up under this program. While the KVDA (previously KVTDC) had exerted great efforts to prioritise this program it is a symptomatic treatment of an existing situation and from an urban planning perspective just band aids the problem.
<i>Land Pooling</i>	Land pooling has encouraged partnerships between landowners, private sector and the national and local governments in the development of urban land. In these projects undeveloped/underdeveloped privately owned land is pooled together and readjusted through a land use planning study and provisions of infrastructure, public utilities and services are provided by KVDA through readjusted plots. By the late 1990s, more than 14,000 developed plots were developed within Kathmandy Valley under this program, benefitting more than 10,000 people.

(For more information see: JICA, et al. 2012; MoUD, 2016; Shambhu K. C., 2010; ADB, 2010; ICIMOD/UNEP/MoEST, 2007)

Urban Development

Appendix 7: A breakdown of all major Nepal regulations and policies reviewed for this research.

To note: The legislation in Appendix 8 and Appendix 9 have not been included in this table to ensure no double-ups.

Please see **Appendix 8** for a detailed assessment of Urban Development legislation and **Appendix 9** for detailed assessment of all Disaster Management Legislation.

Year	Law	Category	Still active?	Details from Text
Early 1960s	Master plans and physical development plans for the Kathmandu Valley			Marked the beginning of conventional practice of urban planning in the country.
1969	Kathmandu Valley Regional Plan 1969		Dissolved	Never properly enforced. Document is in Nepalese, unable to obtain a copy.
1970-1975	The Fourth Plan	Legislation	Replaced	Envisaged the development of regional centres and other nodal points of road networks as urban centres.
1976	Kathmandu Development Plan			This plan brought in zonings of 3 categories: Zone A city core (Kathmandu and Lalitpur), Zone B city fringe and Zone C planned settlements in rural villages of the region (Thapa et al., 2008) which led to the development of the Ring Road.
1975-1980	The Fifth Five Year Plan	Legislation	Replaced	This Plan detailed that there needed to be a separate policy for urban development in Nepal.
1977	The Land Acquisition Act	Legislation-Planning		The Land Acquisition Act, 1977 empowered the government to acquire, through public notification, private land for public welfare, protection of public property or coping with emergency situation, among others
Late 1970s	Site-and-services schemes	Projects	Dissolved	Introduced in late 1970s to improve infrastructure and prevent urban sprawl. They were much wider scope than the GLD projects (in 1988) but they become very unpopular as original land-owners would get displaced from their places and the cost of land acquisition made it difficult also.
1980s	Land Pooling Projects	Projects	Still in place	So far (2015) 12 land pooling projects have been implemented within Kathmandu Valley (260 hectares, 12,160 housing plots) another 10,000 housing plots coming up through 11 projects.
1985-1990	The Seventh Plan	National Plan	Replaced	Put urbanisation and housing policy in the national development agenda for the first time. This plan envisaged the concept of rural-urban linkage as a strategy to achieve the objectives of urban-based rural development policy
1986	Basic Housing Plan	Plan	Not enforced	Recognised housing as the basic need of people.
1988	Guided Land development Projects (GLD)	Projects	Still happening	In response to early signs of haphazard urban development these projects were initiated to improve existing infrastructure by reorganising road network in and around the project areas. GLD project contributed in preventing urban sprawl by ensuring a better use of the land and by ensuring a better use of the land and by encouraging infill development.
1992-1997	The Eighth Plan	National Plan	Replaced	Highlights the need for making urban and rural areas complementary of each other Carried on the rural-urban linkage policies and objectives that existed in the Seventh plan. Called for establishment of national urban system and market-oriented urban sub-systems.
1974	The Public Roads Act	Legislation-Transport	Still enforced	The Public Roads Act, 1974 made provisions to acquire lands required for the development, maintenance, expansion, or improvement of the public roads, and to collect development tax from beneficiary landholders in order to maintain convenience and economic interest of the general public.
1988	The Town Development Act	Legislation-Planning	Still in power	The Town Development Act, 1988 defined provisions for land development to increase supply of planned urban plots. Town Development Committees (TDCs) are formed under this Act to facilitate planned urbanization by channelling policy- and technical support from the government to the local level. The TDC Act, 1988 replaced the TDC Act, 1963 and Town Planning Implementation Committee Act, 1973. The Town Development Act and Regulations gives Municipalities authority to: • Prepare and enforce land use plans and building by-laws • Demolish any non-permitted works • Freeze land transfers for period of up to 2 years • Undertake guided land development, land pooling and sites and services programs with the consent of 51% of property owners.
1988	Kathmandu Valley Development Authority Act	Legislation-Development	Still in power	Kathmandu Valley Development Authority (KVDA) Act, 1988 led to the formation of KVDA in 2012 as an umbrella planning body for Kathmandu Valley
1994	National Building Code	Legislation-Building		The Nepal National Building Code (NBC), 1994 was prepared to improve building design and construction methods, learning lessons from destruction caused by the 1988 earthquake This legislation was brought in to mitigate the effects of earthquakes in urban areas. It has originally 20 volumes with 3 volumes added in 2003. The Nepal NBC was first drafted in 1994 following the lessons learned from the 1988 M6.8 earthquake which killed more than 700 people in eastern Nepal. Approved by the government in 2003, the NBC is a legally binding document in all municipalities since 2005. Nonetheless, because of the lack of resources, only a handful of municipalities have implemented the NBC. However, considerations for safety against earthquake have received wide attention after the devastating earthquakes of April/May 2015. The number of municipalities implementing the NBC and building Bye-laws is increasing.
1996	The National Action Plan on Disaster Management in Nepal	Disaster Management	Still in Power	Has four key plans: Disaster preparedness, disaster response, disaster reconstruction and rehabilitation, and disaster mitigation.
1997	The Town Development Fund Act	Legislation-Development projects		The Town Development Fund (TDF) Act, 1997 led to the establishment and operation of TDF. TDF provides economic and technical support to municipalities to undertake development projects.

Year	Law	Category	Still active?	Details from Text
1997	United Nations Development Programme launched Rural Urban Partnership Programme			Objective was to improve people's livelihoods. (mostly rural focus).
1997	Ownership of Joint Housing Act	Legislation-Housing	Current	The Ownership of Joint Housing Act, 1997 put forward the concept of group housing to promote high density in urban areas
1998	The Building Act	Legislation-Building	Current	The Building Act, 1998 made provisions for the regulation of building construction works in order to Third United Nations Conference on Housing and Sustainable Urban Development (Habitat III) – Nepal National Report 48 protect building against earthquake, fire, and other natural calamities. This act tries to promote and enforce safer building practice in Nepal with four types of buildings: i) International state of the art, ii) professionally engineered buildings; iii) mandatory rule of thumbs and iv) rural buildings. The Nepal National Building Code was developed along these lines. The Building Act 1999 was designed and enforced a year after the development of the NBC and provides the NBC with the legal framework required to mandate the building construction requirements (Parajuli et al., 2000; MoUD, 2015).
1999	Local Self-Governance Act	Part 2 and Part 4		20% reservation for women in ward-level election. This act also gives emphasis to interrelationships among development, environment, and disaster explicitly and implicitly. defined the role of local bodies for effective and efficient provision of service and facilities to people. The Act also empowered local bodies to collect revenue, and to undertake planning and execution of local development schemes. This legal framework is designed to try and provide for more citizen engagement in decision making. It is under this Act that local bodies have been developed: Village Development Committee (VDC), District Development Committee (DDC) and Municipality. These bodies are responsible for development planning and budgeting. They are to build local infrastructure and provide basic services. They are also important for mobilizing and coordinating local development partners. It is within these local bodies that implementation of land use policy, enforcing and monitoring building codes and ensuring the construction of disaster resilient infrastructure at a local level. Department of Urban Development and Building Construction is recognised as the central institution for implementation of the building code and monitoring of local bodies
1997-2002	Ninth Plan	Strategy	Dissolved	Gave strength to disaster management means
2000	Build-Operate-Transfer Policy	Policy- Building		The Build-Operate-Transfer (BOT) Policy, 2000 initiated the concept of public-private partnership for urban infrastructure development.
2001	National Transport Policy			Integrated land use planning (planning and transport)
2002-2007	Tenth Plan	Strategy	Dissolved	Gave special attention to disaster management while developing infrastructure and making construction and development projects sustainable. It designated the Ministry of Home Affairs as the national focal point for disaster management.
2007	National Urban Policy	Policy	Current	Initiatives in it that manage rapid urbanisation of major towns. The NUP has put forward the concept of sustainable urbanization through, among others, natural disaster management. In order to achieve one of its three objectives – healthy, secure and economically vibrant urban environment – the NUP has adopted a strategy to establish a system of development and implementation of 'disaster management plan' by the local bodies to minimize loss of lives and properties due to the probable natural disasters. Aims for (i) settlement and economic activities in stressing balanced national urban structure, (ii) development of safe and prosperous settlement areas by increasing the resilience against environmental shocks and stresses, and (iii) effective urban management through capacity development of local bodies. The 2012 National Land-use Policy has classified different categories of land, with a focus on optimal utilization and enforcement of land use control. It encourages relocation of settlements from hazard- prone areas and settlement development in safer locations that are hazard free. However, its implementation is being strengthened with a three-tier participatory land use planning exercise with UN- Habitat support.
2007-2010	Three Year Interim Plan			Recognised disaster as one of the major impediments to national development
2009	National Strategy for Disaster Risk Management	Disaster Management		The NSDRM prepared by the Government of Nepal, follows a paradigm shift from merely responding to post disaster situations to disaster prevention through development. Its main vision is to have disaster resilient communities in Nepal. To do this, the long-term strategies include: <i>Development and restructuring of institutional structures; Strengthen policy-wide and legal arrangements to ensure stakeholders' participation while adhering to integrated policy and decentralized implementation process. Create enabling environment from the central to household level within the State to prepare and implement disaster risk reduction and preparedness plans. Ensure mainstreaming disaster reduction into overall development process along with sectoral development and poverty reduction plans.</i>
2009	National Adaptation Program of Action (NAPA) for Climate Change			A Program of Action by the GoN to address direct impacts of climate change such as those related to disastrous floods and reduced freshwater supplies. Indirect impacts of climate change could be experienced due to extreme events that may increase food prices and /or damage livelihood, assets of the vulnerable communities.
2009/2010	Janata Awas Karyakram (People's Housing Program)	Project (not. Leg).	Only 1 year project.	The Government of Nepal initiated the Janata Awas Karyakram (People's Housing Program) in FY 2009/10 (see Case Study 1). DUDBC also initiated to construct multistorey housing as a part of resettlement program for the people living in informal settlements. The program aims to provide planned housing by relocating families living along the river bank as an effort towards the improvement of the unplanned settlements.
2010	National Strategy for Disaster Risk Management			This plan has given the importance to the disaster risk management and sets the long term vision for developing the capacity of the country for coping with any type of natural and human-induced disasters. It has clearly mentioned in the policy and actions under the section 6.3 (Disaster Risk Management), that the preparation of risk sensitive land use plan and following the building code will be made compulsory in urban and urbanizing areas. It has also mentioned about the minimization of the impacts of climate change by protecting environment and availing opportunities; increasing the access of people in water induced disaster prevention services; developing safe, low cost and environment friendly housings; developing appropriate settlements and cities from the environmental and social perspective

Year	Law	Category	Still active?	Details from Text
Sept. 2010	National Adaptation Programme of Action	Resilience		Designed to address immediate and urgent adaptation needs.
2011	Nepal's Climate Change Policy	Resilience	Current	Aims to improve people's livelihoods through climate change impact mitigation and adaptation activities and obliges to channel over 80% of total climate finance to grassroots level activities.
2011	Local Disaster Risk Management Planning Guideline	Disaster Management	Current	Developed by MoFALD
2012	The National Shelter Policy	Policy- Shelter	Current	The National Shelter Policy, 1996 was revised in 2012 to address the new challenges of slums, rental housing and housing for internally displaced and economically weaker section. The National Shelter Policy, 2012 has projected that around 1,364,000 housing units will be required in urban areas by 2023 with around 900,000 new constructions. stresses the right to safe and adequate housing to all
2013	Gender equality and social inclusion (GESI) Operational Guidelines	Social	Current	Developed by Min of Urban Development to provide increase access of women, poor and excluded communities in urban development and its resources and benefits. Deals with water, sanitation and hygiene (WASH) and housing, urban development and building (HUB) sectors.
2015	New Constitution	Constitution	Current	Development of federal states coming into three tiers of government- central, provincial and local. Give more power to local governments. Have guaranteed every person the right to live in a clean environment as a fundamental right. Under the local level, there will be Village Council, Municipal Council and District Assembly. The local governments will be able to make their own laws without being inconsistent to federal or provincial laws. The country is now in the process of implementing the new constitution.
2015	Guideline for Settlement Development, Urban Planning and Building Construction 2015	Reconstruction Phase post Disaster	Current	<i>The policy regulates open space for buildings, right of way, and confirms the suitability of the land for settlements development (Sangraula, 2015). The objective of the policy is to regulate the 28 uncontrolled and risky urban housing on unsystematically plotted land to promote earthquake resistant buildings (MoUD, 2015).</i> Through the Guideline buildings have been categorised as <u>A,B</u> , C or D according to the size, purpose and location. provisioned to be done by a civil engineer with at least five years long experience but the design of 'C' and 'D' category buildings can be prepared by a civil Sub-Engineer.
2015	Design CatLog for Earthquake Resistance Building Volume-I 2015	Reconstruction Phase post Disaster	Current	Prior to the earthquake very few municipalities were actually following the National Building Codes (only 191 out of 16). Therefore following the disaster the Nepal Government designed the Resistance Building Volume-1 which contains 16 prototypes and four flexible earthquake resistant building designs. The policy aims to enhance the environment, social acceptable and economic viable development while also considering climate change concerns and sustainable resource management. The Catalog is not a 'legislation' per say and is an example of an owner-driven reconcentration approach. It does try to promote 'building back better' for long-term sustainability of the community.
2015	Guideline for Distributing the Subsidy to Reconstruct the Damaged Residence due to Earthquake	Reconstruction Phase post Disaster	Current	This is around the control of loans to affected families for reconstruction of earthquake resistant housing. he Ministry or Urban Development delegates the authority to local bodies so that Village Development Committees and municipalities disburse the amount to the affected families. Originally it was around 2000 USD however I believe this has been increased to 6000 USD. There are different rules around getting the loan like you can't be receiving financial support from anywhere else and you only get one loan for one house. Another rule is that you can get an extra 3000 USD loan with 2% interest rate pay back on top of the original loan. It does not however help people who have already built back their houses at their own cost. Another issue is that poor families will not be able to get the bank loan as they lack the collateral capacity.
2015	Post-Earthquake Recovery and Reconstruction Policy 2015	Reconstruction Phase post Disaster	Current	Under this policy the National Reconstruction Authority is established (under the chairmanship of the Prime Minister of Nepal). The primary goal of the policy is to enforce safer residences and livelihoods through earthquake reconstruction development projects. It might be important to highlight that the role of the National Reconstruction Authority is still uncertain due to political tensions from the opposition political party in power currently. The Authority aims to dictate the development resilience building by integrating social, economic and physical developments. This is through policy development, allocating budgets, approving projects and plans and prioritize tasks.
2015	National Urban Development Strategy (draft)	Reconstruction Phase post Disaster	Current	Initiatives in it that manage rapid urbanisation of major towns. Document aimed to guide and regulate urbanisation process in Nepal, deconcentrate economic activities outside Kathmandu Valley, redefine conceptual issues on regional planning and development, achieve balanced regional and sub-regional development, and create off-farm employment opportunities and serve hinterland. Seeks to promote resilience in urban development including climate change adaptation, development in safer locations, the review and enforcement of building codes, regulations, guidelines and planning bye-laws, and capacity building of the government institutions and local bodies.
2015	Fundamental Construction Bye-Laws on Settlement Development	Reconstruction Phase post Disaster	Current	Formulated following the earthquake to promote seismic risk-resilient urban development.
2015	Urban Planning and Building Construction	Reconstruction Phase post Disaster	Current	Formulated following the earthquake to promote seismic risk-resilient urban development.
2016	Reconstruction and Rehabilitation Policy	Policy	Current	Provides policy instruments for steering reconstruction and rehabilitation.
2017	National Urban Development Strategy	Strategy	Current	Seeks to promote resilience in urban development including climate change adaptation, development in safer locations, the review and enforcement of building codes, regulations, guidelines and planning bye-laws, and capacity building of the government institutions and local bodies.

Appendix 8: Review of Physical Planning and Policies relevant to urban development in Kathmandu Valley

Year	Document	Review
2002	<u>Development Plan 2020 for Kathmandu Valley</u>	- Also called Kathmandu Valley Long-Term (2002-2020) Development Concept. - Prepared by the MOPPW (now MOUD) and KVTDC (now KVDA) in 2002 (as they were then). - The plan lacks necessary implementation and financing plans and therefore has been limited in its capacity to plan development with Kathmandu Valley (World Bank, 2013; ICIMOD/UNEP/MoEST, 2007). - The Plan in implementation as it lacked necessary implementation and financing plans. It was also not formally adopted by the Cabinet and published in the Nepali official register. Lack of enforcement and monitoring power and political instability are the key factors that impeded implementation of the plan (World Bank, 2013; ICIMOD/UNEP/MoEST, 2007).
2007	<u>2007: National Urban Policy</u>	- Updated June 2007 and prepared by MOPPW and DUDBC- main aim is to address the considerable challenges apparent in the urban sector in an integrated and well-coordinate manner. - NUP with its three main objectives; to develop a balanced national structure, to develop a clean and safe urban environment and to make an urban management effective, sets out a number of different strategies and supporting policies for achieving this (reference the NUP). There are strategies that are directly relevant to the KV urban sectors. These include (summarise) - The NUP highlights the need for all local bodies to develop a Disaster Risk Management Plan and also inter-alia community mobilisations for local bodies' implementation (MoUD, 2016; UNDP, 2012). - As the primary document for urban development in Nepal, it has experienced significant problems with implementation and financing (MoUD, 2015). Which has resulted in plans and strategies attached to this document to also lack implementation and financing.
2007	<u>KVBB Land Use Plan</u>	- This Plan sits under the Kathmandu Valley By-Laws (2007) which is detailed below. - The prevailing land use plan was originally planned in 1976 with only minor updates and amendments in early 1990s (JICA, et al. 2012; ICIMOD/UNEP/MoEST, 2007). The document is quite outdated and also combined with weak development control doesn't reflect the development on the ground (JICA, et al. 2012). The situation on the ground is more haphazard, low

		density, mixed use pattern of land use and mostly undertaken by private individual land and building owners (JICA, et al. 2012; ICIMOD/UNEP/MoEST, 2007). - Before this plan was designated in the KVBB there had been no land use regulations outside of the Ring Road (JICA, et al. 2012). This impacted negatively with no mitigations around forest or agricultural land being changed into residential and other urban land uses (JICA, et al. 2012). Evidence of this is agricultural land declined from 64% in 1984 to 52% in 1994 and down to less than 42% by 2000 and is evidently to have continued downwards to date (JICA, et al. 2012). - This base map is for all municipalities of Kathmandu valley and purpose is to be the baseline for all land use plan, zonings, development control guidelines and spatial dimension of all component (including transportation) to try and provide an integrated sustainable development through Kathmandu Valley (Shambhu, K. C. 2010). - Worth noting that although KVBB has a land use zoning map, Kathmandu Metropolitan City have different land use zoning and land use plans and their own municipal budge to review those plans (JICA, et al. 2012). They also have their own regulations of their own. This was a major issue due to overlapping of plans and no coordination until the formal adoption of the KVDA Regulations in March 2012 which empowered the KVDA to assume its mandate and responsibilities (World Bank, 2013). - The Land use map does include hazardous areas or no build zones in landslide and flood prone areas.
2008:	KMC Risk Sensitive Land Use Plan (2008-2012)	- This document is still in draft form and not been adopted. - To implement valley RSLUP, it needs a better institutional arrangement in horizontal as well as vertical level (KVDA, 2015).

Appendix 9: National Building Code Volumes

Code Number (and year introduced)	Code Title
NBC 000: 1994	Requirements for State-of-the-Art Design: An Introduction
NBC 101: 1994	Materials Specifications
NBC 102: 1994	Unit Weight of Materials
NBC 103: 1994	Occupancy Load
NBC 104: 1994	Wind Load
NBC 105: 1994	Seismic Design of Buildings in Nepal
NBC 106: 1994	Snow Load
NBC 107: 1994	Provisional Recommendation on Fire Safety
NBC 108: 1994	Site Consideration for Seismic Hazards
NBC 109: 1994	Masonry: Unreinforced
NBC 110: 1994	Plain and Reinforced Concrete
NBC 111: 1994	Steel
NBC 112: 1994	Timber
NBC 113: 1994	Aluminium
NBC 114: 1994	Construction Safety
NBC 201: 1994	Mandatory Rules of Thumb: Reinforced Concrete Buildings With Masonry Infill
NBC 202: 1994	Mandatory Rules of Thumb: Load Bearing Masonry
NBC 203: 1994	Guidelines for Earthquake Resistant Building Construction: Low Strength Masonry
NBC 204: 1994	Guidelines for Earthquake Resistant Building Construction: Earthen Building (EB)
NBC 205: 1994	Mandatory Rules of Thumb: Reinforced Concrete Buildings Without Masonry Infill
NBC 206: 2003	Architectural Design Requirements
NBC 207: 2003	Electrical Design Requirements for (Public Buildings)
NBC 208: 2003	Sanitary and Plumbing Design Requirements

(Source: NBC; UNCRD, 2008; MPPW, 2009)

Appendix 10: Building Approval Levels of Application

Four levels of application

International State-of-Art ○ The major thrust of the code is aimed at typical and most common currently erected in Nepal. However, you can produce more sophisticated designs and techniques from development countries then what appears in the code- provided they still meet minimum standards set for Nepal.
Professional-Engineered Structures ○ These are standard code requirements that all professionally qualified engineers will follow when designing structures in Nepal. Covers all structures (including hospitals, meetings halls, factories, etc.)
Rules of Thumb (Pre-engineered for defined building types): ○ For classes of buildings constructed of modern materials (concrete, steel) but not exceeding certain simple criteria as to height, configuration, number of storeys and flood area, mandatory rules-of-thumb are provided. ○ The requirements are in terms of max spans, min reinforcing, positions of earthquake-resisting elements etc. ○ The documents are such that experienced overseer or mason should be able to understand them and present sufficient details at the time of permit application to prove to a non-technical appraiser at the local authority that the requirements are being met.
Advisory Guidelines (Non-engineered construction employing traditional materials and skills): ○ 2 guidelines developed through study of 50 prevailing building types in Nepal. One deals with low-strength masonry and another with earthen structures. ○ These are for remote buildings.

Source: (copied from Parajuli et al., 2000; UNCRD, 2008; KVDA, 2015b)

Appendix 11: Analysis of Nepal's Plans and Implementation Tools for Urban Development

Nepal's' Plans and Implementation Tools	Relevant Policies, Plans and Laws.	Their purpose (Application of Plans and Tools to Mitigation)	What's missing
<u>Planning</u> Comprehensive or land use plan. Hazard component of comprehensive plan Recovery/reconstruction plan	2007: KVBB Land Use Plan	Identify hazardous areas and adopt mitigation policies that guide development and redevelopment in hazardous areas	The Land Use Plan does not include hazardous land (flooding, landslide or earthquake). And some of the 'Urban Expansion' areas are within flooding areas (according to the flood mapping done by JICA, 2013). This is still a major gap in the link between reducing risks and urban planning within Kathmandu Valley there is no current guide on development in hazardous areas (before 2015). Earthquake Vulnerability Maps have only been prepared for 5 Municipalities out of 58 (Kathmandu/Bhaktapur, Dhanusa, Siraha and Mahottari) by the Earthquake Risk Reduction and Recovery Preparedness Program for Nepal (Humanitarian Response, Unknown; Kumar and Pradha, 2015; UNDP, 2008). Risk Hazard Maps have only been prepared for 4 Municipalities since 2014 (Kumar and Pradha, 2015). Neither of these maps have been integrated into planning legislation or land use plans.
<u>Development regulations</u> Zoning ordinance Subdivision ordinance Fault setback ordinance	(DRAFT) KV risk-sensitive Land Use Plan (DRAFT) 2010: Kathmandu City: Risk Sensitive Land Use Plan (draft) Kathmandu Valley Bylaws (2007) Each Municipality has requirements around land use construction, setbacks and height restrictions.	Control the type, location and density of development in hazardous areas	Until the KV Risk Sensitive Land Use Plan is approved, there is currently no zonings mitigating from disasters that cover all of Kathmandu Valley. There is currently no penalties or fines for violators who don't comply with zoning ordinances. There is no 'fault' setbacks, or 'no build' zones or 'high risk' zones within Kathmandu Valley. The Municipality Building By-Laws do not have subdivision codes or any fault setbacks from hazardous areas. The KVBB come into effect in 2007, however prior the development prior to this was split across 5 other Municipalities which were contradicting and not in alignment. In most cases zoning not even a part of their Building-Codes.

			The impact of the KVBB is still hard to determine as it was only enforced for 7 years before the earthquake. Within the KVBB there are nine different categories and three sub-categories of land use but there are no provisions for governing change of approved use (JICA, et al. 2012). KVBB does define set-back distances from the road centre for new buildings but repairs and renovations are exempt from these provisions which causes issues on its implementation and purpose.
<u>Building Standards</u> Building code Special hazard resistance building standards Retrofit standards for existing buildings	Town Development Act (1988 & Regulations (2004) Building Act (1999) National Building Code (1996) Kathmandu Valley Building by-Laws (2007) Before 2007:	Strengthen existing development and require new development to withstand hazards	The NBC and Kathmandu Valley By-Laws do not include any measures around retrofitting. This is a big issue with thousands of existing housing units in Kathmandu built in the past without and code inputs or standards and therefore left vulnerable to various forces (MoUD, 2015). There are also lots of retrofitting that takes place as buildings change their use such as schools, hospitals, government offices, malls, etc. which are always filled with large numbers of populations and if collapse cause large casualties and economic losses (HRRP, 2016). The Building Act 1999 does not specify a particular organization that is authorized to follow up and monitor the compliance to NBC by the Municipalities. Therefore, the compliance part falls through the cracks, as seen in many cases across Kathmandu Valley (MPPW, 2009). Therefore, retrofitting codes and guidelines need to be firstly developed and mainstreamed but also need to include hazard risk techniques and criteria. Compliance enforcement is ineffective due to the high workload and limited number of trained professionals. There is also the tracking system of building permit processing is also inefficient. There are missing hazard resistant building standards. E.g. flooding, The NBC 000: 1994 State-Of-The Art Design and NBC 105: 1994 Seismic Design of Buildings focus on seismic design to reduce impact of earthquakes. The issues for the Nepalese with these two codes is Municipalities' lack of enforcement

			capacities within their planning processes, with Lalitpur being the first Municipality within Kathmandu Valley to enforce the NBC but not until 2003. The Town Development Act and Regulations gives Municipalities authority to: • Prepare and enforce land use plans and building by-laws • Demolish any non-permitted works • Freeze land transfers for period of up to 2 years • Undertake guided land development, land pooling and sites and services programs with the consent of 51% of property owners.
<u>Land and property acquisition</u> Transfer of development potential from one site to another Acquisition of undevelopable lands Acquisition of development rights building relocation Acquisition of damaged buildings	<i>Land Acquisition Act, 2034 (1977)</i> Gives power to Nepal that if Nepal so deems necessary, acquire any land at any place for any public purpose. (GoN, 1977)	Remove existing development or prevent future development in hazardous areas	The Act only allows acquisition of land if for the following reasons: - <i>To construct residential quarters for the staff, workers or labourers of the institution, or otherwise provide for their welfare, or undertake any function in the interest of the general public.</i> - <i>To operate a project connected with an institution fully owned by Government of Nepal</i> These two issues are incredibly vague and don't give rights to GoN for relocation of buildings in undevelopable lands or acquisition of buildings that are damaged and not in a habitable state. Lack of legislation to move buildings in risk areas is a major issue for Kathmandu Valley which over the years has had major construction in hazardous areas due to lack of planning and implementation of building codes and by-laws.
<u>Policies for critical public facilities</u> Local requirements for critical facilities (hospitals, schools) Location of capital facilities		Direct new development away from Hazardous areas	

(streets, water) in less hazardous areas			
<u>Taxation and fiscal policies</u> Impact tax to cover additional public costs of building in hazardous areas Reduces or below-market taxation for open space or non- intensive uses in hazardous areas	<i>Local Self- Governance Act 1999</i> , gives Municipalities and VDCs the authority to apply taxes.	Maintain low density in hazardous areas	Although, the LGSA gives local governments authority to apply taxes, they have been found to have limited capacity to do so (especially prior to 2015). Therefore, they do not have much ability to get taxes to cover public costs in building in hazardous areas. There is also no additional fee for developers in building in these areas or within urban fringe areas where providing infrastructure is more expensive for the local governments.
<u>Information dissemination</u> Public information program Hazard disclosure requirements		Informs the public and those involved in real estate transactions about hazards	Dissemination of disaster's preparedness and response messages to communities are done through the Disaster Management initiatives of Nepal (these are detailed in the next section). There is found to be no dissemination requirements that informs public or those involved in real estate about issues with building in hazard risk areas or constructing building that reduce risks to hazards (until after the 2015 earthquake).

Source: The above table was developed building off Philip Berke, Timothy Beatley and Richard Eisner's growth management tools for natural hazard mitigation (Berke, P. et. al., 1992) and off 'land use management tool' to guide development in hazard-prone areas from Olshansky and Kartez, 2998: 170-174).

Details in Table: Source: WB, 2013; ADB, 2010.

Appendix 12: Disaster Management Regulations and Policies in Nepal

Disasters

The biggest historically measured earthquake in Nepal was in 1934 with a measured 8.4 on the Richter scale, followed by the April 2015 earthquake with a measured 7.8 on the Richter scale (Kumar and Pradha, 2015; MoUD, 2015). Both resulted in widespread destruction and large casualty numbers (MoUD, 2015). However, *The Nepal Disaster Report 2013*, with the exception of epidemics in 2009, flood and landslides have caused the most casualties since 2000 (Nepal Disaster Report, 2013). This is especially prevalent in urban areas like Kathmandu Valley (Kumar and Pradha, 2015).

Agency

The Ministry of Home Affairs (MoHA) is the national agency for disaster management, with the Department of Narcotics, Drug Control and Disaster Management (under the MHA) formulate and implement the national policy for disaster mitigation and preparedness (MoUD, 2015). 'All activities in disaster management are guided and directed by the Central Disaster Relief Committee chaired by the Home Minister in accordance with the *Disaster (Relief) Act 1982*', which also coordinates emergency and preparedness activities (MoUD, 2015). The responsibilities linked to action and implementation is however passed down to the Local Natural Disaster Management Relief Committee (Municipalities) through the Regional and District Level Relief Committees. This whole structure is complex, time consuming and only focused on post disaster scenarios of emergency rescue and relief (Kumar and Pradha, 2015). *The Natural Calamity Relief Act 1982* which is the most prevalent legislation for disaster management in Nepal, is outdated, focuses on responses and not on how to mitigate disasters (Kumar and Pradha, 2015).

Legislation

While the *National Urban Policy 2007* requires Disaster Risk Management (DRM) Plans to be prepared by local bodies, the *Local Self-Governance Act 1998* is the legislation that mandates local bodies must implement DRM activities (Kumar and Pradha, 2015). These two regulations sit under different Ministries who don't have good coordination or relationships (MoUD, 2017; WB, 2013). In 2009, Periodic Plans started incorporating disaster management components for municipalities to address (Kumar and Pradha, 2015). Even though the DRM plans are incorporated into Periodic Plans of Municipalities, the internalisation of DRM activities in Municipality planning is quite slow and lacks a much needed support system (Kumar and Pradha, 2015). This was quite evident before KVDA was developed however was not addressed completely prior to the earthquake.

Earthquake Vulnerability Maps have only been prepared for 5 Municipalities out of 58 (Kathmandu/Bhaktapur, Dhanusa, Siraha and Mahottari) (Humanitarian Response, n.d.; Kumar and

Pradha, 2015; UNDP, 2008). Risk Hazard Maps have only been prepared for 4 Municipalities since 2014 (Kumar and Pradha, 2015). When the maps and plans have been prepared, they are not utilised for reducing vulnerabilities and not considered at all in the process of construction (Kumar and Pradha, 2015). The lack of disaster mitigation efforts combined with lack of coordination with the built environment left Nepal in a very vulnerable position. Maps are developed by international organisations or donors in collaboration with Nepal though local bodies have no ownerships over them.

Year	Regulation or Policy or Strategy	Description	Analysis
1982	National Calamity Relief Act 1982	Makes arrangements for the operation of relief and the maintenance of people and public property. Gives power for GoN to declare disaster areas and direct different agencies for relief work. Gives mandates to MoHA is the implementing agency and leading agency for disaster rescue, relief and preparedness activities. Committees were established central level through institutional arrangements down to local level. All focused on the emergency phase.	This act primarily focuses on response and relief following a disaster and provides no foundation for preparedness or adaptation. MoHA are regulated as being entirely responsible for disaster management.
1982	Disaster Relief Act 1982	The Act focuses on coordination during the Relief of disasters.	This Act is not relevant in disaster management or preparedness and only focuses on emergency phase of the disaster.
1999	Local Self-Governance Act (1999)	The act has provisioned for disaster related management works under the domain of the VDCs, Municipalities, and DDCs under Clause 28 (k), (4), 96 (j) (7) and 189 (1) (p) (2). Ex. <i>To carry out or cause to be carried</i>	The act requires local authorities to play the role of disaster preparedness. This

		<i>out necessary works in respect of controlling natural calamities or to carry out necessary functions on the controlling of natural calamities or to carry out necessary acts in respect of controlling natural calamities.</i>	
2002	Kathmandu Valley Earthquake Mitigation Study GoN, JICA	This study Produced and Earthquake Risk Assessment which provided recommendations on risk reduction. This included possible temporary or permanent spaces for evacuation and settlement.	This assessment never translated into mapping or planning legislation for Kathmandu Valley.
2005- 2006	Disaster Risk Management Master Plan (development)	KMC, EMI and NSET developed a disaster risk management master plan for Kathmandu city.	This plan however was never given legislative weight in parliament and therefore didn't result in limiting development in areas of risk or changing any by-laws to mitigate risk.
2009 – 2010	Kathmandu Metropolitan City Risk Sensitive Land Use Plan (KMC- RSLUP) (Source: UNDP, 2012)	Designed to guide the future development of Kathmandu City. Integrates hazard, vulnerability and disaster risk assessment into local land use planning by using hazard and risk information and including emergency management parameters (evacuation risks). Development of Zoning Ordinance, Socio Economic Profile and Emergency Management Plan were done in accordance with this plan. A three-year plan was proposed on how Kathmandu Valley plans and local land use plans in	This plan is still not yet adopted. No record found on what stage it is at.

		the Valley can be made risk sensitive.	
2009	NAPA- National Strategy Action Plan for Climate Change (Source: UNDP, 2012)	GoN developed a Program of Actions on how to address impact climate changes including mitigating floods. It is a strategic tool assesses Nepal's vulnerability to climate change and provides a framework for developing adaptation measures. It looks at risk reduction themes of disaster mitigation, prevention and preparedness. There are 6 themes to which one is Urban Settlements and infrastructure. This includes: improving governance structures of effective and pro-poor, reducing threat through prevention, improving coping capacity of vulnerable communities formulating and implementing climate change adaptation measure and finally providing and enabling conditions to ensure resilient infrastructures. Source: NAPA to Climate Change, Ministry of Environment 2010.	This report looks into Climate Change Adaptation measures but does not bring together the Disaster Risk Reduction agenda well to support the sustainable development of Kathmandu Valley (UNDP, 2012). It also does not provide strategies for incorporating adaptation into building codes, by-laws or land use plans in Kathmandu Valley. It sits silo to those regulations and does not address incorporation issues.
2010	National Strategy for Disaster Risk Management (Source: UNDP, 2012)	Developed by GoN marks the transition from only addressing post disaster situations to disaster preventions through mainstreaming disaster reduction in the development process. Main vision is to have disaster resilient communities. Does highlight that there is a need to ensure mainstreaming disaster reduction into overall development process along	The Plan, though a national framework, does important work in developing programs for national development it does not address inherent issues within construction of communities only on management of a disaster. The main aims are in disaster resilient communities

		with sectoral development and poverty reduction plans. However, no strategy or process is given on how this would happen.	yet strategies are all about instructional structures and stakeholder participation through decentralised implementation process.
2010 - 2013	Three Year National Plan (Source: UNDP, 2012) Periodic Plan	This was the first plan to primarily focus on disaster risk management. It set long term visions for developing the capacity of the country to cope with any natural disaster. 6.3 <i>Disaster Management</i> highlights that a risk sensitive land use plan and following the building code will be made compulsory in urban areas. The plan also touches on developing appropriate settlements and cities from the environmental and social perspective.	The proper policy implementation and the lack of coordination between Ministry of Physical Planning and Works and the Ministry of Local Development resulted in the three-year plan.
2013	<i>Draft Kathmandu Valley Risk Sensitive Land Use Plan</i>		This Plan is still yet to be adopted, it went to cabinet prior to the earthquake in 2015 but was put on hold. It has since been through another round of reviewing and updating and is with the GoN for adoption. As quoted by an interviewee 'if it's not adopted now, I don't see it ever happening. There would be a reason we are not aware of as to why it has been shelved.

(Source (for Description): UNDP, 2012; MoUD, 2015; MoHA, 2013; ASD, 2012; HRRP, 2016; World Bank, 2013)

Urban Governance

Appendix 13: Major Organisations involved in Urban Development in Nepal

The purpose of forming KVDA in 2011 was to have one entity legally able to expand and develop the urban areas in a planned and coordinated way that would address the primary services and facilities required to the people (UNHabitat, 2015). Prior to the establishment of KVDA the procedures of building permissions were split between different authorities from all levels of government (see **Appendix 14**). This resulted in different implementation techniques and massive miscommunications. As seen in the table, the introduction of KVDA improved significantly the urban development of planning processes. It created one body responsible for coordination and managing Kathmandu Valley. This reinforces the principal factor that relative success of resilience is around institutionalisation or embedment in local government structure (Jones et al., 2013; Larson, 2003; Tadele and Manyena, 2009).

It is required under the NBC that buildings get a building permit to build (National Building Code). Prior to the earthquake in 2015, more than 98% of buildings in Nepal were built by owner-builders who follow the advice of local craftsmen (Parajuli et al., 2000). This included areas that had access to engineers. This was a direct result of poor institution and technical capacity within urban local authorities to implement the provisions (Parajuli et al., 2000). There was no system to control professional standards of engineers and designers through reference to professional qualifications and membership or by legal means. It is quite prevalent that permit application documents are 'lost' and manipulated (Parajuli et al., 2000).

Coordination among KVDA and local governments is lacking, the division of functional responsibilities between the KVDA and local governments also needs clarification. Currently KVDA's broad mandate can potentially overlap with those of local bodies in several areas (WB, 2013). Urban management capacities of local governments are limited, and their mandates are under-funded, this is where KVDA sometimes stands in, however these roles are not yet defined.

Political Level	Power	Role	Responsibility for Urban Development
Central	The Constitution, 2015		Knowledge creation, standardisation, knowledge transferring & capacity building, mapping, recording, coordination, enforcement and monitoring of safer building construction.

			(MoUD, 2015) MoHA, MoFALF, MoUD/ DUDBC
National Planning Commission (NPC)		National level policy formulation	
Ministry of Federal Affairs and Local Government (previously Ministry of Local Government)		Monitor the role of municipalities under the local Self Governance Act 1999.	
Ministry of Land Reform and Management		Land Reform Office- give registration of property and completion certificate. Department of Natural Land Use Project and Department Land Reform Management.	
Ministry of Urban Development (Urban development fell under Ministry of Housing and Physical Planning before 2012)		Response for urban infrastructure.	The Ministry of Urban Development (MoUD) is responsible for: urban and regional planning; developing and managing basic urban infrastructure services such as water supply, sanitation, solid waste management; and housing. It has two implementing agencies, the Department of Water Supply and Sewerage (DWSS), and the Department of Urban Development and Building Construction (DUDBC), as well as other organisational entities such as the Town Development Fund. It has oversight over regional planning authorities such as the KVDA and other Town Development Committees (MoUD, 2015: pp. 47-48). The MoUD shares responsibility for the urban sector with other central government agencies such as the National Planning Commission (NPC) and the Ministry of Federal Affairs and Local Development

			(MoFALD) (see figure 6) (MoUD, 2015: pp. 47-48). Prior to the establishment of MoUD the responsibility for Urban Development fell into the hands of MoPPW.
Ministry of Physical Planning and Works Now Ministry of Urban Development	Founded 2000	Development national strategic transport networks Improvements housing and urban environment	
Department of Urban Development and Building Construction (DUDBC)		Supporting municipalities in preparing plans and base maps Preparing building codes (and updating it) Partnering with NGOs in addressing slum and squatter issues Has Planning Division Housing Division Building Division Implement urban development plans and programs in Nepal.	“Custodian” of the code and responsible for its updates and modification. Also expected to check the Municipalities are incorporating the code into their building permit process (KVDA, 2015b).
Ministry of Home Affairs		Disaster management?	
District		Districts are governed by District Councils that oversee the District Development Committees and also adopt resolutions relating to taxes, charges, fees, tariffs, loans or borrowings (UNHabitat, 2015). Safer Building Coordination Subcommittees	Coordination, Capacity/Skill Development, Awareness Raising and Education, Recording and Monitoring (MoUD, 2015). District development committees are involved in the physical development of municipalities (Choe and Pradhan, 2010: p.38)
Kathmandu Valley Town Development Committee	Town Development Act 1988	Under MOPPW Basic objectives are to develop the entire KV in a planned manner. Formulate and implement physical	

		development plans, and to provide people in valley basic services Under Nepal Institution it has more authority than one district. (World Bank, 2013b)	
Local Government	Local Self-Governance Act 1991	The main aim was to decentralise and carry out local level activities with full responsibilities and authority within District, Municipality and VDCs level.	Enforcement of National Building Code and ensuring safe construction within their areas.
Municipality	Municipalities function under the Local Self-Governance Act, 1999 which was originally enacted to promote decentralisation in an attempt to develop local self-governance systems (UNHabitat, 2015).	The Act dictates that Municipalities are responsible for approving budgets, plans and programmes.	Municipalities have many different platforms in which to engage with their people. However, it is complicated and involves different forums and community groups. All of the communication must also go through a Ward chairperson also before getting to their local government (Municipality government).
Kathmandu Valley Development Authority (formally established 13 April 2012) Previously Kathmandu Valley Development Committees The Kathmandu Valley Development Authority Act 1988/89 was also prepared to expedite the development of the valley through an authoritative body which was the KVTDC.	Kathmandu Valley Development Authority Act 2045 Before 2012 under Established through the Town Development act 1988	KVDA Act gives KVDA authority at the level of the District level and therefore takes priority over Municipalities in Kathmandu Valley. Carry out's Urban development-related activities.	Primary mandate to prepare and implement an integrated physical development plan for Kathmandu Valley. The enactment of the KVDA Regulations of 2012 empowered the KVDA to take on the functions identified under the KVDA Act. The regulations, however, do not reaffirm the KVDA's broader authority for metropolitan management and coordination, as stipulated in the KVDA Act. This is an issue currently (World Bank, 2013) Prior to 2012, KVTDC was responsive for development

			of the valley, however they didn't have as much power as KVDA now and therefore didn't implement much. The KVDA Regulations of 2012 refer to Articles 29 and 29.1 of the 1988 Act. However, the Act itself does not give the KVDA power. To give power to KVDA either the formal adoption of the 1988 KVDA Act needs to happen or efforts to harmonize the 1988 KVDA Act with the 2012 KVDA Regulations (World Bank, 2013).
Private Sector			
Real Estate Developers		Development of houses and apartments	
Land Brokers		Provision of services land for construction	People have evidently turned to informal land brokers who operate at local level without license, his is because the amount of formal land market is limited to a few land-pooling project areas and developable housing plots not available to some buyers. This was very common between 1990 – 2000. There is no restriction on land subdivision (except that the minimum size of a plot should be 80 sqm) and therefore land brokers designate themselves that a particular plot of land will be subdivided. Land brokers commonly advertise land subdivision in newspapers that are not the actual on-site situation and vary drastically. People have been often cheater by this process.

Source: KVDA, 2015; ASD, 2012; World Bank, 2013; UNHabitat, 2015; ICIMOD/UNEP/MoEST, 2007; UN Habitat, 2010; JICA, et al. 2012; MoUD, 2015)

Appendix 14: Procedure of Building Permission

Process	Authority before KVDA	KVDA forward	Remarks (Comments)
Planning Permit	KVTDC	KVDA	KVTDC (now KVDA) were in charge of issuing planning permits
Initial Environmental Examination (IEE)	Related ministry (for urban development MOPPW	KVDA	
Environment Impact Assessment (EIA)	Ministry of Environment	Ministry of Population and Environment	Only if required (depends on project size)
Structure safety permit	Division office of DUDBC		Technical requirements are listed under the National Building Code (year).
Building permit	Municipality or VDC	KVDA	As each Municipality had its own process, the development of KVDA brought together one consistent building permit process.
Completion certificate	Municipality or VDC	KVDA	Only required for IEE or EIA and Structure safety permit.
Registration of property	Land Revenue Office	Land Revenue Office	Landowners avoid registering their properties to avoid taxation and therefore don't get safety permits or completion certificates (as only need them if going to register their buildings). This is why it's possible for people to construct and live in without permission and why local governments cannot control all development or construction activities in the valley.

Source: JICA, et al. 2012; UNDP, 2012, World Bank, 2013.

Interviews

Appendix 15: Interview Questions

This is an indicative list of questions that were asked to Interviewees.

Theme	Research Question	Interview Question	Factors
Urban Land Use	How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?	- In the past, has data been collected and produced around growth predictions (health, education, infrastructure, housing demands) to feed into long term strategy for Kathmandu Valley? - How has the lack of statistical data effected the planning legislation, policy and mapping before the earthquake	Geographical & Demographic
		- How are environmentally constrained areas mapped in Kathmandu Valley (if any)? Are those mapped areas considered during the planning process for new development in KV? - How has Kathmandu's development been planned over time?	Change of land use in valley
		- What are the disaster impacts on the land use changes in Kathmandu Valley?	Disaster impacts on urban resilience of Kathmandu
Urban Development	What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?	- Have land use zonings, urban growth boundaries, restricted development areas, or risk mapping been used to control growth especially in disaster risk land within Kathmandu Valley historically?	Unplanned urban development

		- Has there been any attempt to zone areas within Kathmandu Valley as not-urban or disaster risk areas? - How effective is the building permit system within Nepal? Primarily within Kathmandu Valley?	
		- How effective do you think urban growth boundaries would be for controlling urban sprawl and mitigating disasters within Kathmandu Valley? - How has urban expansion and growth within Kathmandu Valley been guided by hazard risk information?	Urban Growth Boundaries
		- From your knowledge, how has development in Kathmandu Valley addressed urban planning legislation?	Urban Planning ability to reduction damages in the 2015 earthquake
		- What work was done for earthquake resilience for Kathmandu Valley prior to the earthquake? Especially in field of development and regulations. Was there any focus on other disasters (e.g. for fires, floods, landslides)? - There are quite a lot of Disaster Management policies (Disaster Relief Act, National Action Plan on Disasters etc.) how have then been linked back to urban	Effectiveness of Disaster Management

		planning and development plans and strategies. Which have priority when it comes to development of homes, schools, roads etc.? - Prior to the earthquake, was there any site selection process that considered geological investigations to build and promote disaster risk planning, especially around deciding where development and infrastructure was built (not just houses, but also roads, hospitals, schools, train tracks, etc.)?	
		- Any extra comments?	The Bigger Picture Thinkers
Urban Governance	What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?	- How has historic governance of Nepal played a role in the development of the Valley?	Historical governance and political structures and their impacts
		- Who are the key government actors relating to urban development and urban resilience for Kathmandu Valley?	Key government actors
		- How involved are the community in the decision making of development within their neighbourhoods or even city wide?	Decision making and implementation

		- How effective have the National Building Code and Building By-Laws been? How much have they been implemented in urban development?	
		- What is situation of monitoring of urban development in Kathmandu Valley?	Monitoring
		- How connected and coordinated are the different sectors? Especially transport, environment, infrastructure, disaster risk management and urban planning? - Historically how have different sectors worked together to plan for disaster risk management, has any of it been through urban development strategies? - Before KVDA and even now, how do Municipalities work together in Kathmandu Valley to plan for growth and implement Building codes through similar planning processes?	Coordination
		- What is the knowledge, financial capacity and resources for local levels to design, plan and implement building codes and contingency plans for building critical infrastructure within Kathmandu Valley?	Resources and capabilities

Lessons for urban planning of KV	What is the role for planners in reducing disaster risks within a city? How can they help make a city resilient? (especially within context of developing countries)	- In your opinion, what has been the biggest issue for urban planning and development in mitigating disasters within Kathmandu Valley?	Lessons Learnt
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Appendix 16: Interview Quotes

Theme Urban Land Use: How urban resilient is Kathmandu and how did this impact on the extent of the disaster of the earthquake in 2015?

Throughout all factors, a key theme arising is the importance of social and cultural norms within the Valley that have impacted the resilience of the area. Some quotes around this are provided below:

Interviewee	Quote
iCC 1	"Nepal has a really strong cultural history as no foreigners were allowed in until 1952. The social pressure is therefore huge. It is complex and at every point it is reiterated at what you place is and what your position should be. Even if it is not explicit it is reiterated at every single point."
GArchP 2	"The Right to Land Act is very strong and overrides many planning regulations (not legislatively but culturally)"

Factor One- Geography and Demographics

Factor One, following two questions (some comments are also provided below):

- In the past, has data been collected and produced around growth predictions (health, education, infrastructure, housing demands) to feed into long term strategy for Kathmandu Valley?
- How has the lack of statistical data effected the planning legislation, policy and mapping before the earthquake?

Interviewee	Quote
iCC 1	"There is no long term strategic thinking"
Arch 2	"Only population data but they are quite out of date, they have enough issues dealing with the demands of today, let alone looking into future demands"
PM 2	"Statistical data on natural disaster events and damage impacts and are lacking, this has led to lack of capacity building around other disasters not just earthquake, as flooding causes just as many deaths in the Valley." – PM (ADRA)
UNP 1	"Statistics and data have always been lacking in Nepal, this is a major issue but won't be addressed as there is no capacity or funding to do so" –

Factor Two- Change of Land Use Patterns in Kathmandu Valley

Factor Two

Some comments from **PM 1** on informal settlements were:

- "Urban settlements has historically gone unnoticed in the cities and therefore have grown and expanded in high risk areas, especially to flooding, and constantly in risk of further damage to the people and the infrastructure in the settlement... the government refuses to plan for them or incorporate them into any of their policies... they are completely ignored."
- "We need to change cultural perceptions- government is not including settlements into their planning"

The following questions were asked to most of the interviewees: *How are environmentally constrained areas mapped in Kathmandu Valley (if any)? Are those mapped areas considered during the planning process for new development in KV?* The respondents highlighted a lack of any mapping

being undertaken in the past let alone mapping determining development in the valley that has mitigated development in disastrous areas.

Interviewee	Quote
UNP 1	"There has been no detailed mapping incorporated into any development process, with no standard methodology for hazard mapping that seems to ever be agreed on, no mapping will ever be undertaken and Kathmandu will continue to be impacted by disasters."
Eng 2	"The problem is that [the] maps are not strictly followed by development settlements or housings or other things. Enforcement of those risks mapping is a serious problem. The mapping itself is there. It may not be in much detail. But it exists. Enforcement is the weak point"
Arch 1	"KV is a unique combination of rivers/hills/ forests/ lakes, this has impacted how the land use has changed as some of the areas have been changed to development, as this has been repeated over the years, it has made Kathmandu Valley more at risk each time."

To provide an initial collectively image of the interviewees' opinions of the planning of Kathmandu over time, the following question was asked to most respondents: *How has Kathmandu's development been planned over time?*

Interviewee	Quote
Arch 3	'There was an earthquake Impact Study done in 2002 and a Flood Study done in 2009 however these are not considered in the approval process for buildings and they also have no legislative power'
UNP 1	'Land changes in Kathmandu Valley has resulted in limited developable land, this has left the attitude to build higher and in some cases on top of already poor designed houses.' – iUNP (Prem UNRCO) Therefore no planning has really happened.
GArchP 2	"Urban Management within KV is difficult. Due to plotting and fragmentation of land. Also because of Institutional memory keeps changing"
UrbREM 1	"What is the actual scenario of urban context, what urban assessment tools are available and why types of risks are there in urban, are all things that have not being considered or addressed through the planning and development of Kathmandu Valley."
GArchP 1	"Kathmandu has not been planned, it has just happened. In 1966 building by-laws were brought in to try and make some control, before then all that existed was social norms and standards and practices, for example 2-3 story, but that was all. This is because it can't be higher than the temple."
GEng 1	"In Nepal, we don't have urban planning. In many stances, we just have ad hoc and reactive kind of planning. In many of Municipalities they don't even have urban planning, theory planning document. So, even if municipalities do have some kind of documents. It's not the documents municipalities' prepared themselves"

Factor Three- Disaster impacts on urban resilience of Kathmandu

Factor Three, Interviewee's were asked the following question: *What are the disaster impacts on the land use changes in Kathmandu Valley?* Most agree that the focus from disaster resilience has shifted and it's not been acknowledged that hydrological disasters are causing more impact on the valley than earthquakes.

Interviewee	Quote
PM 2	'Other disasters need to also be considered as part of disaster management, as a reactive country, the current focus is on Earthquakes but there is more damage from other hydrological disasters.'
Eng 2	In terms of damage to property and economic losses it is far greater than the earthquake. In terms of life loss, it may be earthquake that killed 9,000 people in one event. But hydrological landslide hazard is killing more than 300 people each year. In time frame of 20-30 years it becomes more people. Flood landslide, these events are actually killing more people and damaging more infrastructure and people's livelihoods. Cost is more."

Theme Urban Development: What role has urban development and urban planning played in reducing the impact of disasters for Kathmandu Valley?

Factor One: Unplanned urban development

Have land use zonings, urban growth boundaries, restricted development areas, or risk mapping been used to control growth especially in disaster risk land within Kathmandu Valley historically?

Interviewee	Quote
Eng 2	'Some of the Historical areas, whole heritage sites. On those areas construction and expansion of settlements or even dismantling the whole architecture and building something new is not allowed. The policy was here to preserve the historical, archeologically sites. Zonings for residential and commercial... those terms are only now coming up and being used. Earlier those things are not designated. But for example there is a flood risk mapping of the rivers in Kathmandu and certain areas on either side of these rivers are not allowed to be built... however evidence shows if you walk around these areas, there is lots of development still happening"
GArchP 2	"The Kathmandu Valley Risk Sensitive Land Use Plan aims to apply zonings and restricted development areas due to hazards risk. It will also give authority to power to plan for land uses. At the moment Kathmandu Valley's Municipalities don't have the power to do that.... Previously there has been lots of studies and plans to do this but no implementation.'
GArchP 1	"There needs to be some sort of Risk Sensitive land use plan in the valley for any resilience to happen... this would include Restricted land for developed propose development land and land use and zoning mapping."
Arch 4	"A risk sensitive land use plan... that is what we need. It has been prepared but no risk sensitive land use plan was initiated before the earthquake, but due to cabinet reasons it was delayed, and because of the earthquake it has been further delayed... if it was approved earlier, it might have made a difference to the disaster we had in Kathmandu"
GEng 1	"Ministry of Federal Affairs which is the line ministry for Municipalities and development committees, they selected a few municipalities to prepare the periodic planning for 5 years. What happened was it was not local thoughts, it was something

	supplied by central government. So what happened was...the consultant came and they formulated the plan, but nobody looked back into the document and this example is exemplary to all other urban areas. In many stances, they don't have any kind of planning at municipality level, if they have it, again in many stances it is not produced by local level, it is produced by ministries or departments, consultants etc. It is therefore ignored on the ground and a big issue for urban resilience of Kathmandu. "
GEng 1	"Yes, it is because one is the ownership issue, another is government's priority and the enforcement. Enforcement is because of the priority. So very few municipalities do have land use plans... They are not, none of them have been enforced, I don't see any."
GArchP 2	"Restricted areas do not exist, only in government areas... there is some setbacks only from rivers, but these are not usually enforced in all cases"
Arch 2	"Political instability over the years has caused many documents to not be passed in time to mitigate disasters."

Has there been any attempt to zone areas within Kathmandu Valley as not-urban or disaster risk areas?

Interviewee	Quote
CC 1	"Kathmandu Valley has always been a 'rent seeking' behaviour, just build and build with no checks. It wouldn't matter if there was risk areas identified, they would be ignored. There has also been no change in this behaviour since the earthquake"
PM 3	"There are zonings for heritage. Temples and other sites are protected. But zonings restricting development has not occurred in the Valley"
Arch 2	"There has been minimal attempt to reduce development occurring in risk areas. An example within Kathmandu City (including Latlipur) as there are 5 rivers across the city and yet loss of life and properties in last 5 years is significantly high and not changed the rate of development in that area."
iPM 1	"This sort of planning is very new to Kathmandu Valley, they are still working out what it does. The RSLUP will do this."

How effective is the building permit system within Nepal? Primarily within Kathmandu Valley?

Interviewee	Quote
Eng 1	"Checks are irregular and poor"
GArchP 1	"Until there is an Urban Planning Act, the effectiveness of the building permit system will be poor and history will just keep repeating itself"
HD 3	"In theory setting building permits were always required in Municipalities. However, whether the government was able to enforce it or not was another question. But it was always in place"
HD 3	"Before the earthquake, the MOUD and the DoUDBC and different local level authorities only wanted back of the envelop sketches, now they want actual detailed technical drawing"
CC 3	"There was limited support and enforcement from Municipalities around monitoring building permits, the result of this can be seen today with the damage around us (in Kathmandu)"

Factor Two: Urban Growth Boundaries

Factor Two, How effective do you think urban growth boundaries would be for controlling urban sprawl and mitigating disasters within Kathmandu Valley?

Interviewee	Quote
GArchP 1	‘They don’t exist in Kathmandu Valley and probably won’t. An example is even now the government is demolishing and building 250 wide on both sides of the outer ring road. As part of a land pooling project... Government has no resources to build its own infrastructure so this is the only way. It has had land pooling projects in KV since 1990. 25 currently in KV... there is no more land in Kathmandu to do these sorts of projects however, therefore the infrastructure will remain the same include the uneven cut up of land”
GEng 1	“This is something that has never existed in Kathmandu Valley.... Now that I think about it, it should have... it would have made a big difference on where development occurred, and we might not have so many buildings in areas of high risk.”
Arch 1	“All you need to do is look at maps of Kathmandu Valley dating back 30 years and watch the development spread out over the Valley in no coordinated or organised manner. This is one of the biggest problems. If there had been some sort of control over how it spread. Maybe we wouldn’t see such devastation from flooding near rivers or even the destruction from the earthquake.... Yes, I think it would have made a difference.”
Arch 2	“The urban sprawl as you call it, has caused major problems in how to deal with disasters. Mostly in retrospect to access. With narrow streets, if there is a fire or an emergency, no one can get in or out. This has just spread and spread outside the initial ring road. Now it’s spreading even further into the valley. The area is becoming more and more at risk... a boundary would help with lowering these risks.”
GArchP 2	“Growth boundaries have never been used in Kathmandu Valley... this is due to the lack of planning in general... if a growth boundary had been used, it wouldn’t have solved all our problems, but it might have helped.”

How has urban expansion and growth within Kathmandu Valley been guided by hazard risk information?

Interviewee	Quote
Eng 2	“No, not like that, but there has been some attempt by government to have planned settlements. Government sets up a project. Government temporarily takes the land from private owners and it allocates some part of the land for roads, public spaces, and other necessary infrastructure. It is kind of urban planning. After that they hand back to the owner. Maybe their land is reduced, but it is much better land when they get back. This is Department of Urban Development and housing that looks after those projects... however hazard risk information is not considered or factored into these projects’
Arch 2	“Government doesn't have any urban disaster risk reduction plans and therefore risk reduction hasn’t been incorporated into urban development. There also are no new policies expected to happen”
GEng 1	“No, no, no, in policy levels a few years back, the government was talking about having satellite cities, especially within Kathmandu, but I don't know

	the progress. It's the only thing they are doing, but satellite cities won't solve all the problems, there is other issues... the cities also weren't looking at being located in areas of no hazards... it wasn't even a consideration in the planning"
GArchP 1	"Government has attempted small scale Strategic Planning as you call it, where it gathers plots of land and develops up a neighbourhood, providing infrastructure, public park, roads... however during this process no hazard mapping is done and they don't find out if the site is at risk before constructing.... There is no link between these two."

Factor Three: Urban Planning ability to reduce damages in the 2015 earthquake

From your knowledge, how has development in Kathmandu Valley addressed urban planning legislation?

Interviewee	Quote
HD 2	'It is now compulsory to have setbacks from the road and some height restrictions however it wasn't before the earthquake... Certain housing designs now require approval from municipalities. Yet to see how effective that has been as the Monitoring and Evaluation is still to occur... if this had been the case before the earthquake, we probably would have seen significantly less damage."
GArchP 1	"All Municipalities in Kathmandu Valley have separate by-laws, only a few recently adopted the Government modern standard by-laws and Building Code, however most of the construction that was damaged in 2015 was done prior to that adoption... this shows the type of construction that was happening."
UrbREM 1	"What is the actual scenario of urban context, what urban assessment tools are available and why types of risks are there in urban, are all things that have not being considered or addressed through the planning and development of Kathmandu Valley."
GArchP 1	"The Core area history goes back 2000 years. Many old and historic buildings. The Last 20 years most development in the valley doesn't comply building code and only a few municipalities actually enforced the code. This directly affected the amount of damage that happened. The Government is very serious after this earthquake."
HD 2	"The Policy and the practical are very separate. Even down to the cultural differences of construction between masons and engineers are different... policy is not reflective of what's happening on the ground and therefore has not been enforced by the people who are main respondents for enforcing it."
GArchP 1	"Kathmandu needs to seriously consider an urban regeneration of Kathmandu as most buildings are still in trouble of future disasters.... This is a direct result of the lack of enforcement of building standards in the past... we need to do something now, or we will be in even further trouble."
GEng 1	"They do have planning documents, but very few things are happening according to their planning documents. In terms of, talk about planning. We do limited urban planning and even if we have some planning documents, these documents are not owned or enacted by local governments or municipalities. Another is the development activities, the planning documents are the guiding documents for the development. But this never happens nobody bothers to do the planning documents, even the city

	councils other councils, and all councils, they endorse the whole projects at city level, but I was part of city council when I was working for Municipality but I never realised nobody was bothered to look into the planning documents.”
iNGOP 1	“A study recently done by Ministry of Urban Development and collaboration with JICA showed that 94% of buildings in Nepal are unsafe and not up to building standards. The building standards are complex and the knowledge is also missing.”

Factor Four: Effectiveness of Disaster Management

What work was done for earthquake resilience for Kathmandu Valley prior to the earthquake?

Especially in field of development and regulations. Was there any focus on other disasters (e.g. for fires, floods, landslides)?

Interviewee	Quote
iCC 1	“Local Disaster Risk Management Plan which is done by MoFALD. It is owned by MoFALD and it is implemented by NGOs. This is supposed to happen at Local development level, last count there was 1000 out of 4000. However, they have done them and they are done for projects only and information is not shared and not shared in meaningful way and is not stored. No one has a copy. Local Disaster Risk Management Plans are supposed to identify the priority areas for that local development area on disaster management.”
CC 3	“Disaster Management has always been just for response, very little ever focused on preparedness or even risk reduction”
Eng 2	“We need to develop some guidelines some regulations for that. Even we can include that into building codes itself. Into the foundations when they construct the building- they have to look into the soil type and whether the area itself is a risk zone or not. There could be flood, earthquake, landslide risks. That is not integrated into the building code. Building code is structural only. Now mainly focusing on earthquake resistance. Now they have revised the building code based on learning from last disaster.”
Arch 2	“There has been some work on earthquake mapping and flood mapping to try and then go into disaster risk reduction policies... but these were a while ago and we didn’t see anything happen with them..”
GEng 1	“Urban resilience is a new word. This has not been practiced in Nepal. You won’t find anything in our policies or legislation on resilience... there are policies on disaster risk management, but they don’t sit under urban planning and don’t therefore have to be considered in urban development...”

There are quite a lot of Disaster Management policies (Disaster Relief Act, National Action Plan on Disasters etc.) how have then been linked back to urban planning and development plans and strategies. Which have priority when it comes to development of homes, schools, roads etc?

Interviewee	Quote
CC 3	“The Policies exist for disaster management but they don’t have mechanisms to support it. Plans or capacities to deliver them are never drafted or enforced... if they aren’t enforced... we won’t ever see change.”
DisMEe 1	“A lot of the policies are not binding and therefore not enforced by local governments, let a lot looked at when development occurs.”

PM 2	"The National Calamity Relief Act and Disaster Relief Act are over 30 years old but is still running with that act, it needs to be updated... this is the case with a lot of legislation around disaster management."
PM 2	"Disaster Risk Management plans were a thought, and 96 were developed but nothing happened to them....so we begin the journey again. This is a common thought on most policies to do with urban development also." – PM (ADRA)
GArchP 2	"Disaster Management and the Acts around that sit under a different department to Planning. Those two Ministries don't talk and don't like each other, therefore limited integration happens and in most cases their projects and plans are in silo of each other."
GEng 1	"There is no link. The damage that happened in the earthquake shows this. There never has thought to be a link. Even disaster risk reduction is a new theory and it doesn't consider urban development in its plans."

Prior to the earthquake, was there any site selection process that considered geological investigations to build and promote disaster risk planning, especially around deciding where development and infrastructure was built (not just houses, but also roads, hospitals, schools, train tracks, etc.)?

Interviewee	Quote
Eng 2	With regards to new settlements do they look into risks? "Fire risks is built into building code. They do not look into the mapping of flooding, they do not look into the earthquake risk zone. But genuinely. They do try to avoid. They do not build the city in the flood plain they go for other areas. In some cases if there is a flood plain they try and put those areas as a park. They maintain some open spaces."

Factor Five: The Bigger Picture Thinkers

Factor Five, Some of the key comments are provided below:

Interviewee	Quote
iCC 1	'Min of Urban Department is enforcement now of only the building codes. They have not enforced the land use planning. They could enforce those things, but it's not built into their regulation. There is a lack.'
Eng 2	"Biggest lesson learnt is that in preparedness. Need to therefore focus more on preparedness and risk reduction. Not only reduction of the losses, but reduction of risk themselves. But reduction is a very tricky part because it should be integrated with development process and it's a continuing process. The important thing is how to mainstream disaster risk reduction into development plans. That is a very critical thing. Mainstreaming in form of risk assessment, in governance, in investment, in enhancing preparedness"
UrbREM 1	"Planning in urban areas for resilience is different as people are dependent compared to rural as rural is independent. Rural have no movement issue and can usually rely on themselves. Urban movement is difficult due to the very high density, which makes urban areas highly vulnerable to disasters through the sheer amount of infrastructure and access issues (getting out, and getting help in). Their vulnerability is also increased during an urban crisis as they are so dependent on other areas and can very quickly lose livelihood. Multi-hazard risk. Preparation and mitigations should be focused on that and not on rescue phase."

UrbEM 1	"There are many other disasters that Kathmandu Valley needs to be addressing,, not just earthquakes, there is flooding and landslide issues that need to be built into the building codes and any future development"
PM 2	"There was a paradigm shift in 2012 from response to disaster risk reduction within Nepal, however the impacts of this shift couldn't be reflected during the 2015 earthquake as not enough had happened to make a major change."
GEng 1	"It was like we nowadays, we have this vocal bottom up planning approach but again it is very reactive, because annually, before city council, we do some formalities at local level, like ward level, and we collect the schemes and projects that the community has prioritised, but it is not the plan. This is just the, the kind of needs assessment. You assess the means from the lames of the local people. For me it's not planning, it is a tool you can use while you prepare your planning document, but it is not a plan. But it has been termed as a kind of planning approach."

Theme Urban Governance: What lessons can be learned from urban development management within Kathmandu Valley with regards to urban disaster risk reduction?

Factor one: Historical Governance and Political Structures and their impacts

How has historic governance of Nepal played a role in the development of the Valley?

Interviewee	Quotes
iNGOP 1	"If [urban development or disaster risk legislation is] not passed, it probably won't get passed. The parliament passes things very quickly when it wants something passed and it will sit on things it doesn't..."
iCC 1	"Nepalese congress currently governs the Central district which is super conservative and therefore everything is run from Central which is Kathmandu. All the money and decisions come from central government. At a District Level you only get 20% of your budget to decide what you can do with it based on the needs of what you want to do. The rest is dictated by Central Government."
iCC 1	"Within Kathmandu Valley those who are CEOs of Municipal Governments which is highest position here are arguably more powerful then district CEOs and therefore there is a clash. Those positions are allocated based on political interest. When you have an overturn in Prime Minister or overturn in Ministers which happens every 5-6 months, those CEOs change. Therefore, the ability to do long-term planning is greatly accredited."
Eng 2	"Before it was central government delegated power. That means they can control that."
UNP 1	"Due to the way government has formed in Nepal, it is very reactive and not proactive, for all things from planning to disaster management."
HD 2	"Change of government every 9 months doesn't help the implementation of urban development in a resilient way."
Eng 2	"The government instability is an issue... this has led to development not being planned, due to regular changes in government and therefore a high decision turnover."
CC 2	"With the historic political system we have, the government and the people of Nepal are very reactive, it's going to be a hard push to move this to proactive for all disasters in Nepal."

PM 1	"Decentralise power from central to local community, needs to actually happen and not be a talked about topic for the last 20 years... without this change of power, we won't see any change in the way our areas are built."
PM 2	"There has been progress on coordination of all sectors before the earthquake, but Government changes are always making the process slower."
GArchP 1	"A good example of how long it takes to get things passed in parliament is the Disaster Risk Management Bill has been prepared and in 'discussion' for 7 years."
iNGOP 1	"Previously the constitution made no explicit mention of local government bodies and their roles, this was why they had no power to do very much at all, especially when it came to urban governance"
iNGOP 1	"Here they have simulation problem of being really quick and amazingly fast at doing things, and things go wrong because they are too quick and then just turning tables and taking too long to do anything, they don't have a middle ground. I think it is (cultural), but then politically I find it fascinating how they do expect decisions to be implemented straight away, there is no lead in time or anything, looking at consequences of this policy, it is like from tomorrow we are doing this."
iNGOP 1	"A lot of policy for past 5 years has just been waiting for the new constitution and then elections to be put forward to cabinet... this has been something that's been repeated over the years for all sorts of reasons."

Factor two: Key Government Actors

Who are the key government actors relating to urban development and urban resilience for Kathmandu Valley?

Interviewee	Quotes
ICC 1	"MinHome Affairs, don't have any money and don't have a budget for it but they have a plan. MoFALD have a budget for disaster management and don't have a plan. They don't like each other and don't talk to each other."
Eng 2	"Problem is here in disaster management process, there are two aspects. One is Disaster Risk Reduction the other is Response and reduction of losses. One if risk management another is crisis management. The focus till now is more on crisis management. Loss reduction. We have now in new policy to focus more on risk reduction. But people do not understand its importance, because risk reduction has to be built into the development process. It is not a single department that can do risk reduction. All departments, all development partners, have to build in risk reduction into their development. They have to think how they can reduce the risks with their development. One- disaster may impacting their development. Second their development may be enhancing disaster. They have to think their development should not be increasing risk. And their redevelopment should reduce risks. They should think of exisiting risks and taking their actions further."
PM 2	"Governments of all levels are not directly linking with ground realities they are trying to tackle... therefore there is no actual key government actors"
GEng 1	"I don't know why they named them Min of UD as they don't have any relation with the Municipalities so I don't know where they are going to develop. Municipalities, line min of Municipalities could be Min of UD or they could have some sort of dotted arrangement."

Factor three: Key decision makers and implementation

How involved are the community in the decision making of development within their neighbourhoods or even city wide?

Interviewee	Quotes
Eng 2	"Not in urban development no. For earthquake preparedness, there has been community groups established."
GArchP 2	"Field survey done by UNDP for community integration found that there has been no serious engagement from Municipality representation to their people, this is a representation of historically contexts to which development has been occurring in the Valley."
PM 2	"Communities are very unaware of government policies and strategies for Kathmandu Valley. There is a huge gap between what the government regulates and what the people do."
GEng 1	"No they don't. If you want to construct a house, or build some apartments, if you want to do construction you have to get permission from Municipality but it does not require you to get permission from neighbours, unless it's industry or bigger infrastructure that falls under category of EIA or IRE guidelines, they have to do that, in principle, but there are ways around it."

How effective have the National Building Code and Building By-Laws been? How much have they been implemented in urban development?

Interviewee	Quotes
Eng 2	"Approvals are given by Municipalities. While giving approval they want to see structural design for earthquake resilient building and there should be supervision of engineer. This however doesn't happen much in practice and there always seems to be ways around it"
UNP 1	"There is no implementation or enforcement of the building codes from Municipalities as they are more focused on getting more taxes so the more buildings and residents the better. They are more motivated by revenue then risks"
HD 2	"Government has been trying to have standards/codes but very strict standards, therefore people just building own as they can't meet the standards"
CC 2	"All Municipalities have been entitled to implement it [NBC]. But capacity is only within DoUDBC. This is due to fragmentation within Government"
GArchP 2	"The biggest issue is that local authorities can apply the Building Code whatever way they see fit and feel is appropriate in their jurisdiction... in some cases they just don't apply it"
GEng 1	"Yes they prepared it [DoUDBC]. And they are responsible for building standards but for the implementation they don't have any capacity to implement because again it is municipalities and VDCs....And the line Ministries for local authorities is MinFALD. If they don't own it, or initiate it, nothing will happen..."
iNGOP 1	"A study recently done by Ministry of Urban Development and collaboration with JICA showed that 94% of buildings in Nepal are unsafe and not up to building standards. The building standards are complex and the knowledge is also missing."
GEng 1	"The big municipalities, a few small municipalities, have some kind of land use plan, but they all obsolete and never have been used...I think there are some enforcement issues and development issues as well. The ultimate property

	rests with government but we have individual property rights... so it means I'm free to use my land the way I want to use it. Even though the ultimate authority rests with Government, we have been through lots of troubles and political changes, government was not that strong, to implement that. The land use plans. And it was not a priority at all, in another way, and you know the organisation chain, and there was zoning and according to a zone some particular is allocated for farming, but they don't get a high land price... the people are getting it for the new settlements or built up area, there is no incentive, there is a plan, but there is no incentive for each cluster. So people are tempted to sell their land for housing or business complexes, whatever comes, and gives them more money."
GEng 1	"If you (as local authority or government) don't have anything properly planned, you can give money to your follower whenever you want to construct road, construct building, public amenities, if you don't have a planning document. You are more powerful. It is a risk if you have to be guided by the planning document you have very limited say in status-quo..... The interest is another thing that the employees interest, political interest, followers' interest, they are there, so I don't know how sensibly they will proceed this planning thing. But authority wise it, before it was various authorities various departments but now all public development related authorities and departments are under one umbrella one municipality so the possibility does exist. And legally, there are policies that all Urban Development should be guided by planning document. They are there, since long I guess, but it's not happening."

Factor four: Monitoring

Factor four, What is situation of monitoring of urban development in Kathmandu Valley?

Interviewee	Quotes
HD 2	"Very few buildings we have dealt with in Urban setting previously got approval to build their houses"
DisMEe 1	"The overall issue with urban development is around monitoring and evaluation of urban projects, the governments lack capacity and don't undergo required monitoring and evaluation"
GArchP 1	"Previously there has been no inspection done in the field even following a building permit being granted... in most cases this means what was approved wasn't what was actually constructed."

Factor five: Coordination

How connected and coordinated are the different sectors? Especially transport, environment, infrastructure, disaster risk management and urban planning?

Interviewees	Quotes
ICC 1	There is a tension between MoFALD and Min of Urban development. They both say they do similar things and have been for years, and nothing ever really changes. Urban Development work with Department of Urban Development and Building Codes. MinFALD have a budget for Disaster Management but then you have Ministry of Home Affairs and they are the disaster response agency they have responsibility and they have a plan which is called the DERP."

GArchP 2	"MoUD do all the technical planning for the cities- base maps included. Municipalities however fall under a different Min. This is where the problem lies."
Arch 4	"No and that is really a problem. The coordination is really a problem "
Eng 2	"A problem here with new settlements is that government doesn't build the settlements, settlements are built by private sector. And the land owner themselves or some housing profits. Department of building construction build only government officers. "
GArchP 1	"The Valley has many municipalities with interrelated infrastructure, however due to urbanisation of unplanned settlements and lack of coordination, major issues have arisen."
GEng 1	"This is the case, you can see everywhere, it's very vivid this is the case, an example is you go out to the road, it is black topped and the immediate day after the telecommunication come and start digging the whole for themselves, they complete it, and then electric authority comes in and they do their part, then water supply authority comes in and they do their part. We don't have this cross- departmental who have an organised planning approach. We are very reactive. The activities we do are basically, are isolated in themselves."
GEng 1	"Municipality is one authority, road department is still there with another authority, water supply is another authority, electricity is another authority, land is another authority, industry is another authority, so there were multiple power centres, which were not actually integrated. So you know they were not integrated and that itself was a problem."

Historically how have different sectors worked together to plan for disaster risk management, has any of it been through urban development strategies?

Interviewees	Quotes
CC 2	"Coordination between agencies and organisations (public and private) is a major issue when it comes to reducing risks in urban contexts"
PM 1	"Platforms in the past have been set up to try and coordinate and share resources between governments, agencies and NGOs, such as the Urban Environment Platform which existed for 2-3 years but never had any meetings... all attempts at coordination in the past has failed."
GArchP 1	"There has always been segregated authority of who looks after urban development from housing, water, education, infrastructure, disaster risk planning etc. within Nepal"

Before KVDA and even now, how do Municipalities work together in Kathmandu Valley to plan for growth and implement Building codes through similar planning processes?

Interviewees	Quotes
GArchP 1	"KVDA now Coordinates with all departments and sectors and is independent... however, before they became established in 2012 it was a mess"
GEng 1	"No, not exactly, they have their own rules, it's very ironic you have Ministry of Urban Development and Ministry of Federal Affairs and Local Development. Ministry of Federal Affairs and Local Development is line Ministry for Municipalities... You can see the discrepancy already as Municipalities doesn't have any accountability to Ministry of Urban

	Development. They have enforcement problem, they cannot do anything even if they wanted to do it. If they want, for example Ministry of Federal Affairs and Local Development they started promoting environment friendly urban development and Min UD couple years back, they started promoting clean cities approach, because clean city approach, there was a green city approach, they were all jargons, the city council can donor and development partners, they have their own set of kitchens where they cook their own food, they don't coordinate while developing this environment friendly local government. Ministry of Federal Affairs and Local Development, didn't bother to invite Ministry of Environment or Ministry of Urban Development they just did it. If Ministry of Environment has to do something, now we have some legislative, but because of the institutional arrangement they cannot perform well, so they cannot do anything at all, except endorsing, having endorsement of the EIA or IRE reports. That is all they can do. Ministry of Urban Development don't have any institutional accountability as well to the Ministry of Federal Affairs and Local Development or Municipalities, it is not happening you can see that."
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Factor six: Resource and Capabilities

What is the knowledge, financial capacity and resources for local levels to design, plan and implement building codes and contingency plans for building critical infrastructure within KV?

Interviewee	Quote
iCC 1	"[In Disaster management], there is only a district level budget, there is no funding down to the ground level. Who keeps that is the LDO which is MoFALD."
iCC 1	"Biggest factor regards to long term planning, is political system and lack of revenue raising and ability to raise revenue."
Eng 2	"At this moment, the biggest problem is they don't even have the basic infrastructure. They don't even have a building to work in or house in. It takes time."
HD 2	"There has always been a lack of resources for local governments. Without engineering, how can we ensure quality of the construction?"
CC 2	"Local Authorities have significant lack of resources and therefore lack of preparation for disasters, like the earthquake in 2015"
UrbREM 1	"There is a great need to enhance the capacities of local bodies. Government has recently included systems and guidelines in place to build a level of urban resilience but the capacity of local government staff is very low. There are no financial resources or technical logistics items also. This has always been one of the biggest issues for Kathmandu Valley and resulted in the urban crisis we saw in 2015"
PM 2	"There is a need to improve Municipalities technical skills and capabilities as they are not even fulfilling basic requirements for their people."

Lessons for Urban Planning of Kathmandu Valley

In your opinion, what has been the biggest issue for urban planning and development in mitigating disasters within KV?

Interviewees	Quotes
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iCC 1	"Biggest factor regards to long term planning, is political system and lack of revenue raising and ability to raise revenue. Corrupt is so pervasive and therefore it is difficult to have long-term projection of how much revenue you could raise. Lots of example of big companies not paying tax as they are politically aligned. A good example is Electricity situation in Nepal. Over last 10 years there has been severe power cuts down to only having power for 8 hours a day and the city would be on a schedule. You would get a schedule of when you did and would not have power. This led to massive increase in generators being bought and backup batteries being bought. This was just blamed on increase in population size. What it was in fact is that industries were buying off power and getting power meant for Kathmandu residents converted to Industries. In Dec last year there was a change in Government, this person wanted to prove he was able to do something. He therefore appointed someone new in the electricity board. The problem was basically fixed in three days. And the prime minister then fired three key ministers, and now we have power 24 hours a day. It was solved. So looking at those systems and how much power is in politics."
Eng 1	"Implementation and monitoring of the building codes. Most people in KV don't go get building permits..... Construction techniques are also a major issue with most masons not being trained in how to construct houses."
HD 2	"Monitoring and Evaluation implementation and ownership are the 3 main issues with planning in KV"
Arch 2	"The biggest issues is that the growth happened so quickly and then development occurred without control and now you have unplanned, poorly designed, in risk buildings all close together, they can't even mitigate fires right now let alone a major disaster like an earthquake, massive flooding or landslides."
HD 3	"Engagement and coordination at implementing level and not policy level"
GEng 1	"The way the planning documents have been prepared is a problem. So the document actually, should be. Considering we have just restructured, I am not aware of how the authority will recently restructured will be practiced by the municipalities, now they are more powerful than earlier. They are independent then earlier. Earlier they were only local institution, now they are local governments."

Photos

Appendix 17: Photos of buildings following the 2015 Earthquake



Collapsed adobe (mud brick) houses located a few blocks Bhaktapur, Kathmandu Valley. (Credit: Satoru Nishikawa. Source: World Economic Forum, 2015)



A man walking through the devastation in Bhaktapur, Kathmandu Valley.

Source: <http://www.abc.net.au/news/2015-04-29/earthquake-damage-in-bhaktapur-near-kathmandu-nepal/6429298>



Kathmandu City. Source: <https://makezine.com/2015/09/09/the-great-nepali-radio-shake-up/>



Kathmandu City. Source: <http://www.circleofblue.org/2015/world/nepal-earthquake-damages-at-least-14-hydropower-dams/>

Conclusion

Appendix 18: A list of recommendations

For Kathmandu to achieve a more resilient urban space there needs to be overall a more integrated approach for achieving disaster management through the conditions of physically, socially, economically, environmentally and culturally safe and resilient urban spaces.

There is a strong disconnect between the policies and realities. This is very prevalent in the case of politics and a neoliberalism attempt to decentralise that went wrong. As politics are the driving force of implementation and change when it comes to building a resilient city.

Importantly no single mitigation strategy can be relied upon to solve all the hazard problems; it requires a combination of approaches (land use planning and management, warning and evacuation provisions, relief measures, structural protection etc). This paper however aims to highlight that land-use planning and management processes and provisions have a very important role to play in enabling communities to mitigate disasters in a way that is appropriate to their situation.

Recommendations:

- Creation of integrated digital data base system involving each agency concerning disaster management.
- Integrated of disaster management plans with all future master plans.
- More frequent reviews and updates of the building code, building regulations and guidelines and planning by-laws on basis of lessons learnt with mechanisms to enforce and monitor them in all urban areas.
- Increase technical capacity of municipalities (local bodies)) to ensure building code/regulations/guidelines and planning-by-laws.
- Change the process of approval to ensure that more detailed drawings are submitted and check-ups are undertaken periodically through the construction process by engineers/architects/planners.
- Prepare a simplified building guideline/by-laws that can be understood by people on the ground who are constructing their own buildings (this should be phased out over time).
- Facilitate mandatory enforcement and periodic monitoring of land use regulation, building code/regulations/guidelines and planning by-laws in all urban areas.
- Something this report didn't address but was noted through field trip and interviews is that Nepal lacks significant retrofitting regulations and requirements, not just relevant following a disaster like the earthquake but also for any retrofitting of older houses. There needs to be capacity building around safer building practices for retrofitting given to municipalities but also technical support in regulations on what this is.
- More monitoring needs to be done by KVDC on structural and functional change of buildings and land use. Private buildings originally designed for residential purposes like public schools, hospitals, restaurants and department stores are exposing large populations to risks when the building use changes. It is important to therefore have a comprehensive guideline and process for changing building use (some of this idea is from MoUD, 2015) but not it all.
- Preparation of soil capacity map and risk sensitivity land use plans for all urban areas.
- Importance placed on site location especially against floods and landslides
- Accessibility factors of the site during emergencies.
- New codes needed for fire, time, retrofitting, non-structural safety, (Look at BCC code and add others).

- Proper coordination with the municipalities by the sectoral organizations like DUDBC, department of urban development, responsible organizations for drinking water, real estate, sewerage etc. is essential while planning for the urban development. Building and strengthening relationships between various actors and stakeholders through more effective communication is necessary.
- Local government, especially the municipality needs to be made competent enough to regulate urbanization.
- Urban lands in Nepal's cities have been used in construction purposes mainly. There should be multiple use of land so as to optimize its value.
- Proper urban planning using land-use planning tool can be used for infrastructure management. There should be clear rules and policies about building homes and there should be a strong regulatory body to make sure that the policies and rules are implemented.\
- Disaster mitigation techniques and strategies needs to be woven into all parts of planning. All planning documents need to be centred around building resilience and mitigating risks. Kathmandu Valley has a history of flooding and earthquakes, there should be more integration between mitigating these risks and deciding where urban development occurs and at what standard it is constructed.